

Discuss and recall — the Grade 5 course

An entry diagnostic: what Grade 5 left behind, and what Grade 6 will need on the first day.

LESSON 1

1 × 40 MIN

MATEMATIKA 6, KIRISH

STAGE 6 DIAGNOSTIC

LESSON CLOCK

6'

20'

10'

4'

What Grade 5 covered	6 min
The diagnostic	20 min
Answers and diagnosis	10 min
The year ahead	4 min

LEARNING OBJECTIVES

- Recall the four operations on whole numbers and on fractions.
- Recall the order of operations and the use of brackets.
- Recall perimeter, area and the units used for each.
- Identify, from the diagnostic, what needs revising this month.

TEXTBOOK

National	pp. 3–8
Cambridge	Stage 7, pp. 1–6
Workbook	Entry test

01

Explanation

What Grade 5 covered

BLOCK	WHAT YOU SHOULD BE ABLE TO DO
whole numbers	add, subtract, multiply and divide in columns
order of operations	brackets, then $\times$ and $\div$, then $+$ and $-$
fractions	simplify, compare, add and subtract with a common denominator
decimals	read, order, add and subtract
measurement	perimeter and area of rectangles, and the units
angles	measure with a protractor, and name acute, right and obtuse

NOTHING HERE IS FINISHED

Every one of these blocks returns this year in a longer form: fractions are divided, decimals are multiplied, and area moves from rectangles to circles.

The diagnostic — 20 marks, 20 minutes

Q	TASK	MARKS
1	$348 + 2\,976$ and $5\,004 - 1\,897$	2
2	$47 \cdot 26$ and $918 \div 27$	3
3	$12 + 3 \cdot (8 - 5)$	2
4	Simplify $\frac{18}{24}$ and compare $\frac{3}{5}$ with $\frac{5}{8}$	3
5	$\frac{2}{3} + \frac{1}{4}$	3
6	$4.7 + 2.85$ and $9.1 - 3.62$	3
7	A rectangle 7 cm by 4 cm: its perimeter and area	4

THE ANSWERS

$3\,324$; $3\,107$; $1\,222$; 34 ; 21 ; $\frac{3}{4}$ and $\frac{3}{5} < \frac{5}{8}$; $\frac{11}{12}$; 7.55 and 5.48 ; 22 cm and 28 cm^2 .

Diagnosis

IF Q... WAS WRONG	REVISE	BEFORE
1–2	column arithmetic	the first week
3	the order of operations	algebraic expressions, lesson 8
4–5	simplifying and adding fractions	dividing fractions, lesson 22
6	decimals	the Cambridge decimal block, lesson 76
7	perimeter and area	the circle, lesson 92

A GAP LEFT OPEN IN SEPTEMBER CLOSES NOTHING BY ITSELF

Each row above names the lesson where the missing skill is needed again. That lesson is the deadline — not the end of the year.

The year ahead

QUARTER	MAIN BLOCKS	HOURS
I	directed numbers, expressions and equations, dividing fractions, ratio	54
II	percentages, decimals, angles, the circumference of a circle	42
III	the area of a circle, compound figures, speed, volume	60
IV	data and pie charts, probability, solids and nets, revision	48

SIX HOURS A WEEK, NOT FIVE

This class has one lesson a week more than the national plan needs. That hour carries Cambridge Stage 7 work — negative numbers, decimals, data and probability — which the national Grade 6 programme does not cover at all.

02 Worked examples

EXAMPLE 1 Compute $12 + 3 \cdot (8 - 5)$.

1 Brackets first: $8 - 5 = 3$.

2 Then multiply: $3 \cdot 3 = 9$.

3 Then add: $12 + 9 = 21$.

ANSWER

21

EXAMPLE 2 Compute $\frac{2}{3} + \frac{1}{4}$.

① The lowest common denominator is 12.

② $\frac{8}{12} + \frac{3}{12}$

③ $= \frac{11}{12}$

ANSWER

$$\frac{11}{12}$$

EXAMPLE 3 A rectangle is 7 cm by 4 cm. Find its perimeter and area.

① $P = 2(7 + 4) = 22 \text{ cm}$
A length.

② $S = 7 \cdot 4 = 28 \text{ cm}^2$
An area — square units.

ANSWER

22 cm and 28 cm²

03 Interactive model

Mark the diagnostic in the lesson and hand back the diagnosis table rather than a score; a number tells a pupil nothing they can act on.

What Grade 5 left behind

$12 + 3 \cdot (8 - 5)$ equals:

45

21

27

15

18 24 simplifies to:

$\frac{3}{4}$

$\frac{9}{12}$

$\frac{2}{3}$

$\frac{6}{8}$

$\frac{2}{3} + \frac{1}{4}$ equals:

$$\frac{3}{7}$$

$$\frac{11}{12}$$

$$\frac{2}{12}$$

$$\frac{3}{4}$$

Which is larger, $\frac{3}{5}$ or $\frac{5}{8}$?

$$\frac{3}{5}$$

$$\frac{5}{8}$$

equal

cannot say

$4.7 + 2.85$ equals:

7.55

6.55

7.12

4.98

A 7×4 rectangle has perimeter:

11

22

28

14

And area:

11

22

28

44

Area is measured in:

cm

cm²

cm³

no units

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Natural number	Natural son	Натуральное число
Fraction	Kasr	Дробь
Numerator	Surat	Числитель
Denominator	Maxraj	Знаменатель
Order of operations	Amallar tartibi	Порядок действий
Perimeter	Perimetr	Периметр
Area	Yuza	Площадь
Diagnostic	Tashxis ishi	Диагностическая работа

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The order of operations puts first:

addition

brackets

multiplication

subtraction

18 24 simplifies to:

$$\frac{3}{4}$$

$$\frac{2}{3}$$

$$\frac{1}{2}$$

$$\frac{4}{3}$$

To add fractions you need:

the same numerator

a common denominator

nothing

a decimal

Perimeter is measured in:

cm

cm²

cm³

degrees

$918 \div 27$ equals:

32

34

36

27

This year has how many hours of mathematics?

170

204

136

240

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $348 + 2\,976$

ANSWER $3\,324$

2. $5\,004 - 1\,897$

ANSWER $3\,107$

3. $47 \cdot 26$

ANSWER $1\,222$

4. $918 \div 27$

ANSWER 34

5. $12 + 3 \cdot (8 - 5)$

ANSWER 21

6. $\frac{18}{24}$ simplified

ANSWER $\frac{3}{4}$

7. $4.7 + 2.85$

ANSWER 7.55 **Medium · the standard question**

7 PROBLEMS

1. $\frac{2}{3} + \frac{1}{4}$

ANSWER $\frac{11}{12}$

2. Compare $\frac{3}{5}$ and $\frac{5}{8}$

ANSWER $\frac{3}{5} < \frac{5}{8}$

3. $9.1 - 3.62$

ANSWER 5.48

4. A 7×4 rectangle: perimeter and area

ANSWER $22\text{ cm}, 28\text{ cm}^2$

5. $100 - 4 \cdot (7 + 8)$

ANSWER 40

6. $\frac{5}{6} - \frac{1}{3}$

ANSWER $\frac{1}{2}$

7. A square of side 9: its area

ANSWER 81 cm^2

Hard · stretch and reason

7 PROBLEMS

1. $(24 + 36) \div (10 - 4)$

ANSWER 10

2. $2 \cdot 3^2 + 4$

ANSWER 22

3. A rectangle of area 48 and width 6: its length

ANSWER 8

4. A rectangle of perimeter 30 and width 7: its length

ANSWER 8

5. $\frac{3}{4}$ of 60

ANSWER 45

6. Order $\frac{1}{2}$, $\frac{2}{5}$, $\frac{3}{5}$

ANSWER $\frac{2}{5} < \frac{1}{2} < \frac{3}{5}$

7. How many hours a week does this class have?

ANSWER 6

07 Homework

Homework — 5 tasks

Rework, in full, every diagnostic question you lost a mark on.

1. Rework every diagnostic question you got wrong.
2. Compute $45 + 5 \cdot (12 - 8)$ and $(45 + 5) \cdot 12 - 8$, and say why they differ.
3. Simplify $\frac{24}{36}$ and $\frac{45}{60}$.

4. Compute $\frac{3}{4} + \frac{1}{6}$.

5. A rectangle is 9 *cm* by 5 *cm*. Find its perimeter and area with units.

Directed numbers: adding and subtracting integers

A Cambridge insert: numbers below zero, the number line as a road, and what a double minus does.

LESSON 2–4

3 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 1.1

LESSON CLOCK

20'

25'

30'

30'

15'

Numbers below zero	20 min
Ordering integers	25 min
Adding	30 min
Subtracting	30 min
Where they are used	15 min

LEARNING OBJECTIVES

- Read and place negative numbers on a number line.
- Compare and order integers.
- Add and subtract integers using the number line and the sign rules.
- Use directed numbers for temperature, height and money.

TEXTBOOK

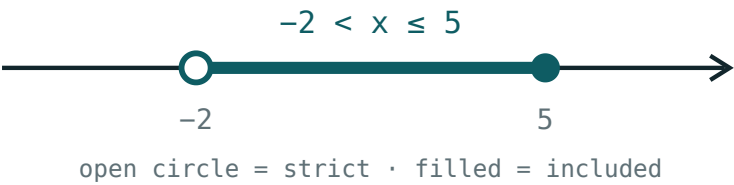
National	pp. 9–14
Cambridge	Stage 7, pp. 8–13
Workbook	Exercise 1.1

01

Explanation

Numbers below zero

Counting numbers stop at zero; the world does not. Temperature, debt, depth and floors below ground all need numbers on the other side.



The integers, positive and negative, on one line

SITUATION	POSITIVE	NEGATIVE
temperature	above freezing	below freezing
height	above sea level	below sea level
money	saved	owed
a lift	upper floors	the basement

EVERY NUMBER HAS AN OPPOSITE

5 and -5 sit the same distance from zero on opposite sides, and they add to zero. Zero is its own opposite, and is neither positive nor negative.

Ordering integers

On the number line, the number further to the right is the larger. Among negatives that reverses the everyday feeling: -2 is **larger** than -7 .

COMPARISON	TRUE?	WHY
$5 > 3$	yes	further right
$-2 > -7$	yes	-2
$-7 > -2$	no	the digits mislead
$0 > -1$	yes	zero beats every negative
$-100 > 1$	no	every negative is below every positive

THE BIGGER THE DIGITS, THE SMALLER A NEGATIVE NUMBER IS

-100 is far below -2 . Reading the digits and forgetting the sign is the error this whole block is designed to prevent.

Adding

Adding a positive moves right; adding a negative moves left.

SUM	ON THE LINE	ANSWER
$-3 + 5$	from -3 , five right	2
$3 + (-5)$	from 3 , five left	-2
$-3 + (-5)$	from -3 , five left	-8
$-5 + 5$	back to the start	0

SAME SIGNS ADD, DIFFERENT SIGNS SUBTRACT

With the same sign, add the digits and keep the sign. With different signs, subtract the smaller digits from the larger and keep the sign of the larger. The number line proves both.

Subtracting

$$a - b = a + (-b)$$

Subtracting is adding the opposite, so a minus in front of a negative becomes a plus.

EXPRESSION	REWRITTEN	ANSWER
$7 - 10$	$7 + (-10)$	-3
$-4 - 6$	$-4 + (-6)$	-10
$5 - (-3)$	$5 + 3$	8
$-5 - (-3)$	$-5 + 3$	-2

TWO MINUSES NEXT TO EACH OTHER MAKE A PLUS

$5 - (-3)$ is 8 , not 2 . Taking away a debt leaves you better off — which is exactly what the rule says in words.

Where they are used

PROBLEM	WORKING	ANSWER
from -7°C the temperature rises 12°	$-7 + 12$	5°C
from 4°C it falls 9°	$4 - 9$	-5°C
the difference between 6°C and -3°C	$6 - (-3)$	9°
a diver at -12 m rises 5 m	$-12 + 5$	-7 m
a lift from floor -2 up 7	$-2 + 7$	floor 5

A DIFFERENCE IS ALWAYS A SUBTRACTION

From -3° to 6° the rise is $6 - (-3) = 9$ degrees — and the double minus is exactly why the answer is bigger than either number.

02 Worked examples

EXAMPLE 1 Compute $-3 + 5$, $3 + (-5)$ and $-3 + (-5)$.

1 Different signs: $5 - 3 = 2$, sign of the larger — positive.
2

2 Different signs: $5 - 3 = 2$, sign of the larger — negative.
 -2

3 Same signs: add and keep the sign.
 -8

ANSWER

2, -2 , -8

EXAMPLE 2 Compute $5 - (-3)$ and $-5 - (-3)$.

① Subtracting is adding the opposite.

② $5 + 3 = 8$

③ $-5 + 3 = -2$

ANSWER8 and -2 **EXAMPLE 3** The temperature is -7°C and rises by 12° . What is it now?

① A rise means adding.

② $-7 + 12$

③ $= 5^{\circ}\text{C}$

ANSWER 5°C **03 Interactive model**

Draw a vertical number line as a thermometer on the board and keep it there all week; every question in this block is answered by pointing at it.

Signs on the number line

Which is larger, -2 or -7 ?

-2

-7

equal

cannot say

$-3 + 5$ equals:

-8

2

-2

8

$3 + (-5)$ equals:

8

2

-2

-8

$-3 + (-5)$ equals:

-8

-2

2

8

$7 - 10$ equals:

3

-3

17

-17

$5 - (-3)$ equals:

2

8

-8

-2

$-5 - (-3)$ equals:

-8

-2

2

8

From -7° a rise of 12° gives:

5°

-5°

19°

-19°

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Integer	Butun son	Целое число
Negative number	Manfiy son	Отрицательное число
Positive number	Musbat son	Положительное число
Number line	Koordinata to'g'ri chizig'i	Числовая прямая
Opposite number	Qarama-qarshi son	Противоположное число
Below zero	Noldan past	Ниже нуля
Rise	Ortish	Повышение
Fall	Kamayish	Понижение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

On the number line the larger number is:

further left

further right

nearer zero

the one with more digits

The opposite of -9 is:

9

-9

0

$\frac{1}{9}$

$-4 - 6$ equals:

2

-2

10

-10

$6 - (-3)$ equals:

3

9

-9

-3

Every negative number is:

above zero

below zero

equal to zero

a fraction

A diver at -12 m who rises 5 m is at:

-17 m

-7 m

7 m

17 m

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $-3 + 5$

ANSWER 2

2. $3 + (-5)$

ANSWER -2

3. $-3 + (-5)$

ANSWER -8

4. $7 - 10$

ANSWER -3

5. $-4 - 6$

ANSWER -10

6. $5 - (-3)$

ANSWER 8

7. The opposite of -9

ANSWER 9

Medium · the standard question

7 PROBLEMS

1. $-5 - (-3)$

ANSWER -2

2. Order $-7, 2, -2, 0, 5$

ANSWER $-7 < -2 < 0 < 2 < 5$

3. From -7° a rise of 12°

ANSWER 5°

4. From 4° a fall of 9°

ANSWER -5°

5. The difference between 6° and -3°

ANSWER 9°

6. A diver at -12 m rises 5 m

ANSWER -7 m

7. $-8 + 8$

ANSWER 0

Hard · stretch and reason

7 PROBLEMS

1. $-7 + 12 - (-5)$

ANSWER 10

2. $-3 - (-8) + (-4)$

ANSWER 1

3. The difference between -15° and -4°

ANSWER 11°

4. A lift goes from floor -3 up 9, then down 4

ANSWER floor 2

5. Which is larger, -100 or -1 ?

ANSWER -1

6. Write a subtraction whose answer is -6

ANSWER e.g. $2 - 8$

7. A debt of 50 000 is cancelled: the change in what you own

ANSWER $+50\ 000$

07 Homework

Homework — 5 tasks

Draw a number line beside every question until the signs feel automatic.

1. Compute $-6 + 9$, $6 + (-9)$ and $-6 + (-9)$.
2. Compute $4 - 11$ and $-4 - (-11)$.
3. Order -12 , 3 , -3 , 0 , 8 , -20 from smallest to largest.
4. The temperature falls from 2°C by 11° . What is it now?
5. Find the difference between 8°C and -6°C , and say why the answer is larger than either.

Directed numbers: multiplying and dividing integers

A Cambridge insert: two signs make a third, and one rule covers both operations.

LESSON 5–7

3 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 1.2

LESSON CLOCK

20'

25'

30'

30'

15'

Why a minus times a minus is a plus	20 min
The rule of signs	25 min
Dividing	30 min
Mixed expressions	30 min
Squares and powers	15 min

LEARNING OBJECTIVES

- Multiply and divide integers using the sign rule.
- Explain the rule with repeated addition and with a pattern.
- Evaluate expressions mixing all four operations on integers.
- Square a negative number correctly.

TEXTBOOK

National	pp. 15–20
Cambridge	Stage 7, pp. 14–18
Workbook	Exercise 1.2

01

Explanation

Why a minus times a minus is a plus

Start from repeated addition, then continue the pattern downwards.

PRODUCT	VALUE
$3 \cdot (-4)$	-12
$2 \cdot (-4)$	-8
$1 \cdot (-4)$	-4
$0 \cdot (-4)$	0
$(-1) \cdot (-4)$	4
$(-2) \cdot (-4)$	8

THE PATTERN DECIDES IT

Each row adds 4 to the one above. Continuing past zero forces $(-1) \cdot (-4) = 4$ — the rule is not an arbitrary convention but the only way to keep the arithmetic consistent.

The rule of signs

$++ = +$
 $+- = -$
 $-+ = -$
 $-- = +$

PRODUCT	SIGNS	ANSWER
$6 \cdot 7$	same	42
$6 \cdot (-7)$	different	-42
$(-6) \cdot 7$	different	-42
$(-6) \cdot (-7)$	same	42

COUNT THE MINUS SIGNS

An even number of negative factors gives a positive product; an odd number gives a negative one. $(-1)(-2)(-3)$ has three, so the answer is -6.

Dividing

Division follows the same rule, because it is multiplication by the reciprocal.

QUOTIENT	ANSWER	CHECK
$-12 \div 3$	-4	$3 \cdot (-4) = -12$
$12 \div (-3)$	-4	$(-3)(-4) = 12$
$-12 \div (-3)$	4	$(-3) \cdot 4 = -12$
$0 \div (-3)$	0	$(-3) \cdot 0 = 0$

DIVIDING BY ZERO IS STILL IMPOSSIBLE

$0 \div (-3)$ is 0 , but $-3 \div 0$ has no answer at all: no number multiplied by 0 gives -3 . Negative numbers change nothing about that.

Mixed expressions

EXPRESSION	WORKING	ANSWER
$-3 \cdot 4 + 5$	$-12 + 5$	-7
$-3 \cdot (4 + 5)$	$-3 \cdot 9$	-27
$20 \div (-4) - 3$	$-5 - 3$	-8
$(-2)(-3)(-4)$	three minus signs	-24

THE ORDER OF OPERATIONS DOES NOT CHANGE

Brackets, then $\times$ and $\div$, then $+$ and $-$ — exactly as with positive numbers. Only the signs are new.

Squares and powers

EXPRESSION	MEANING	ANSWER
$(-5)^2$	$(-5) \cdot (-5)$	25
-5^2	$-(5 \cdot 5)$	-25
$(-2)^3$	$(-2)(-2)(-2)$	-8
$(-1)^{100}$	an even number of minuses	1

$(-5)^2$ AND -5^2 ARE DIFFERENT

The bracket says the whole -5 is squared; without it, only the 5 is, and the minus stays outside. This one pair of brackets is worth a mark in every paper from now on.

02 Worked examples**EXAMPLE 1** Compute $(-6) \cdot (-7)$ and $6 \cdot (-7)$.

① Same signs give a positive product.
42

② Different signs give a negative one.
-42

ANSWER

42 and -42

EXAMPLE 2 Compute $(-2)(-3)(-4)$.

① $(-2)(-3) = 6$
Two minuses.

② $6 \cdot (-4) = -24$
One more minus.

③ Three negative factors — an odd number — so the answer is negative.

ANSWER

-24

EXAMPLE 3 Compute $(-5)^2$ and -5^2 .

① $(-5)^2 = (-5)(-5) = 25$

The bracket squares the sign too.

② $-5^2 = -(5 \cdot 5) = -25$

Here only the 5 is squared.

ANSWER

25 and -25

03 **Interactive model**

Build the descending pattern table on the board line by line and let the class call out the next value; they state the rule themselves before it is written down.

Count the minus signs

$6 \cdot (-7)$ equals:

42

-42

13

-13

$(-6) \cdot (-7)$ equals:

42

-42

-13

13

$-12 \div (-3)$ equals:

4

-4

36

-36

$12 \div (-3)$ equals:

4

-4

9

-9

$(-2)(-3)(-4)$ equals:

24

-24

9

-9

$(-5)^2$ equals:

25

-25

10

-10

-5^2 equals:

25

-25

10

-10

$(-1)^{100}$ equals:

1

-1

100

-100

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Product	Ko'paytma	Произведение
Quotient	Bo'linma	Частное
Sign	Ishora	Знак
Rule of signs	Ishoralar qoidasi	Правило знаков
Repeated addition	Takroriy qo'shish	Повторное сложение
Square	Kvadrat	Квадрат
Pattern	Qonuniyat	Закономерность
Order of operations	Amallar tartibi	Порядок действий

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$- \cdot -$ gives:

+

-

0

either

$+ \cdot -$ gives:

+

-

0

either

An odd number of negative factors gives:

a positive product

a negative product

zero

either

$-12 \div 3$ equals:

4

-4

-9

9

$(-2)^3$ equals:

8

-8

6

-6

$-3 \div 0$ equals:

0

-3

nothing — it is undefined

3

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $6 \cdot (-7)$

ANSWER -42

2. $(-6) \cdot (-7)$

ANSWER 42

3. $(-6) \cdot 7$

ANSWER -42

4. $-12 \div 3$

ANSWER -4

5. $12 \div (-3)$

ANSWER -4

6. $-12 \div (-3)$

ANSWER 4

7. $(-5)^2$

ANSWER 25 **Medium · the standard question**

7 PROBLEMS

1. -5^2

ANSWER -25

2. $(-2)^3$

ANSWER -8

3. $(-2)(-3)(-4)$

ANSWER -24

4. $-3 \cdot 4 + 5$

ANSWER -7

5. $-3 \cdot (4 + 5)$

ANSWER -27

6. $20 \div (-4) - 3$

ANSWER -8

7. $(-1)^{100}$

ANSWER 1 **Hard · stretch and reason**

7 PROBLEMS

1. $(-1)^{101}$

ANSWER -1

2. $-2 \cdot (-3)^2$

ANSWER -18

3. $(-2 \cdot 3)^2$

ANSWER 36

4. $(-36) \div (-4) \div (-3)$

ANSWER -3

5. $-10 + (-4)(-5)$

ANSWER 10

6. $(-3)^2 - (-3)$

ANSWER 12

7. Why is $(-1) \cdot (-4) = 4$ forced?

ANSWER The pattern of products rises by 4 each row

07 Homework**Homework — 5 tasks**

Count the minus signs before you multiply anything.

1. Compute $8 \cdot (-9)$, $(-8) \cdot (-9)$ and $(-8) \cdot 9$.
2. Compute $-45 \div 9$, $45 \div (-9)$ and $-45 \div (-9)$.
3. Compute $(-4)^2$ and -4^2 , and explain the difference.
4. Compute $(-2)(-5)(-3)$.
5. Compute $-6 \cdot 3 + 20 \div (-4)$.

Algebraic expressions

A letter standing for a number — and a sentence in words turned into one line of algebra.

LESSON 8–10

3 × 40 MIN

MATEMATIKA 6, §1

S7 2.1

LESSON CLOCK

20'

30'

30'

30'

10

A letter for a number	20 min
Words into algebra	30 min
Algebra into words	30 min
The shorthands	30 min
Homework	10 min

LEARNING OBJECTIVES

- Explain what a letter stands for in an expression.
- Write a worded rule as an algebraic expression.
- Read an expression back into words.
- Use the standard shorthands of algebraic notation.

TEXTBOOK

National	pp. 21–28
Cambridge	Stage 7, pp. 20–24
Workbook	Exercise 2.1

01

Explanation

A letter for a number

An **algebraic expression** is an arithmetic expression in which some numbers are replaced by letters. The letter stands for a number that may change — which is why it is called a **variable**.

EXPRESSION	WHAT THE LETTER STANDS FOR
$3n$	the number of legs on n tripods
$x + 5$	a number five more than x
$2(a + b)$	the perimeter of a rectangle a by b
$100 - p$	the change from 100 after spending p

WHY LETTERS ARE WORTH THE TROUBLE

One line, $P = 2(a + b)$, replaces a sentence that would have to be rewritten for every rectangle in the world. Algebra is arithmetic said once instead of endlessly.

Words into algebra

IN WORDS	IN ALGEBRA
a number increased by 7	$x + 7$
a number decreased by 7	$x - 7$
7 less than a number	$x - 7$
a number subtracted from 7	$7 - x$
three times a number	$3x$
a third of a number	$\frac{x}{3}$
three times a number, then add 2	$3x + 2$
add 2 to a number, then treble it	$3(x + 2)$

THE ORDER OF THE WORDS DECIDES THE BRACKETS

$3x + 2$ and $3(x + 2)$ are different: for $x = 4$ they are 14 and 18. “Then” tells you which operation happens second, and a bracket is how you write that.

Algebra into words

Reading in the other direction is the better test of understanding — and there is usually more than one right reading.

EXPRESSION	ONE READING	A CONTEXT
$5n$	five times a number	the cost of n pens at 5 each
$n + 3$	three more than a number	an age three years from now
$2n + 1$	double a number and add one	an odd number
$\frac{s}{t}$	one number divided by another	speed, from distance and time

EVERY FORMULA IS AN EXPRESSION WITH A NAME

$P = 2(a + b), S = ab, v = \frac{s}{t}$ — the left side names what the expression on the right computes.

The shorthands

WRITTEN AS	NOT AS	RULE
$3x$	$3 \cdot x$	the multiplication sign is dropped
ab	$a \cdot b$	the same between two letters
x	$1x$	a coefficient of 1 is not written
$-x$	$-1x$	the same for -1
$\frac{x}{3}$	$x \div 3$	division is written as a fraction
x^2	$x \cdot x$	a power counts equal factors

3X MEANS MULTIPLY, NEVER “THIRTY-SOMETHING”

In arithmetic, digits written together mean place value; in algebra, symbols written together mean a product. It is the one place where the two notations disagree.

02 Worked examples

EXAMPLE 1 Write “three times a number, then add 2” and “add 2 to a number, then treble it”.

- ① First: multiply, then add.

$$3x + 2$$

- ② Second: add, then multiply — the addition needs a bracket.

$$3(x + 2)$$

- ③ They differ: for $x = 4$ they give 14 and 18.

ANSWER

$$3x + 2 \text{ and } 3(x + 2)$$

EXAMPLE 2 A pen costs p sum and a pencil q . Write the cost of 3 pens and 4 pencils.

- ① 3 pens cost $3p$.

- ② 4 pencils cost $4q$.

- ③ Together: $3p + 4q$.

ANSWER

$$3p + 4q$$

EXAMPLE 3 Write the perimeter and area of a rectangle with sides a and b .

- 1 Perimeter: two lots of $a + b$.

$$P = 2(a + b)$$

- 2 Area: one side times the other.

$$S = ab$$

ANSWER

$$2(a + b) \text{ and } ab$$

03 **Interactive model**

Read an expression aloud and ask for a story that fits it; the class invents contexts far faster than they translate them, and the translation improves as a result.

Words into algebra

“A number increased by 7”:

$7x$

$x + 7$

$x - 7$

$7 - x$

“7 less than a number”:

$7 - x$

$x - 7$

$x + 7$

$7x$

“A number subtracted from 7”:

$7 - x$

$x - 7$

$x + 7$

$7x$

“Three times a number, then add 2”:

$$3x + 2$$

$$3(x + 2)$$

$$2x + 3$$

$$x + 6$$

“Add 2 to a number, then treble it”:

$$3x + 2$$

$$3(x + 2)$$

$$2(x + 3)$$

$$3x + 6 \text{ only}$$

“A third of a number”:

$$3x$$

$$\frac{x}{3}$$

$$x - 3$$

$$x + 3$$

$1x$ is written:

$$1x$$

$$x$$

$$x^1$$

$$x + 1$$

The perimeter of an a by b rectangle:

$$ab$$

$$a + b$$

$$2(a + b)$$

$$2ab$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Algebraic expression	Harfli ifoda	Буквенное выражение
Variable	O'zgaruvchi	Переменная
Constant	O'zgarmas	Постоянная
Term	Had	Член
Coefficient	Koeffitsiyent	Коэффициент
Sum	Yig'indi	Сумма
Product	Ko'paytma	Произведение
Formula	Formula	Формула

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A letter in an expression stands for:

a word

a number

an operation

nothing

$3x$ means:

$3 + x$

$3 \cdot x$

thirty-something

x^3

“7 less than x ” is:

$7 - x$

$x - 7$

$7x$

$x + 7$

$3(x + 2)$ at $x = 4$ is:

14

18

20

12

A third of x is written:

$3x$

$\frac{x}{3}$

$\frac{3}{x}$

$x - 3$

The area of an a by b rectangle:

$2(a + b)$

ab

$a + b$

$2ab$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. “A number increased by 7”

ANSWER $x + 7$

2. “A number decreased by 7”

ANSWER $x - 7$

3. “Three times a number”

ANSWER $3x$

4. “A third of a number”

ANSWER $\frac{x}{3}$

5. “A number subtracted from 7”

ANSWER $7 - x$

6. The cost of n pens at 5 sum

ANSWER $5n$

7. The perimeter of an a by b rectangle

ANSWER $2(a + b)$

Medium · the standard question

7 PROBLEMS

1. “Three times a number, then add 2”

ANSWER $3x + 2$

2. “Add 2 to a number, then treble it”

ANSWER $3(x + 2)$

3. 3 pens at p and 4 pencils at q

ANSWER $3p + 4q$

4. The change from 100 after spending p

ANSWER $100 - p$

5. An age n now, in 5 years

ANSWER $n + 5$

6. An odd number, written with n

ANSWER $2n + 1$

7. Speed from distance s and time t

ANSWER $\frac{s}{t}$

Hard · stretch and reason

7 PROBLEMS

1. The perimeter of a square of side a

ANSWER $4a$

2. The total of three consecutive numbers starting at n

ANSWER $3n + 3$

3. The cost of n books at p each with 2000 postage

ANSWER $np + 2000$

4. Half of one number added to twice another

ANSWER $\frac{a}{2} + 2b$

5. The area of a rectangle whose length is 3 more than its width w

ANSWER $w(w + 3)$

6. Read $2(n - 1)$ in words

ANSWER One less than a number, then doubled

7. Why do $3x + 2$ and $3(x + 2)$ differ?

ANSWER The bracket changes which operation is done first

07 Homework

Homework — 5 tasks

Say the expression aloud before writing it; the word “then” marks the bracket.

- Write in algebra: a number increased by 12; 12 less than a number; a quarter of a number.
- Write “double a number, then subtract 5” and “subtract 5 from a number, then double it”.
- A notebook costs a and a pen b . Write the cost of 5 notebooks and 2 pens.
- Write the perimeter and area of a square of side a .

5. Read $4(n + 3)$ in words and give its value at $n = 2$.

Simplifying algebraic expressions

Collecting like terms, and taking a common factor out of a bracket.

LESSON 11–12

2 × 40 MIN

MATEMATIKA 6, §2

S7 2.3

LESSON CLOCK

20'

25'

20'

12'

3

Like terms	20 min
Collecting	25 min
Expanding a bracket	20 min
Taking a factor out	12 min
Homework	3 min

LEARNING OBJECTIVES

- Identify like terms and collect them.
- Expand a bracket by multiplying every term inside it.
- Take a common factor outside a bracket.
- Check a simplification by substituting a number.

TEXTBOOK

National	pp. 29–34
Cambridge	Stage 7, pp. 25–29
Workbook	Exercise 2.3

01

Explanation

Like terms

Terms are **like** when their letter parts are identical. Only like terms can be collected — for the same reason that apples and oranges cannot be added into one number of apples.

PAIR	LIKE?	WHY
$5x$ and $-2x$	yes	the same letter
$5x$ and $5y$	no	different letters
$5x$ and $5x^2$	no	different powers
$3ab$ and $7ba$	yes	$ab = ba$

THE COEFFICIENT MAY DIFFER; THE LETTERS MAY NOT

$5x$ and $-2x$ are like terms and collect to $3x$. $5x$ and $5y$ are not, and $5x + 5y$ is already as simple as it gets.

Collecting

EXPRESSION	COLLECTED
$5x + 3x$	$8x$
$7a - 2a$	$5a$
$4x + 3y + 2x - y$	$6x + 2y$
$6m - 9m$	$-3m$
$3a + 4b$	$3a + 4b$

A TERM CARRIES THE SIGN IN FRONT OF IT

In $4x + 3y + 2x - y$ the last term is $-y$, with coefficient -1 . Collecting the y terms gives $3y - y = 2y$, not $4y$.

Expanding a bracket

$$a(b + c) = ab + ac$$

EXPRESSION	EXPANDED
$3(x + 4)$	$3x + 12$
$5(2a - 3)$	$10a - 15$
$x(x + 2)$	$x^2 + 2x$
$2(x + 3) + 4x$	$6x + 6$

EVERY TERM INSIDE THE BRACKET IS MULTIPLIED

$3(x + 4)$ is $3x + 12$, not $3x + 4$. Drawing two arrows from the 3 to both terms stops this error for good.

Taking a factor out

Expanding read backwards: find what every term has in common and write it outside.

EXPRESSION	COMMON FACTOR	FACTORISED
$6x + 9$	3	$3(2x + 3)$
$4a - 10$	2	$2(2a - 5)$
$x^2 + 5x$	x	$x(x + 5)$
$12m + 8n$	4	$4(3m + 2n)$

CHECK BY SUBSTITUTING A NUMBER

Put $x = 2$ into $6x + 9$ and into $3(2x + 3)$: both give 21. One substitution catches almost every slip in this lesson.

02 Worked examples

EXAMPLE 1 Simplify $4x + 3y + 2x - y$.

1 x terms: $4x + 2x = 6x$.

2 y terms: $3y - y = 2y$.
The last term is $-y$.

3 $= 6x + 2y$
Nothing else is like.

ANSWER

$6x + 2y$

EXAMPLE 2 Simplify $2(x + 3) + 4x$.

- ① Expand: $2x + 6 + 4x$.
Both terms multiplied.
- ② Collect: $6x + 6$.
- ③ Check at $x = 1$: $8 = 8$ ✓

ANSWER

$6x + 6$

EXAMPLE 3 Factorise $6x + 9$.

- ① Both terms divide by 3.
- ② $6x \div 3 = 2x$ and $9 \div 3 = 3$.
- ③ $= 3(2x + 3)$
Check by expanding.

ANSWER

$3(2x + 3)$

03 Interactive model

Use coloured chalk for each letter when collecting; the class sees the groups before it counts them, and the sign errors halve.

Collect, expand, factorise

$5x + 3x$ equals:

$8x$

$8x^2$

$15x$

8

$5x$ and $5y$ collect to:

$10xy$

$10x$

they do not collect

$25xy$

$4x + 3y + 2x - y$ equals:

$6x + 4y$

$6x + 2y$

$8x + 2y$

$6x - 2y$

$6m - 9m$ equals:

$3(x + 4)$ equals:

$5(2a - 3)$ equals:

$6x + 9$ factorises to:

$x^2 + 5x$ factorises to:

$$x(x + 5)$$

$$x(x + 5x)$$

$$5x(x + 1)$$

$$x^2(1 + 5)$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To simplify	Soddalashtirish	Упростить
Like terms	O'xshash hadlar	Подобные члены
To collect	Ixchamlash	Приведение подобных
To expand	Qavsni ochish	Раскрыть скобки
Common factor	Umumiy ko'paytuvchi	Общий множитель
Bracket	Qavs	Скобка
Coefficient	Koeffitsiyent	Коэффициент
Substitution check	Tekshirish	Проверка подстановкой

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Like terms have:

the same coefficient

the same letter part

the same sign

nothing in common

$7a - 2a$ equals:

$5a$

$9a$

5

$14a$

$3a + 4b$ simplifies to:

$7ab$

$7a$

it is already simplest

$12ab$

$x(x + 2)$ equals:

$4a - 10$ factorises to:

A simplification is checked by:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $5x + 3x$

ANSWER $8x$

2. $7a - 2a$

ANSWER $5a$

3. $6m - 9m$

ANSWER $-3m$

4. $3(x + 4)$

ANSWER $3x + 12$

5. $5(2a - 3)$

ANSWER $10a - 15$

6. $6x + 9$ factorised

ANSWER $3(2x + 3)$

7. $3a + 4b$

ANSWER Already simplest

Medium · the standard question

7 PROBLEMS

1. $4x + 3y + 2x - y$

ANSWER $6x + 2y$

2. $2(x + 3) + 4x$

ANSWER $6x + 6$

3. $x(x + 2)$

ANSWER $x^2 + 2x$

4. $x^2 + 5x$ factorised

ANSWER $x(x + 5)$

5. $12m + 8n$ factorised

ANSWER $4(3m + 2n)$

6. $5a + 2b - 3a + 4b$

ANSWER $2a + 6b$

7. $3(2x - 1) + 5$

ANSWER $6x + 2$

Hard · stretch and reason

7 PROBLEMS

1. $4(x + 2) - 3(x - 1)$

ANSWER $x + 11$

2. $2x(x + 3) - x^2$

ANSWER $x^2 + 6x$

3. $5(a + b) - 5a$

ANSWER $5b$

4. $3x + 2y - (x - y)$

ANSWER $2x + 3y$

5. Factorise $9x^2 + 6x$

ANSWER $3x(3x + 2)$

6. Check $6x + 9 = 3(2x + 3)$ at $x = 2$

ANSWER Both give 21

7. Why can $3a$ and $3a^2$ not be collected?

ANSWER Their letter parts differ

07 Homework**Homework — 5 tasks**

Substitute one number into both forms at the end; if they disagree, the simplification is wrong.

1. Simplify $7x + 2y - 3x + 5y$.
2. Expand $4(x + 5)$ and $6(2a - 3)$.
3. Simplify $3(x + 2) + 2x$.
4. Factorise $8x + 12$ and $x^2 + 7x$.
5. Check your answer to task 4 by substituting $x = 3$ into both forms.

Finding the value of an algebraic expression

Substituting a number for the letter — including a negative one, where the brackets matter.

LESSON 13–15

3 × 40 MIN

MATEMATIKA 6, §3

S7 2.2

LESSON CLOCK

20'

25'

30'

30'

15'

Substituting a number	20 min
Negative values	25 min
Using formulae	30 min
Tables of values	30 min
Homework	15 min

LEARNING OBJECTIVES

- Substitute a value and evaluate, respecting the order of operations.
- Substitute a negative value using brackets.
- Use a formula to compute a required quantity.
- Build a table of values for an expression.

TEXTBOOK

National	pp. 35–42
Cambridge	Stage 7, pp. 24–28
Workbook	Exercise 2.2

01

Explanation

Substituting a number

Replace the letter by the number, then follow the order of operations exactly as in arithmetic.

EXPRESSION	AT $X = 4$	VALUE
$3x + 2$	$3 \cdot 4 + 2$	14
$3(x + 2)$	$3 \cdot 6$	18
x^2	$4 \cdot 4$	16
$2x^2$	$2 \cdot 16$	32
$(2x)^2$	8^2	64

2X² AND (2X)² ARE DIFFERENT

In $2x^2$ only x is squared; in $(2x)^2$ the whole product is. Powers act before multiplication unless a bracket says otherwise.

Negative values

Substitute a negative number **inside brackets**, and the sign rules do the rest.

EXPRESSION	AT $x = -3$	VALUE
$2x + 5$	$2 \cdot (-3) + 5$	-1
x^2	$(-3)^2$	9
$-x$	$-(-3)$	3
$x^2 - x$	$9 - (-3)$	12
$5 - 2x$	$5 - 2(-3)$	11

-x IS NOT ALWAYS NEGATIVE

It means “the opposite of x ”. When x is negative, $-x$ is positive. Reading the minus sign as “opposite” rather than “negative” removes the confusion entirely.

Using formulae

FORMULA	DATA	VALUE
$P = 2(a + b)$	$a = 7, b = 4$	22 cm
$S = ab$	$a = 7, b = 4$	28 cm^2
$s = vt$	$v = 60, t = 3$	180 km
$S = \frac{1}{2}ah$	$a = 10, h = 6$	30 cm^2
$C = \frac{5}{9}(F - 32)$	$F = 68$	$20\text{ }^\circ\text{C}$

CARRY THE UNITS THROUGH

A formula answers with a quantity, not a bare number: 22 cm , 28 cm^2 , 180 km . The units also check the work — an area that comes out in cm means something went wrong.

Tables of values

A table shows how the value moves as the letter changes — the first step towards a graph.

X	-2	-1	0	1	2	3
$2x + 1$	-3	-1	1	3	5	7
x^2	4	1	0	1	4	9
$10 - 3x$	16	13	10	7	4	1

TWO PATTERNS TO NOTICE

$2x + 1$ climbs by 2 each step and $10 - 3x$ falls by 3, while x^2 does neither — it falls, reaches zero and climbs again. Grade 7 will call the first two *linear*.

02

Worked examples

EXAMPLE 1 Find the value of $x^2 - x$ at $x = -3$.

1

Substitute in brackets: $(-3)^2 - (-3)$.

2

 $(-3)^2 = 9$

3

 $-(-3) = +3$

4

 $9 + 3 = 12$

ANSWER

12

EXAMPLE 2 A car travels at 60 km/h for 3 hours. Use $s = vt$.

① $s = 60 \cdot 3$

② $= 180$

③ With units: 180 km.

ANSWER

180 km

EXAMPLE 3 Convert 68 °F using $C = \frac{5}{9}(F - 32)$.

① $F - 32 = 36$
Bracket first.

② $\frac{5}{9} \cdot 36$

③ $= 20$

④ 20 °C.

ANSWER

20 °C

03 Interactive model

Insist on brackets round every substituted negative number; the sign errors that plague the whole year begin here and end here.

Substitute into $2x + 1$

$f(x) = 2x + 1$

x 2

denominator 1

value 5

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
To substitute	O‘rniga qo‘yish	Подставить
Value of an expression	Ifodaning qiymati	Значение выражения
Formula	Formula	Формула
Table of values	Qiymatlar jadvali	Таблица значений
To evaluate	Hisoblash	Вычислить
Order of operations	Amallar tartibi	Порядок действий
Square	Kvadrat	Квадрат
Units	O‘lchov birligi	Единицы измерения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$3x + 2$ at $x = 4$ is:

14

18

20

12

$3(x + 2)$ at $x = 4$ is:

14

18

20

12

x^2 at $x = -3$ is:

9

-9

6

-6

$-x$ at $x = -3$ is:

$(2x)^2$ at $x = 4$ is:

$s = vt$ with $v = 60$, $t = 3$ gives:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $3x + 2$ at $x = 4$

ANSWER 14

2. $3(x + 2)$ at $x = 4$

ANSWER 18

3. x^2 at $x = 4$

ANSWER 16

4. $2x^2$ at $x = 4$

ANSWER 32

5. $(2x)^2$ at $x = 4$

ANSWER 64

6. $2x + 5$ at $x = -3$

ANSWER -1

7. $-x$ at $x = -3$

ANSWER 3

Medium · the standard question

7 PROBLEMS

1. $x^2 - x$ at $x = -3$

ANSWER 12

2. $5 - 2x$ at $x = -3$

ANSWER 11

3. $P = 2(a + b)$ at $a = 7, b = 4$

ANSWER 22 cm

4. $S = ab$ at $a = 7, b = 4$

ANSWER 28 cm²

5. $S = \frac{1}{2}ah$ at $a = 10, h = 6$

ANSWER 30 cm²

6. $s = vt$ at $v = 60, t = 3$

ANSWER 180 km

7. A table of $2x + 1$ at $x = -1, 0, 1$

ANSWER -1, 1, 3

Hard · stretch and reason

7 PROBLEMS

1. $C = \frac{5}{9}(F - 32)$ at $F = 68$

ANSWER 20°C

2. $C = \frac{5}{9}(F - 32)$ at $F = 32$

ANSWER 0°C

3. $x^2 + 2x - 3$ at $x = -4$

ANSWER 5

4. $(x - 1)(x + 2)$ at $x = -2$

ANSWER 0

5. $\frac{a + b}{2}$ at $a = -7, b = 3$

ANSWER -2

6. For which x does $2x + 1$ equal 9?

ANSWER $x = 4$

7. Why is $2x^2$ not $(2x)^2$?

ANSWER The power acts on x alone unless a bracket says otherwise

07 Homework

Homework — 5 tasks

Put every negative substitution in brackets, without exception.

- Find $4x - 3$ at $x = 5$ and at $x = -5$.
- Find $x^2 + x$ at $x = -4$.
- Find the perimeter and area of a rectangle with $a = 9$ and $b = 6$.
- Convert 86°F to Celsius.

5. Make a table of $10 - 3x$ for $x = -2$ to 3.

Solving equations

Undoing what was done to the unknown — and checking the root in the equation you were given.

LESSON 16–18

3 × 40 MIN

MATEMATIKA 6, §4

S7 2.5

LESSON CLOCK

20'

25'

30'

30'

15'

The balance	20 min
One and two steps	25 min
Brackets and both sides	30 min
Word problems	30 min
Homework	15 min

LEARNING OBJECTIVES

- Solve one-step and two-step equations by inverse operations.
- Solve equations with a bracket and with the unknown on both sides.
- Check a root by substituting into the original equation.
- Write a simple word problem as an equation.

TEXTBOOK

National	pp. 43–50
Cambridge	Stage 7, pp. 32–38
Workbook	Exercise 2.5

01

Explanation

The balance

An equation is a balance: whatever is done to one side must be done to the other, or it tips.

OPERATION ON THE UNKNOWN	UNDONE BY
adding 5	subtracting 5
subtracting 5	adding 5
multiplying by 3	dividing by 3
dividing by 3	multiplying by 3

UNDO IN THE REVERSE ORDER

In $3x + 5$ the x was multiplied first and added to second, so the addition is undone first and the multiplication second — like taking off shoes before socks were put on.

One and two steps

EQUATION	STEP	ROOT
$x + 7 = 12$	subtract 7	$x = 5$
$x - 4 = 9$	add 4	$x = 13$
$5x = 40$	divide by 5	$x = 8$
$\frac{x}{3} = 6$	multiply by 3	$x = 18$
$3x + 5 = 20$	subtract 5, divide by 3	$x = 5$
$2x - 7 = 11$	add 7, divide by 2	$x = 9$

CHECK IN THE EQUATION AS IT WAS GIVEN

For $3x + 5 = 20$: $3 \cdot 5 + 5 = 20$ ✓. Checking in a line of your own working repeats any error you have already made.

Brackets and both sides

EQUATION	FIRST MOVE	ROOT
$2(x + 3) = 14$	expand or divide by 2	$x = 4$
$3(x - 1) = 12$	divide by 3	$x = 5$
$5x = 2x + 12$	subtract $2x$	$x = 4$
$4x + 1 = 2x + 9$	subtract $2x$, then 1	$x = 4$

GATHER THE UNKNOWNNS ON THE SIDE WHERE MORE OF THEM ARE

In $4x + 1 = 2x + 9$ taking $2x$ from both sides leaves $2x$ on the left, which is positive. Working the other way gives $-2x$ and an extra chance to slip.

Word problems

PROBLEM	EQUATION	ANSWER
a number trebled and increased by 5 is 26	$3x + 5 = 26$	7
a rectangle of perimeter 30 with width 6	$2(6 + b) = 30$	$b = 9$
5 pens cost 20 000	$5p = 20\,000$	4000
Ali is 4 years older than Sardor; together 26	$x + (x + 4) = 26$	11 and 15

THREE LINES: LET, EQUATION, ANSWER

Say what the letter stands for, write the equation, solve it, and answer the question in words. The first and last lines carry as many marks as the algebra between them.

02 Worked examples

EXAMPLE 1 Solve $3x + 5 = 20$.

- 1 Subtract 5 from both sides: $3x = 15$.
- 2 Divide both sides by 3: $x = 5$.
- 3 Check: $3 \cdot 5 + 5 = 20$ ✓

ANSWER

$x = 5$

EXAMPLE 2 Solve $4x + 1 = 2x + 9$.

① Subtract $2x$: $2x + 1 = 9$.

② Subtract 1 : $2x = 8$.

③ Divide by 2 : $x = 4$.

④ Check: $17 = 17$ ✓

ANSWER

$$x = 4$$

EXAMPLE 3 A rectangle has perimeter 30 cm and width 6 cm . Find its length.

① Let the length be b .

② $2(6 + b) = 30$

③ $6 + b = 15$

Divide by 2 .

④ $b = 9\text{ cm}$

ANSWER

$$9\text{ cm}$$

03 Interactive model

Draw a pair of scales on the board and physically remove the same weight from both pans; every later step of algebra is that picture.

One step at a time

$x + 7 = 12$ gives:

$x = 19$

$x = 5$

$x = 7$

$x = 12$

$5x = 40$ gives:

$x = 35$

$x = 45$

$x = 8$

$x = 200$

$\frac{x}{3} = 6$ gives:

$x = 2$

$x = 18$

$x = 9$

$x = 3$

In $3x + 5 = 20$ you first:

divide by 3

subtract 5

add 5

multiply by 3

It gives:

$x = 5$

$x = 15$

$x = 8$

$x = 25$

$2(x + 3) = 14$ gives:

$x = 4$

$x = 7$

$x = 11$

$x = 8$

$5x = 2x + 12$ gives:

$x = 4$

$x = 12$

$x = 3$

$x = 6$

A root should be checked in:

your last line

the original equation

a calculator

nothing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Equation	Tenglama	Уравнение
Root	Ildiz	Корень
Unknown	Noma'lum	Неизвестное
Inverse operation	Teskari amal	Обратное действие
To balance	Muvozanatlash	Уравновешивать
Both sides	Ikkala tomon	Обе части
To check	Tekshirmoq	Проверить
Solution	Yechim	Решение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An equation is like:

a list

a balance

a graph

a table

Multiplying by 3 is undone by:

adding 3

subtracting 3

dividing by 3

squaring

$2x - 7 = 11$ gives:

$x = 2$

$x = 9$

$x = 18$

$x = 4$

$3(x - 1) = 12$ gives:

In $4x + 1 = 2x + 9$ you gather the unknowns:

A word problem needs first:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $x + 7 = 12$

ANSWER $x = 5$

2. $x - 4 = 9$

ANSWER $x = 13$

3. $5x = 40$

ANSWER $x = 8$

4. $\frac{x}{3} = 6$

ANSWER $x = 18$

5. $3x + 5 = 20$

ANSWER $x = 5$

6. $2x - 7 = 11$

ANSWER $x = 9$

7. $x + 12 = 12$

ANSWER $x = 0$

Medium · the standard question

7 PROBLEMS

1. $2(x + 3) = 14$

ANSWER $x = 4$

2. $3(x - 1) = 12$

ANSWER $x = 5$

3. $5x = 2x + 12$

ANSWER $x = 4$

4. $4x + 1 = 2x + 9$

ANSWER $x = 4$

5. A number trebled and increased by 5 is 26

ANSWER 7

6. 5 pens cost 20 000: one pen

ANSWER 4000

7. A rectangle of perimeter 30 and width 6

ANSWER Length 9

Hard · stretch and reason

7 PROBLEMS

1. $4(x - 2) = 2(x + 3)$

ANSWER $x = 7$

2. $5x - 3 = 3x + 7$

ANSWER $x = 5$

3. $\frac{x}{2} + 4 = 10$

ANSWER $x = 12$

4. $7 - x = 2$

ANSWER $x = 5$

5. Ali is 4 years older than Sardor and together they are 26

ANSWER 11 and 15

6. Three consecutive numbers add to 42

ANSWER 13, 14, 15

7. Why check in the original equation?

ANSWER A check on your own working repeats your own error

07 Homework**Homework — 5 tasks**

Write the check as its own line beneath every answer.

1. Solve $x + 9 = 21$, $6x = 54$ and $\frac{x}{4} = 7$.

2. Solve $4x + 3 = 27$ and $5x - 8 = 12$.

3. Solve $3(x + 2) = 21$.

4. Solve $6x + 2 = 4x + 14$.

5. A number doubled and increased by 7 is 31. Find it, writing all three lines.

Control work 1 — expressions and equations

Directed numbers, algebraic expressions and one-step and two-step equations.

LESSON 19

1 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 1

S7 1–2 REVIEW

LESSON CLOCK

2'

28'

8'

2'

Instructions	2 min
The paper	28 min
Answers and diagnosis	8 min
What comes next	2 min

LEARNING OBJECTIVES

- Work with the sign rules accurately under time.
- Translate between words and algebra.
- Simplify and evaluate an expression.
- Solve an equation and check the root.

TEXTBOOK

National	pp. 9–50
Cambridge	Stage 7, pp. 8–38
Workbook	Control paper 1

01

Explanation

The paper — 20 marks, 28 minutes

Q	TASK	MARKS	FROM
1	$-3 + 8$, $4 - 9$ and $5 - (-6)$	3	L2–4
2	$(-7) \cdot 4$, $(-36) \div (-9)$ and $(-3)^2$	3	L5–7
3	Write “four times a number, then subtract 3” and “subtract 3 from a number, then multiply by 4”	3	L8–10
4	Simplify $6x + 2y - 4x + 3y$	3	L11–12
5	Find $x^2 - 2x$ at $x = -3$	3	L13–15
6	Solve $4x - 5 = 19$ and check	3	L16–18
7	Solve $3(x + 2) = 5x$	2	L16–18

THE ANSWERS

5, -5, 11; -28, 4, 9; $4x - 3$ and $4(x - 3)$; $2x + 5y$; 15; $x = 6$; $x = 3$.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
double minus not applied	$5 - (-6) = -1$	11
sign rule reversed	$(-36) \div (-9) = -4$	4
$(-3)^2$	-9	9
bracket omitted in a translation	$4x - 3$	$4(x - 3)$
unlike terms collected	$6x + 2y = 8xy$	$6x + 2y$
negative substituted without brackets	$-3^2 = -9$	$(-3)^2 = 9$
no check	a root and nothing else	substitute into the original

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	REVISE BEFORE
Q1–Q2	the next Cambridge insert — inequalities, lesson 20
Q3–Q4	ratio, lesson 42, where expressions return
Q5	percentages, lesson 59, where formulae are evaluated
Q6–Q7	every word problem for the rest of the year

LOOKING FORWARD

The next block leaves algebra for a while and returns to fractions — but every one of the equations above will be needed again in Quarter III, when speed and distance are found from formulae.

02 Worked examples

EXAMPLE 1 Model answer, Q5: find $x^2 - 2x$ at $x = -3$.

① Substitute in brackets: $(-3)^2 - 2(-3)$.

② $(-3)^2 = 9$

③ $-2 \cdot (-3) = +6$
Two minuses.

④ $9 + 6 = 15$

ANSWER

15

EXAMPLE 2 Model answer, Q6: solve $4x - 5 = 19$.

- ① Add 5: $4x = 24$.
- ② Divide by 4: $x = 6$.
- ③ Check: $4 \cdot 6 - 5 = 19$ ✓

ANSWER

$x = 6$

EXAMPLE 3 Model answer, Q7: solve $3(x + 2) = 5x$.

- ① Expand: $3x + 6 = 5x$.
- ② Subtract $3x$: $6 = 2x$.
- ③ $x = 3$
Check: $15 = 15$ ✓

ANSWER

$x = 3$

03 Interactive model

Mark Q1 and Q2 first and return them within the lesson; sign errors corrected the same day rarely come back.

The block in eight questions

$5 - (-6)$ equals:

$(-7) \cdot 4$ equals:

$(-36) \div (-9)$ equals:

$(-3)^2$ equals:

9

-9

6

-6

“Subtract 3 from a number, then multiply by 4”:

$4x - 3$

$4(x - 3)$

$4x + 3$

$3 - 4x$

$6x + 2y - 4x + 3y$ equals:

$2x + 5y$

$10x + 5y$

$2x - y$

$7xy$

$x^2 - 2x$ at $x = -3$ is:

3

15

-15

9

$4x - 5 = 19$ gives:

$$x = 6$$

$$x = 3.5$$

$$x = 24$$

$$x = 4$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Integer	Butun son	Целое число
Expression	Ifoda	Выражение
Like terms	O'xshash hadlar	Подобные члены
To substitute	O'rniga qo'yish	Подставить
Equation	Tenglama	Уравнение
Root	Ildiz	Корень
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 tests:

multiplying integers

adding and subtracting integers

equations

expressions

Q3 loses marks most often through:

the numbers

the missing bracket

the letters

the sign

Q4 answer is:

 $2x + 5y$ $10x + 5y$ $7xy$ $2xy$

Q5 needs the substitution written:

without brackets

in brackets

as a decimal

as a fraction

Q6 must end with:

the root

the root and a check

a graph

a table

Q7 begins by:

dividing

expanding the bracket

squaring

checking

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $-3 + 8$

ANSWER 5

2. $4 - 9$

ANSWER -5

3. $5 - (-6)$

ANSWER 11

4. $(-7) \cdot 4$

ANSWER -28

5. $(-36) \div (-9)$

ANSWER 4

6. $(-3)^2$

ANSWER 9

7. $6x + 2y - 4x + 3y$

ANSWER $2x + 5y$ **Medium · the standard question**

7 PROBLEMS

1. “Four times a number, then subtract 3”

ANSWER $4x - 3$

2. “Subtract 3 from a number, then multiply by 4”

ANSWER $4(x - 3)$

3. $x^2 - 2x$ at $x = -3$

ANSWER 15

4. $4x - 5 = 19$

ANSWER $x = 6$

5. $3(x + 2) = 5x$

ANSWER $x = 3$

6. $2x + 7 = x + 12$

ANSWER $x = 5$

7. $(-2)^3$

ANSWER -8

Hard · stretch and reason

7 PROBLEMS

1. $-4 - (-9) + (-2)$

ANSWER 3

2. $(-2)(-3)(-1)$

ANSWER -6

3. $2x^2 - x$ at $x = -2$

ANSWER 10

4. $5(x - 1) = 3(x + 3)$

ANSWER $x = 7$

5. Simplify $4(a + 2b) - 3a$

ANSWER $a + 8b$

6. A number trebled and decreased by 4 is 17

ANSWER 7

7. Why must -3 be substituted in brackets?

ANSWER Otherwise the square loses its sign

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Compute $-8 + 15$, $3 - 11$ and $7 - (-4)$.
3. Compute $(-5) \cdot 6$ and $(-48) \div (-6)$.
4. Simplify $8a + 3b - 5a + b$ and find its value at $a = 2$, $b = -1$.
5. Solve $5x - 4 = 26$ and check your root.

Think — inequalities

A Cambridge insert: when two quantities are not equal, and how to write and read that.

LESSON 20

1 × 40 MIN

MATEMATIKA 6, O'YLAB KO'R

S7 2.6

LESSON CLOCK

10'

12'

12'

6'

Four symbols	10 min
On the number line	12 min
Listing the integers	12 min
In the world	6 min

LEARNING OBJECTIVES

- Read and write the four inequality symbols.
- Show an inequality on a number line.
- List the integers satisfying a simple inequality.
- Use inequalities to describe a real restriction.

TEXTBOOK

National	pp. 51–53
Cambridge	Stage 7, pp. 38–41
Workbook	Exercise 2.6

01

Explanation

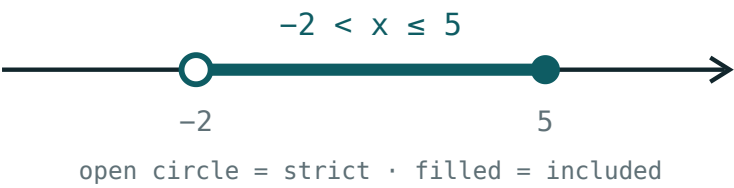
Four symbols

SYMBOL	READ AS	EVERYDAY PHRASE
>	is greater than	more than
<	is less than	fewer than
≥	is greater than or equal to	at least
≤	is less than or equal to	at most, no more than

THE WIDE END FACES THE LARGER NUMBER

7 > 3 and 3 < 7 say the same thing. Reading the symbol aloud from left to right, every time, is how it stops being guessed at.

On the number line



An inequality shown as a ray on the number line

INEQUALITY	CIRCLE AT THE END	ARROW POINTS
$x > 3$	open	right
$x \geq 3$	filled	right
$x < 3$	open	left
$x \leq 3$	filled	left

THE CIRCLE SAYS WHETHER THE END POINT COUNTS

Filled means the number itself is included; open means it is not. That single dot is the whole difference between $>$ and $\geq$.

Listing the integers

CONDITION	THE INTEGERS	HOW MANY
$2 < x < 7$	3, 4, 5, 6	4
$2 \leq x \leq 7$	2, 3, 4, 5, 6, 7	6
$-3 < x \leq 1$	-2, -1, 0, 1	4
$x \geq 5$	5, 6, 7, ...	infinitely many

$<$ EXCLUDES THE END, $\leq$ INCLUDES IT

$2 < x < 7$ has four integers; $2 \leq x \leq 7$ has six. Two symbols, two extra numbers — and it is a mark in every paper that asks for a list.

In the world

STATEMENT	IN SYMBOLS
a lift carries at most 8 people	$n \leq 8$
the class has at least 20 pupils	$n \geq 20$
the speed limit is 60 km/h	$v \leq 60$
water is liquid between 0 and 100 °C	$0 < t < 100$
a triangle's angles	$0^\circ < \alpha < 180^\circ$

“AT MOST” IS $\leq$, NOT $<$

A lift that carries at most eight can carry eight. Everyday phrases translate into the symbols with the equality included far more often than pupils expect.

02

Worked examples

EXAMPLE 1 List the integers satisfying $-3 < x \leq 1$.

1

-3 is excluded — the symbol is $<$.

2

1 is included — the symbol is $\leq$.

3

So: $-2, -1, 0, 1$.

ANSWER

$-2, -1, 0, 1$

EXAMPLE 2 Write “a lift carries at most 8 people” in symbols.

① Let n be the number of people.

② “At most” includes 8 itself.

③ $n \leq 8$

ANSWER

$$n \leq 8$$

EXAMPLE 3 How many integers satisfy $2 \leq x \leq 7$?

① Both ends are included.

② 2, 3, 4, 5, 6, 7

③ Six integers.

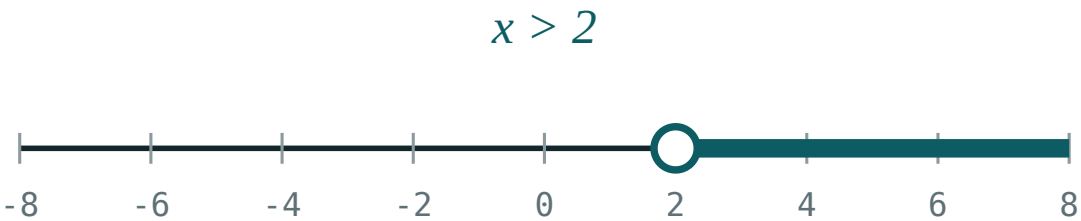
ANSWER

6

03 Interactive model

Ask for the number of pupils the room may hold and write it as an inequality on the board; the class supplies the $\leq$ themselves once they see that the full room is allowed.

An inequality on the number line



inequality $x > 2$

boundary **2 (open)**

$x = 0$ works? **no**

$x = 6$ works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Inequality	Tengsizlik	Неравенство
Greater than	Katta	Больше
Less than	Kichik	Меньше
At least	Kamida	Не менее
At most	Ko'pi bilan	Не более
Number line	Sonlar o'qi	Числовая прямая
Solution set	Yechimlar to'plami	Множество решений
Integer	Butun son	Целое число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\geq$ means:

greater than

at least

at most

less than

$\leq$ means:

at least

at most

greater than

not equal

An open circle on the line means the end point is:

included

excluded

negative

zero

The integers with $2 < x < 7$ number:

The integers with $2 \leq x \leq 7$ number:

“At most 8” is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Read $7 > 3$

ANSWER Seven is greater than three

2. Read $3 < 7$

ANSWER Three is less than seven

3. Read $x \geq 5$

ANSWER x is at least five

4. Read $x \leq 5$

ANSWER x is at most five

5. Is $-2 > -7$?

ANSWER Yes

6. Is $0 < -1$?

ANSWER No

7. The circle for $x > 3$

ANSWER Open

Medium · the standard question

7 PROBLEMS

1. The integers with $2 < x < 7$

ANSWER 3, 4, 5, 6

2. The integers with $2 \leq x \leq 7$

ANSWER 2, 3, 4, 5, 6, 7

3. The integers with $-3 < x \leq 1$

ANSWER -2, -1, 0, 1

4. "A lift carries at most 8"

ANSWER $n \leq 8$

5. "The class has at least 20"

ANSWER $n \geq 20$

6. "The speed limit is 60"

ANSWER $v \leq 60$

7. Water is liquid between 0 and 100 °C

ANSWER $0 < t < 100$

Hard · stretch and reason

7 PROBLEMS

1. How many integers satisfy $-5 \leq x < 4$?

ANSWER 9

2. Write “more than 3 but no more than 10”

ANSWER $3 < x \leq 10$

3. The angles of a triangle satisfy

ANSWER $0^\circ < \alpha < 180^\circ$

4. Which integers satisfy both $x > 2$ and $x \leq 5$?

ANSWER 3, 4, 5

5. Can $x < 3$ and $x > 5$ both hold?

ANSWER No

6. Write the ages of a Grade 6 pupil as an inequality

ANSWER e.g. $11 \leq a \leq 13$

7. Why does “at most” use $\leq$ and not $<$?

ANSWER The stated number is itself allowed

07 Homework

Homework — 5 tasks

Read every symbol aloud from left to right before writing anything.

1. Write in symbols: at least 12; at most 30; more than 7.
2. List the integers satisfying $-4 \leq x < 2$.
3. Show $x > -1$ on a number line, with the right kind of circle.
4. Write three real restrictions from your own day as inequalities.
5. How many integers satisfy $-10 < x \leq 0$?

Discuss and recall — fractions

What Grade 5 established about fractions, gathered on one page before the division begins.

LESSON 21

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 7 ENTRY

LESSON CLOCK

8'

10'

12'

10'

The vocabulary	8 min
Simplifying and converting	10 min
Comparing	12 min
The three operations you already have	10 min

LEARNING OBJECTIVES

- Simplify a fraction to its lowest terms.
- Convert between mixed numbers and improper fractions.
- Compare and order fractions.
- Add, subtract and multiply fractions.

TEXTBOOK

National	pp. 54–57
Cambridge	Stage 7, pp. 66–70
Workbook	Exercise 7.1

01

Explanation

The vocabulary

KIND	EXAMPLE	TEST
proper	$\frac{3}{5}$	numerator smaller
improper	$\frac{7}{5}$	numerator larger
mixed	$1\frac{2}{5}$	a whole part and a proper fraction
equal to one	$\frac{5}{5}$	the two are equal

A FRACTION IS A DIVISION WAITING TO HAPPEN

$\frac{7}{5}$ means $7 \div 5$, which is 1 remainder 2 — that is, $1\frac{2}{5}$. The two forms are the same number written for different purposes.

Simplifying and converting

TASK	METHOD	EXAMPLE
simplify	divide both parts by their HCF	$\frac{18}{24} = \frac{3}{4}$
mixed to improper	whole $\times$ denominator, plus numerator	$2\frac{3}{4} = \frac{11}{4}$
improper to mixed	divide, keep the remainder	$\frac{17}{5} = 3\frac{2}{5}$
equivalent fractions	multiply both parts by the same number	$\frac{2}{3} = \frac{8}{12}$

SIMPLIFY AT THE END, ALWAYS

An answer of $\frac{12}{18}$ is not wrong but is not finished; $\frac{2}{3}$ is. Marks for “lowest terms” are lost far more often than marks for the arithmetic itself.

Comparing

PAIR	METHOD	RESULT
$\frac{3}{7}$ and $\frac{5}{7}$	same denominator	$\frac{3}{7} < \frac{5}{7}$
$\frac{3}{5}$ and $\frac{3}{8}$	same numerator — bigger denominator, smaller parts	$\frac{3}{8} < \frac{3}{5}$
$\frac{3}{5}$ and $\frac{5}{8}$	common denominator 40	$\frac{24}{40} < \frac{25}{40}$
$\frac{4}{9}$ and $\frac{1}{2}$	compare with $\frac{1}{2}$	$\frac{4}{9} < \frac{1}{2}$

BIGGER DENOMINATOR, SMALLER PIECES

A cake cut into 8 gives smaller slices than the same cake cut into 5. That single picture settles every comparison with equal numerators.

The three operations you already have

$$\frac{a}{b} + \frac{c}{b} = \frac{a + c}{b} \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

TASK	WORKING	ANSWER
$\frac{2}{3} + \frac{1}{4}$	$\frac{8}{12} + \frac{3}{12}$	$\frac{11}{12}$
$\frac{5}{6} - \frac{1}{3}$	$\frac{5}{6} - \frac{2}{6}$	$\frac{1}{2}$
$\frac{2}{3} \cdot \frac{3}{5}$	cancel the 3s	$\frac{2}{5}$
$\frac{3}{4}$ of 20	$20 \div 4 \cdot 3$	15

ONE OPERATION IS STILL MISSING

Adding, subtracting and multiplying are done. Division is the whole of the next four lessons — and it turns out to be the easiest of the four.

02 Worked examples

EXAMPLE 1 Simplify $\frac{18}{24}$ and convert $\frac{17}{5}$ to a mixed number.

- ① HCF of 18 and 24 is 6.

$$\frac{3}{4}$$

- ② $17 \div 5 = 3$ remainder 2.

$$3\frac{2}{5}$$

ANSWER

$$\frac{3}{4} \text{ and } 3\frac{2}{5}$$

EXAMPLE 2 Which is larger, $\frac{3}{5}$ or $\frac{5}{8}$?

- ① Common denominator 40.

② $\frac{3}{5} = \frac{24}{40}$

③ $\frac{5}{8} = \frac{25}{40}$

- ④ $\frac{5}{8}$ is larger.

ANSWER

$$\frac{5}{8}$$

EXAMPLE 3 Compute $\frac{2}{3} \cdot \frac{3}{5}$.

① Cancel the 3 in the numerator against the 3 in the denominator.

② $\frac{2}{5}$

Cancel before multiplying — it is always quicker.

ANSWER

$$\frac{2}{5}$$

03 Interactive model

Ask which is bigger, a fifth or an eighth, before anything else; the answer that comes back tells you exactly how much recall this class needs.

Simplifying to lowest terms

$$\frac{18}{24}$$

① $18 = 2 \cdot 3^2$

, $24 = 2^3 \cdot 3$

Factorise
both.

② The HCF is

6 .

The
highest
one
—
anything
less
leaves
work.

③ $\frac{3}{4}$

Divide
both
parts
by
it.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Proper fraction	To'g'ri kasr	Правильная дробь
Improper fraction	Noto'g'ri kasr	Неправильная дробь
Mixed number	Aralash son	Смешанное число
Lowest terms	Qisqarmas ko'rinish	Несократимый вид
Common denominator	Umumiy maxraj	Общий знаменатель
Reciprocal	Teskari son	Обратное число
To simplify	Qisqartirish	Сократить
To compare	Taqqoslash	Сравнить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

7 $\frac{5}{4}$ is:

proper

improper

mixed

equal to one

2 $\frac{3}{4}$ as an improper fraction:

$$\frac{9}{4}$$

$$\frac{11}{4}$$

$$\frac{5}{4}$$

$$\frac{23}{4}$$

18 24 simplifies to:

$$\frac{9}{12}$$

$$\frac{3}{4}$$

$$\frac{2}{3}$$

$$\frac{6}{8}$$

Which is larger, 3 5 or 3 8 ?

$$\frac{3}{5}$$

$$\frac{3}{8}$$

equal

cannot say

2 3 + 1 4 equals:

$$\frac{3}{7}$$

$$\frac{11}{12}$$

$$\frac{2}{12}$$

$$\frac{3}{12}$$

3 4 of 20 is:

5

15

16

60

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\frac{18}{24}$ simplified

ANSWER $\frac{3}{4}$

2. $\frac{45}{60}$ simplified

ANSWER $\frac{3}{4}$

3. $2\frac{3}{4}$ as improper

ANSWER $\frac{11}{4}$

4. $\frac{17}{5}$ as mixed

ANSWER $3\frac{2}{5}$

5. $\frac{3}{7}$ or $\frac{5}{7}$: larger

ANSWER $\frac{5}{7}$

6. $\frac{3}{5}$ or $\frac{3}{8}$: larger

ANSWER $\frac{3}{5}$

7. $\frac{2}{3} + \frac{1}{4}$

ANSWER $\frac{11}{12}$

Medium · the standard question

7 PROBLEMS

1. $\frac{5}{6} - \frac{1}{3}$

ANSWER $\frac{1}{2}$

2. $\frac{2}{3} \cdot \frac{3}{5}$

ANSWER $\frac{2}{5}$

3. $\frac{3}{4}$ of 20

ANSWER 15

4. $\frac{3}{5}$ or $\frac{5}{8}$: larger

ANSWER $\frac{5}{8}$

5. $\frac{4}{9}$ or $\frac{1}{2}$: larger

ANSWER $\frac{1}{2}$

6. $1\frac{1}{2} + \frac{3}{4}$

ANSWER $2\frac{1}{4}$

7. $\frac{2}{5} \cdot \frac{10}{3}$

ANSWER $1\frac{1}{3}$

Hard · stretch and reason

7 PROBLEMS

1. Order $\frac{2}{3}, \frac{3}{5}, \frac{5}{8}$

ANSWER $\frac{3}{5} < \frac{5}{8} < \frac{2}{3}$

2. $2\frac{1}{3} - 1\frac{1}{2}$

ANSWER $\frac{5}{6}$

3. $\frac{5}{8}$ of 64

ANSWER 40

4. A fraction equal to $\frac{3}{4}$ with denominator 28

ANSWER $\frac{21}{28}$

5. $\frac{3}{4} \cdot \frac{2}{9}$

ANSWER $\frac{1}{6}$

6. Which is nearer to 1, $\frac{7}{8}$ or $\frac{9}{10}$?

ANSWER $\frac{9}{10}$

7. Why is $\frac{12}{18}$ an unfinished answer?

ANSWER Both parts still divide by 6

07 Homework

Homework — 5 tasks

Every answer in lowest terms, and mixed numbers where the fraction is improper.

1. Simplify $\frac{24}{36}$, $\frac{30}{45}$ and $\frac{16}{40}$.
2. Convert $3\frac{2}{5}$ to an improper fraction and $\frac{23}{6}$ to a mixed number.
3. Order $\frac{1}{2}$, $\frac{3}{7}$, $\frac{5}{9}$.
4. Compute $\frac{3}{4} + \frac{2}{5}$ and $\frac{7}{8} - \frac{1}{4}$.
5. Compute $\frac{3}{5} \cdot \frac{10}{9}$, cancelling first.

Dividing a proper fraction by a natural number

Sharing a part into equal pieces — the denominator grows, the numerator does not.

LESSON 22–23

2 × 40 MIN

MATEMATIKA 6, §5

S7 7.4

LESSON CLOCK

20'

25'

20'

12'

3

Sharing a part	20 min
The rule	25 min
Cancelling first	20 min
Problems	12 min
Homework	3 min

LEARNING OBJECTIVES

- Divide a fraction by a whole number and explain the result with a picture.
- Write the division as multiplication by the reciprocal.
- Cancel before multiplying.
- Solve simple sharing problems with fractions.

TEXTBOOK

National	pp. 58–64
Cambridge	Stage 7, pp. 71–75
Workbook	Exercise 7.4

01

Explanation

Sharing a part

Half a cake shared between three people: each gets a sixth. The number of pieces the whole is cut into has trebled, and the number taken has not.

$\frac{1}{2} \div 3 = \frac{1}{6}$

DIVISION	IN WORDS	ANSWER
$\frac{1}{2} \div 3$	half shared among three	$\frac{1}{6}$
$\frac{2}{3} \div 2$	two thirds shared between two	$\frac{1}{3}$
$\frac{3}{4} \div 5$	three quarters among five	$\frac{3}{20}$
$\frac{4}{5} \div 4$	four fifths among four	$\frac{1}{5}$

DIVIDING MAKES THE PIECES SMALLER

The answer is always less than the fraction you started with. That is the check to make before writing anything down.

The rule

$$\frac{a}{b} \div n = \frac{a}{b \cdot n} = \frac{a}{b} \cdot \frac{1}{n}$$

Multiply the denominator by the whole number — or, in the form that will cover every case from now on, multiply by the reciprocal of the divisor.

DIVISION	AS MULTIPLICATION	ANSWER
$\frac{3}{4} \div 5$	$\frac{3}{4} \cdot \frac{1}{5}$	$\frac{3}{20}$
$\frac{5}{7} \div 2$	$\frac{5}{7} \cdot \frac{1}{2}$	$\frac{5}{14}$
$\frac{2}{9} \div 3$	$\frac{2}{9} \cdot \frac{1}{3}$	$\frac{2}{27}$

THE NUMERATOR IS UNTOUCHED

$\frac{3}{4} \div 5$ is $\frac{3}{20}$, not $\frac{15}{4}$ and not the whole fraction turned upside down. Only the denominator grows.

Cancelling first

When the whole number divides the numerator, cancel — the answer comes out in lowest terms with no extra step.

DIVISION	CANCEL	ANSWER
$\frac{4}{5} \div 4$	$4 \div 4 = 1$	$\frac{1}{5}$
$\frac{6}{7} \div 3$	$6 \div 3 = 2$	$\frac{2}{7}$
$\frac{9}{10} \div 3$	$9 \div 3 = 3$	$\frac{3}{10}$
$\frac{2}{3} \div 2$	$2 \div 2 = 1$	$\frac{1}{3}$

CANCELLING FIRST KEEPS THE NUMBERS SMALL

$\frac{9}{10} \div 3$ done without cancelling gives $\frac{9}{30}$, which then has to be simplified anyway. One step is better than two.

Problems

PROBLEM	WORKING	ANSWER
$\frac{3}{4}$ of a litre of juice shared between 3 glasses	$\frac{3}{4} \div 3$	$\frac{1}{4}$ litre each
$\frac{5}{6}$ of a field ploughed in 5 days, equally	$\frac{5}{6} \div 5$	$\frac{1}{6}$ a day
$\frac{7}{8}$ of a metre of ribbon cut into 7 pieces	$\frac{7}{8} \div 7$	$\frac{1}{8}$ m each
a bottle $\frac{4}{5}$ full poured into 2 equal jugs	$\frac{4}{5} \div 2$	$\frac{2}{5}$ each

“SHARED BETWEEN” MEANS DIVIDE

Every one of these problems contains that phrase or one like it. Finding it in the sentence is the whole difficulty; the arithmetic afterwards is one line.

02 Worked examples

EXAMPLE 1 Compute $\frac{3}{4} \div 5$.

① Multiply by the reciprocal of 5.

② $\frac{3}{4} \cdot \frac{1}{5}$

③ $= \frac{3}{20}$

Smaller than $\frac{3}{4}$ ✓

ANSWER

$$\frac{3}{20}$$

EXAMPLE 2 Compute $\frac{6}{7} \div 3$.

① 3 divides the numerator 6.

② Cancel: $6 \div 3 = 2$.

③ $= \frac{2}{7}$

ANSWER

$$\frac{2}{7}$$

EXAMPLE 3 $\frac{3}{4}$ of a litre of juice is shared equally between 3 glasses. How much is in each?

① “Shared between” means divide.

② $\frac{3}{4} \div 3$

③ Cancel the 3s.

④ $= \frac{1}{4}$ litre.

ANSWER

$\frac{1}{4}$ litre

03 Interactive model

Fold a paper strip in half, then cut the half into three; the sixth appears in the hand before it appears on the board.

Dividing a fraction by a whole number

$$\frac{3}{4} \div 5$$

① $\frac{3}{4} \cdot \frac{1}{5}$

Multiply
by
the
reciprocal.

② Nothing cancels:

3 and

5 share no factor.

③ $\frac{3}{20}$

Smaller
than
 $\frac{3}{4}$
—
as it
must
be.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To divide	Bo'lish	Делить
Divisor	Bo'luvchi	Делитель
Reciprocal	Teskari son	Обратное число
To share equally	Teng taqsimlash	Разделить поровну
Part	Ulush	Доля
Whole	Butun	Целое
To cancel	Qisqartirish	Сокращать
Lowest terms	Qisqarmas ko'rinish	Несократимый вид

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

1 2 ÷ 3 equals:

$$\frac{1}{6}$$

$$\frac{3}{2}$$

$$\frac{1}{5}$$

$$\frac{2}{3}$$

Dividing a fraction by a whole number makes it:

larger

smaller

the same

negative

3 4 ÷ 5 equals:

$$\frac{15}{4}$$

$$\frac{3}{20}$$

$$\frac{3}{9}$$

$$\frac{4}{15}$$

6 7 ÷ 3 equals:

$$\frac{2}{7}$$

$$\frac{6}{21}$$

$$\frac{18}{7}$$

$$\frac{2}{21}$$

Dividing by n is the same as multiplying by:

$$n$$

$$\frac{1}{n}$$

$$-n$$

$$n^2$$

4 5 ÷ 4 equals:

$$\frac{1}{5}$$

$$\frac{4}{20}$$

$$\frac{16}{5}$$

$$\frac{1}{20}$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{1}{2} \div 3$

ANSWER $\frac{1}{6}$

2. $\frac{2}{3} \div 2$

ANSWER $\frac{1}{3}$

3. $\frac{3}{4} \div 5$

ANSWER $\frac{3}{20}$

4. $\frac{4}{5} \div 4$

ANSWER $\frac{1}{5}$

5. $\frac{6}{7} \div 3$

ANSWER $\frac{2}{7}$

6. $\frac{5}{7} \div 2$

ANSWER $\frac{5}{14}$

7. $\frac{9}{10} \div 3$

ANSWER $\frac{3}{10}$

Medium · the standard question

7 PROBLEMS

1. $\frac{2}{9} \div 3$

ANSWER $\frac{2}{27}$

2. $\frac{8}{9} \div 4$

ANSWER $\frac{2}{9}$

3. $\frac{3}{4}$ of a litre into 3 glasses

ANSWER $\frac{1}{4}$ litre

4. $\frac{5}{6}$ of a field in 5 days

ANSWER $\frac{1}{6}$ a day

5. $\frac{7}{8}$ m of ribbon into 7 pieces

ANSWER $\frac{1}{8}$ m

6. $\frac{4}{5}$ of a bottle into 2 jugs

ANSWER $\frac{2}{5}$

7. $\frac{10}{11} \div 5$

ANSWER $\frac{2}{11}$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{5}{8} \div 10$

ANSWER $\frac{1}{16}$

2. $\frac{12}{13} \div 6$

ANSWER $\frac{2}{13}$

3. $1\frac{1}{2} \div 3$

ANSWER $\frac{1}{2}$

4. $2\frac{2}{3} \div 4$

ANSWER $\frac{2}{3}$

5. A rope $\frac{9}{10}$ m long is cut into 6 equal pieces

ANSWER $\frac{3}{20}$ m

6. Why is the answer always smaller than the fraction?

ANSWER The same part is cut into more pieces

7. Divide $\frac{3}{7}$ by 3 and then by 2

ANSWER $\frac{1}{14}$

07 Homework

Homework — 5 tasks

Cancel before multiplying, and check that the answer is smaller than you started with.

1. Compute $\frac{2}{5} \div 3$ and $\frac{3}{8} \div 2$.
2. Compute $\frac{8}{9} \div 2$ and $\frac{10}{13} \div 5$, cancelling first.
3. Compute $1\frac{1}{3} \div 4$.
4. $\frac{5}{6}$ of a litre of milk is shared between 5 cups. How much is in each?
5. Explain in one sentence why dividing a fraction by 4 makes it smaller.

Dividing a natural number by a proper fraction

How many halves are in three? — the division that makes the answer bigger.

LESSON 24–26

3 × 40 MIN

MATEMATIKA 6, §6

S7 7.4

LESSON CLOCK

25'

25'

30'

30'

10

How many fit?	25 min
The reciprocal	25 min
Why the answer grows	30 min
Problems	30 min
Homework	10 min

LEARNING OBJECTIVES

- Interpret dividing by a fraction as “how many of these fit”.
- Divide a whole number by a fraction using the reciprocal.
- Explain why the answer is larger than the number divided.
- Solve measuring and packing problems.

TEXTBOOK

National	pp. 65–72
Cambridge	Stage 7, pp. 71–75
Workbook	Exercise 7.4

01

Explanation

How many fit?

$3 \div \frac{1}{2}$ asks: how many halves are there in 3? Six — and a picture settles it before any rule is stated.

$$3 \div \frac{1}{2} = 6$$

DIVISION	THE QUESTION IT ASKS	ANSWER
$3 \div \frac{1}{2}$	how many halves in 3	6
$4 \div \frac{1}{4}$	how many quarters in 4	16
$6 \div \frac{2}{3}$	how many two-thirds in 6	9
$5 \div \frac{1}{3}$	how many thirds in 5	15

DIVISION HAS TWO MEANINGS, AND THIS IS THE SECOND

$12 \div 3$ can mean “share 12 among 3” or “how many 3s are in 12”. With a fraction as the divisor, only the second question makes sense.

The reciprocal

$$n \div \frac{a}{b} = n \cdot \frac{b}{a}$$

Turn the divisor upside down and multiply. The **reciprocal** of $\frac{a}{b}$ is $\frac{b}{a}$, and the two multiply to 1.

FRACTION	RECIPROCAL	PRODUCT
$\frac{1}{2}$	2	1
$\frac{2}{3}$	$\frac{3}{2}$	1
5	$\frac{1}{5}$	1
$\frac{7}{4}$	$\frac{4}{7}$	1

DIVISION	AS MULTIPLICATION	ANSWER
$6 \div \frac{2}{3}$	$6 \cdot \frac{3}{2}$	9
$4 \div \frac{4}{5}$	$4 \cdot \frac{5}{4}$	5
$10 \div \frac{5}{6}$	$10 \cdot \frac{6}{5}$	12

TURN OVER THE DIVISOR, NEVER THE DIVIDEND

$6 \div \frac{2}{3}$ is $6 \cdot \frac{3}{2} = 9$. Turning over the 6 instead gives $\frac{1}{9}$ — the reciprocal of the right answer, and a mistake the estimate below catches instantly.

Why the answer grows

Dividing by a number smaller than 1 always gives a larger answer, because the pieces being counted are smaller than the whole.

DIVISOR	COMPARED WITH 1	THE ANSWER IS	EXAMPLE
$\frac{1}{2}$	smaller	larger than 6	$6 \div \frac{1}{2} = 12$
1	equal	unchanged	$6 \div 1 = 6$
$\frac{3}{2}$	larger	smaller than 6	$6 \div \frac{3}{2} = 4$

ESTIMATE BEFORE YOU CALCULATE

Ask first: will the answer be bigger or smaller than the number I started with? One second of thought rules out half the possible mistakes.

Problems

PROBLEM	WORKING	ANSWER
6 m of cloth cut into pieces of $\frac{2}{3}$ m	$6 \div \frac{2}{3}$	9
4 litres poured into $\frac{1}{4}$ -litre cups	$4 \div \frac{1}{4}$	16
10 kg of rice in $\frac{5}{6}$ -kg bags	$10 \div \frac{5}{6}$	12
a wall 9 m long, tiles $\frac{3}{4}$ m wide	$9 \div \frac{3}{4}$	12

“HOW MANY ... FIT” IS THE PHRASE TO LOOK FOR

Cutting, pouring, packing and tiling are all the same question. Once the phrase is spotted, the division writes itself.

02

Worked examples

EXAMPLE 1

Compute $6 \div \frac{2}{3}$.

1

The divisor is less than 1, so expect an answer over 6.

2

$6 \cdot \frac{3}{2}$
Turn over the divisor.

3

$= \frac{18}{2} = 9$

ANSWER

9

EXAMPLE 2 How many $\frac{3}{4}$ m tiles fit along a wall 9 m long?

① The question is “how many fit”, so divide.

② $9 \div \frac{3}{4} = 9 \cdot \frac{4}{3}$

③ $= 12$

ANSWER

12 tiles

EXAMPLE 3 Compute $10 \div \frac{5}{6}$.

① $10 \cdot \frac{6}{5}$

② Cancel: $10 \div 5 = 2$.

③ $2 \cdot 6 = 12$

ANSWER

12

03 Interactive model

Lay a metre rule on the desk and count how many 25-cm cards fit along it; the class answers “four” without hesitation and has just divided by a quarter.

Dividing by a fraction

$$6 \div \frac{2}{3}$$

- ① The divisor is under
1, so the answer beats
6.

Estimate
first.

$$\textcircled{2} \quad 6 \cdot \frac{3}{2}$$

Turn
over
the
divisor.

$$\textcircled{3} \quad 9$$

How
many
two-
thirds
fit in
six.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Reciprocal	Teskari son	Обратное число
How many fit	Nechta joylashadi	Сколько помещается
To turn upside down	Ag'darib qo'yish	Перевернуть
Divisor	Bo'luvchi	Делитель
Quotient	Bo'linma	Частное
Measuring problem	O'lchash masalasi	Задача на измерение
Whole number	Butun son	Целое число
Estimate	Baholash	Оценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$3 \div \underline{1} \ 2$ equals:

$$\frac{3}{2}$$

6

$$\frac{1}{6}$$

1.5

The reciprocal of $\underline{2} \ 3$ is:

$$\frac{3}{2}$$

$$-\frac{2}{3}$$

$$\frac{2}{3}$$

3

Dividing by a number under 1 gives an answer:

smaller

larger

the same

negative

$6 \div \underline{2} \ 3$ equals:

4

9

$\frac{4}{9}$

12

$4 \div \underline{4} \ 5$ equals:

5

$\frac{16}{5}$

$\frac{5}{4}$

3.2

Which fraction is turned over?

the first

the divisor

both

neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $3 \div \frac{1}{2}$

ANSWER 6

2. $4 \div \frac{1}{4}$

ANSWER 16

3. $5 \div \frac{1}{3}$

ANSWER 15

4. $6 \div \frac{2}{3}$

ANSWER 9

5. $4 \div \frac{4}{5}$

ANSWER 5

6. The reciprocal of $\frac{3}{7}$

ANSWER $\frac{7}{3}$

7. The reciprocal of 5

ANSWER $\frac{1}{5}$

Medium · the standard question

7 PROBLEMS

1. $10 \div \frac{5}{6}$

ANSWER 12

2. $9 \div \frac{3}{4}$

ANSWER 12

3. $8 \div \frac{2}{5}$

ANSWER 20

4. 6 m of cloth in $\frac{2}{3}$ m pieces

ANSWER 9

5. 4 litres in $\frac{1}{4}$ -litre cups

ANSWER 16

6. 10 kg in $\frac{5}{6}$ -kg bags

ANSWER 12

7. $6 \div \frac{3}{2}$

ANSWER 4

Hard · stretch and reason

7 PROBLEMS

1. $7 \div \frac{7}{9}$

ANSWER 9

2. $12 \div \frac{8}{9}$

ANSWER $13\frac{1}{2}$

3. $1 \div \frac{4}{7}$

ANSWER $\frac{7}{4}$

4. Which is larger, $8 \div \frac{1}{2}$ or $8 \cdot \frac{1}{2}$?

ANSWER $8 \div \frac{1}{2} = 16$

5. A wall 15 m long, tiles $\frac{5}{8}$ m wide

ANSWER 24 tiles

6. Why does dividing by $\frac{1}{2}$ double a number?

ANSWER Two halves make each whole

7. $20 \div \frac{4}{5} \div 5$

ANSWER 5

07 Homework

Homework — 5 tasks

Estimate first: will the answer be larger or smaller than the number you started with?

1. Compute $8 \div \frac{1}{2}$ and $9 \div \frac{1}{3}$.
2. Compute $12 \div \frac{3}{4}$ and $15 \div \frac{5}{6}$.
3. How many $\frac{2}{5}$ -litre bottles can be filled from 6 litres?
4. A ribbon 8 m long is cut into $\frac{4}{5}$ m pieces. How many pieces?
5. Explain why $6 \div \frac{2}{3}$ is larger than 6.

Dividing a proper fraction by a proper fraction

One rule now covers every case: multiply by the reciprocal of the divisor.

LESSON 27–29

3 × 40 MIN

MATEMATIKA 6, §7

S7 7.4

LESSON CLOCK

20'

30'

30'

30'

10

The single rule	20 min
Cancelling	30 min
Mixed numbers	30 min
Bigger or smaller?	30 min
Homework	10 min

LEARNING OBJECTIVES

- Divide any fraction by any fraction.
- Divide mixed numbers by converting them first.
- Cancel before multiplying, and simplify the answer.
- Decide from the divisor whether the answer grows or shrinks.

TEXTBOOK

National	pp. 73–80
Cambridge	Stage 7, pp. 71–75
Workbook	Exercise 7.4

01

Explanation

The single rule

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$$

The three cases of the last two lessons were the same rule in disguise: a whole number n is $\frac{n}{1}$, whose reciprocal is $\frac{1}{n}$.

DIVISION	AS MULTIPLICATION	ANSWER
$\frac{1}{2} \div \frac{1}{4}$	$\frac{1}{2} \cdot \frac{4}{1}$	2
$\frac{3}{4} \div \frac{1}{2}$	$\frac{3}{4} \cdot \frac{2}{1}$	$\frac{3}{2} = 1\frac{1}{2}$
$\frac{2}{3} \div \frac{4}{5}$	$\frac{2}{3} \cdot \frac{5}{4}$	$\frac{5}{6}$
$\frac{5}{8} \div \frac{5}{8}$	$\frac{5}{8} \cdot \frac{8}{5}$	1

ANY NUMBER DIVIDED BY ITSELF IS 1

The last line is a check worth remembering: if the two fractions are equal, the answer must be 1, whatever the digits look like.

Cancelling

After turning the divisor over, cancel any numerator against any denominator before multiplying.

DIVISION	AFTER TURNING OVER	CANCEL	ANSWER
$\frac{3}{4} \div \frac{9}{8}$	$\frac{3}{4} \cdot \frac{8}{9}$	3	$\frac{2}{3}$
$\frac{5}{6} \div \frac{10}{3}$	$\frac{5}{6} \cdot \frac{3}{10}$	5	$\frac{1}{4}$
$\frac{7}{9} \div \frac{14}{3}$	$\frac{7}{9} \cdot \frac{3}{14}$	7	$\frac{1}{6}$

CANCEL ONLY AFTER THE DIVISOR HAS BEEN TURNED OVER

In $\frac{3}{4} \div \frac{9}{8}$ the 4 and the 8 cannot be cancelled while the $\div$ sign is still there. Turn over first, then cancel — in that order, every time.

Mixed numbers

Convert every mixed number to an improper fraction **before** dividing. Nothing else in the method changes.

DIVISION	CONVERTED	ANSWER
$1\frac{1}{2} \div \frac{3}{4}$	$\frac{3}{2} \cdot \frac{4}{3}$	2
$2\frac{1}{4} \div 1\frac{1}{2}$	$\frac{9}{4} \cdot \frac{2}{3}$	$1\frac{1}{2}$
$3\frac{1}{3} \div \frac{5}{6}$	$\frac{10}{3} \cdot \frac{6}{5}$	4

A MIXED NUMBER CANNOT BE TURNED OVER AS IT STANDS

The reciprocal of $1\frac{1}{2}$ is $\frac{2}{3}$, not $1\frac{2}{1}$. Convert to $\frac{3}{2}$ first and the reciprocal is obvious.

Bigger or smaller?

DIVISOR	THE ANSWER, COMPARED WITH THE FIRST FRACTION	EXAMPLE
less than 1	larger	$\frac{3}{4} \div \frac{1}{2} = 1\frac{1}{2}$
equal to 1	unchanged	$\frac{3}{4} \div 1 = \frac{3}{4}$
greater than 1	smaller	$\frac{3}{4} \div \frac{3}{2} = \frac{1}{2}$

ONE GLANCE AT THE DIVISOR PREDICTS THE ANSWER

Comparing the divisor with 1 takes no calculation at all, and it catches the commonest error of the whole block — turning over the wrong fraction.

02 Worked examples

EXAMPLE 1 Compute $\frac{2}{3} \div \frac{4}{5}$.

① The divisor is under 1, so expect an answer over $\frac{2}{3}$.

② $\frac{2}{3} \cdot \frac{5}{4}$

Turn over the divisor.

③ Cancel 2 against 4.

④ $= \frac{5}{6}$

ANSWER

$$\frac{5}{6}$$

EXAMPLE 2 Compute $2\frac{1}{4} \div 1\frac{1}{2}$.

① Convert: $\frac{9}{4}$ and $\frac{3}{2}$.

② $\frac{9}{4} \cdot \frac{2}{3}$

③ Cancel 9 with 3, and 2 with 4.

④ $= \frac{3}{2} = 1\frac{1}{2}$

ANSWER

$$1\frac{1}{2}$$

EXAMPLE 3 Compute $\frac{5}{6} \div \frac{10}{3}$.

① $\frac{5}{6} \cdot \frac{3}{10}$

② Cancel 5 with 10, and 3 with 6.

③ $= \frac{1}{4}$

Smaller, because the divisor is over 1.

ANSWER

$$\frac{1}{4}$$

03 Interactive model

Put the four divisions of the block side by side — fraction by whole, whole by fraction, fraction by fraction — and ask what is common; the class states the single rule itself.

One rule for every division

$$\frac{2}{3} \div \frac{4}{5}$$

① Divisor under

1 — the answer will be larger.

Estimate
first.

$$\textcircled{2} \quad \frac{2}{3} \cdot \frac{5}{4}$$

Turn
over
the
divisor.

$$\textcircled{3} \quad \frac{5}{6}$$

Cancel
2
against
4.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Division of fractions	Kasrlarni bo'lish	Деление дробей
Reciprocal	Teskari son	Обратное число
Mixed number	Aralash son	Смешанное число
Improper fraction	Noto'g'ri kasr	Неправильная дробь
To cancel	Qisqartirish	Сокращать
Complex fraction	Qavatli kasr	Многоэтажная дробь
Estimate	Baholash	Оценка
Lowest terms	Qisqarmas ko'rinish	Несократимый вид

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\frac{1}{2} \div \frac{1}{4}$ equals:

$\frac{1}{8}$

2

$\frac{1}{2}$

8

To divide by a fraction you:

turn over the first

turn over the divisor

turn over both

turn over neither

2 3 ÷ 4 5 equals:

$$\frac{8}{15}$$

$$\frac{5}{6}$$

$$\frac{6}{5}$$

$$\frac{10}{12}$$

A mixed number must first be:

turned over

converted to an improper fraction

rounded

doubled

1 1 2 ÷ 3 4 equals:

2

$$\frac{9}{8}$$

$$\frac{1}{2}$$

1

A divisor greater than 1 makes the answer:

larger

smaller

the same

zero

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{1}{2} \div \frac{1}{4}$

ANSWER 2

2. $\frac{3}{4} \div \frac{1}{2}$

ANSWER $1\frac{1}{2}$

3. $\frac{5}{8} \div \frac{5}{8}$

ANSWER 1

4. $\frac{2}{3} \div \frac{4}{5}$

ANSWER $\frac{5}{6}$

5. $\frac{1}{3} \div \frac{2}{3}$

ANSWER $\frac{1}{2}$

6. The reciprocal of $\frac{4}{9}$

ANSWER $\frac{9}{4}$

7. The reciprocal of $1\frac{1}{2}$

ANSWER $\frac{2}{3}$

Medium · the standard question

7 PROBLEMS

1. $\frac{3}{4} \div \frac{9}{8}$

ANSWER $\frac{2}{3}$

2. $\frac{5}{6} \div \frac{10}{3}$

ANSWER $\frac{1}{4}$

3. $\frac{7}{9} \div \frac{14}{3}$

ANSWER $\frac{1}{6}$

4. $1\frac{1}{2} \div \frac{3}{4}$

ANSWER 2

5. $2\frac{1}{4} \div 1\frac{1}{2}$

ANSWER $1\frac{1}{2}$

6. $3\frac{1}{3} \div \frac{5}{6}$

ANSWER 4

7. $\frac{3}{4} \div \frac{3}{2}$

ANSWER $\frac{1}{2}$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{4}{5} \div \frac{2}{15}$

ANSWER 6

2. $2\frac{2}{3} \div 1\frac{1}{3}$

ANSWER 2

3. $\frac{9}{10} \div \frac{3}{5} \div \frac{3}{2}$

ANSWER 1

4. Without calculating: is $\frac{5}{7} \div \frac{8}{9}$ larger or smaller than $\frac{5}{7}$?

ANSWER Smaller

5. Without calculating: is $\frac{5}{7} \div \frac{2}{3}$ larger or smaller?

ANSWER Larger

6. A jug holds $\frac{3}{4}$ litre; how many $\frac{1}{8}$ -litre cups fill it?

ANSWER 6

7. Why is $\frac{5}{8} \div \frac{5}{8} = 1$?

ANSWER Any number divided by itself is one

07 Homework

Homework — 5 tasks

Turn over the divisor first, then cancel — never the other way round.

1. Compute $\frac{3}{5} \div \frac{1}{2}$ and $\frac{4}{7} \div \frac{8}{9}$.

2. Compute $\frac{5}{6} \div \frac{5}{12}$.
3. Compute $1\frac{3}{4} \div \frac{7}{8}$.
4. Compute $2\frac{1}{2} \div 1\frac{1}{4}$.
5. Say, without calculating, whether $\frac{2}{3} \div \frac{5}{4}$ is larger or smaller than $\frac{2}{3}$, and why.

Word problems on the division of fractions

Deciding whether to multiply or divide — and finding the whole when a part is given.

LESSON 30–32

3 × 40 MIN

MATEMATIKA 6, §8

S7 7.5

LESSON CLOCK

25'

30'

30'

25'

10

Multiply or divide?	25 min
Finding the whole	30 min
A fraction of a fraction	30 min
Harder problems	25 min
Homework	10 min

LEARNING OBJECTIVES

- Decide from the wording whether a problem needs multiplication or division.
- Find the whole when a fractional part of it is known.
- Solve problems where a fraction of a fraction is taken.
- Check an answer against the sense of the problem.

TEXTBOOK

National	pp. 81–90
Cambridge	Stage 7, pp. 76–80
Workbook	Exercise 7.5

01

Explanation

Multiply or divide?

THE PROBLEM ASKS	OPERATION	EXAMPLE
a fraction of a known amount	multiply	$\frac{3}{4}$ of 20 = 15
how many small amounts fit in a large one	divide	$6 \div \frac{2}{3} = 9$
share an amount into equal parts	divide	$\frac{3}{4} \div 3 = \frac{1}{4}$
the whole, when a part is known	divide	$12 \div \frac{3}{4} = 16$

THE WORD “OF” MEANS MULTIPLY

And “how many fit”, “shared between” and “if this is a third of it” all mean divide. Reading for those phrases is more reliable than reading for the numbers.

Finding the whole

If $\frac{3}{4}$ of a length is 12 m, the whole is $12 \div \frac{3}{4} = 16$ m. Dividing by the fraction undoes taking the fraction.

GIVEN	WORKING	THE WHOLE
$\frac{3}{4}$ of it is 12	$12 \div \frac{3}{4}$	16
$\frac{2}{5}$ of it is 18	$18 \div \frac{2}{5}$	45
$\frac{5}{6}$ of it is 25	$25 \div \frac{5}{6}$	30
$\frac{1}{3}$ of it is 7	$7 \div \frac{1}{3}$	21

THE WHOLE IS ALWAYS BIGGER THAN THE PART

An answer smaller than the given part means the operations were the wrong way round. This single check catches nearly every error in this section.

A fraction of a fraction

PROBLEM	WORKING	ANSWER
$\frac{2}{3}$ of $\frac{3}{5}$ of a field	$\frac{2}{3} \cdot \frac{3}{5}$	$\frac{2}{5}$
half of $\frac{4}{7}$	$\frac{1}{2} \cdot \frac{4}{7}$	$\frac{2}{7}$
$\frac{3}{4}$ of a class of 28 are girls; $\frac{1}{3}$ of the girls wear glasses	$21 \cdot \frac{1}{3}$	7

“OF” CHAINS MULTIPLY

A fraction of a fraction of a whole is one product with three factors. There is no need to work out each stage separately unless the question asks for it.

Harder problems

PROBLEM	WORKING	ANSWER
a tank is $\frac{3}{5}$ full with 24 litres; its capacity	$24 \div \frac{3}{5}$	40
$\frac{2}{3}$ of a job is done in 4 days; the whole job	$4 \div \frac{2}{3}$	6 days
after spending $\frac{2}{5}$, 18 000 sum is left; the original	$18\,000 \div \frac{3}{5}$	30 000
$\frac{3}{8}$ of a road is 15 km; the rest	$40 - 15$	25 km

READ WHAT THE GIVEN NUMBER IS A FRACTION OF

In the third row 18 000 is what is **left**, so it is $\frac{3}{5}$ of the money, not $\frac{2}{5}$. Naming the fraction that matches the given number is the whole difficulty.

02

Worked examples

EXAMPLE 1 $\frac{3}{4}$ of a rope is 12 m. How long is the whole rope?

1

A part is given; the whole is wanted, so divide.

2

$12 \div \frac{3}{4} = 12 \cdot \frac{4}{3}$

3

$= 16$ m.

Bigger than 12 ✓

ANSWER

16 m

EXAMPLE 2 A tank is $\frac{3}{5}$ full and holds 24 litres. What is its capacity?

① 24 litres is $\frac{3}{5}$ of the capacity.

② $24 \div \frac{3}{5} = 24 \cdot \frac{5}{3}$

③ = 40 litres.

ANSWER

40 litres

EXAMPLE 3 After spending $\frac{2}{5}$ of her money, Nilufar has 18 000 sum left. How much had she?

① What is left is $1 - \frac{2}{5} = \frac{3}{5}$ of the money.

The key step.

② $18\ 000 \div \frac{3}{5}$

③ = 30 000 sum.

Check: $\frac{2}{5}$ of 30 000 is 12 000 ✓

ANSWER

30 000 sum

03 Interactive model

Give six problems and ask only for the operation, not the answer; deciding between $\times$ and $\div$ is the skill, and it is worth a lesson on its own.

Multiply or divide?

3 / 4 of 20:

multiply

divide

add

subtract

How many 2 / 3 m pieces fit in 6 m:

multiply

divide

add

subtract

3 / 4 of a rope is 12 m; the whole rope:

multiply

divide

add

subtract

The whole rope is:

9 m

16 m

15 m

48 m

A tank $\frac{3}{5}$ full holds 24 l; its capacity:

14.4 l

40 l

30 l

72 l

$\frac{2}{3}$ of $\frac{3}{5}$:

$\frac{2}{5}$

$\frac{5}{8}$

$\frac{10}{9}$

$\frac{6}{15}$ unsimplified

After spending $\frac{2}{5}$, what is left is:

$$\frac{2}{5}$$

$$\frac{3}{5}$$

$$\frac{1}{5}$$

$$\frac{5}{3}$$

The whole must be:

smaller than the part

larger than the part

equal to it

either

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Word problem	Matnli masala	Текстовая задача
Part of a whole	Butunning qismi	Часть целого
To find the whole	Butunni topish	Найти целое
Fraction of a fraction	Kasrning kasri	Дробь от дроби
Remainder	Qoldiq	Остаток
Per unit	Bir birlikka	На единицу
To interpret	Talqin qilmoq	Истолковать
Sense check	Mantiqiy tekshirish	Проверка по смыслу

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“Of” means:

“How many fit” means:

2 ⁵ of a number is 18; the number is:

5 6 of it is 25; it is:

30

20.8

35

150

2 3 of 3 5 equals:

$\frac{2}{5}$

$\frac{9}{10}$

$\frac{5}{8}$

$\frac{6}{8}$

A whole smaller than its part means:

a correct answer

the wrong operation

a large fraction

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{3}{4}$ of 20

ANSWER 15

2. $\frac{2}{3}$ of 27

ANSWER 18

3. 6 m in $\frac{2}{3}$ m pieces

ANSWER 9

4. $\frac{3}{4}$ of a rope is 12 m: the rope

ANSWER 16 m

5. $\frac{1}{3}$ of it is 7: it

ANSWER 21

6. $\frac{2}{5}$ of it is 18: it

ANSWER 45

7. $\frac{5}{6}$ of it is 25: it

ANSWER 30

Medium · the standard question

7 PROBLEMS

1. $\frac{2}{3}$ of $\frac{3}{5}$

ANSWER $\frac{2}{5}$

2. Half of $\frac{4}{7}$

ANSWER $\frac{2}{7}$

3. A tank $\frac{3}{5}$ full holds 24 l: its capacity

ANSWER 40 l

4. $\frac{2}{3}$ of a job in 4 days: the whole job

ANSWER 6 days

5. After spending $\frac{2}{5}$, 18 000 is left: the original

ANSWER 30 000

6. $\frac{3}{4}$ of 28 are girls; $\frac{1}{3}$ of them wear glasses

ANSWER 7

7. $\frac{3}{8}$ of a road is 15 km: the whole road

ANSWER 40 km

Hard · stretch and reason

7 PROBLEMS

1. $\frac{3}{8}$ of a road is 15 km: the rest

ANSWER 25 km

2. After giving away $\frac{1}{4}$ and then $\frac{1}{3}$ of the rest, 12 are left: the original

ANSWER 24

3. $\frac{4}{5}$ of a number is 4 more than $\frac{3}{5}$ of it: the number

ANSWER 20

4. A jug holds $1\frac{1}{2}$ l; how many $\frac{1}{4}$ l cups fill it?

ANSWER 6

5. $\frac{2}{3}$ of $\frac{3}{4}$ of 40

ANSWER 20

6. Half of a number added to a third of it is 25: the number

ANSWER 30

7. Why must the whole be larger than the part?

ANSWER The part is a fraction of it, under one

07 Homework

Homework — 5 tasks

Write the operation you have chosen before doing any arithmetic.

- $\frac{2}{3}$ of a length is 14 m. Find the whole length.
- A barrel is $\frac{5}{8}$ full with 30 litres. Find its capacity.

3. Find $\frac{3}{5}$ of $\frac{5}{9}$ of 36.
4. After spending $\frac{3}{7}$ of his money, Bekzod has 24 000 sum left. How much had he?
5. How many $\frac{3}{4}$ m planks can be cut from a board 9 m long?

Think — Project 1: mixed-up properties

A Cambridge project: which of the four operations obey which laws, tested rather than assumed.

LESSON 33

1 × 40 MIN

MATEMATIKA 6, O'YLAB KO'R

S7 PROJECT 1

LESSON CLOCK

8'

12'

14'

6'

The three laws	8 min
Testing them	12 min
Counter-examples	14 min
Presenting the findings	6 min

LEARNING OBJECTIVES

- State the commutative, associative and distributive laws.
- Test each law on each operation with examples.
- Produce a counter-example where a law fails.
- Present the findings as a table with a conclusion.

TEXTBOOK

National	pp. 91–93
Cambridge	Stage 7, project pages
Workbook	Project sheet 1

01

Explanation

The three laws

$a + b = b + a$ $(a + b) + c = a + (b + c)$ $a(b + c) = ab + ac$

LAW	WHAT IT SAYS	IN WORDS
commutative	$a + b = b + a$	the order does not matter
associative	$(a + b) + c = a + (b + c)$	the grouping does not matter
distributive	$a(b + c) = ab + ac$	multiplication spreads over addition

A LAW THAT HOLDS FOR ONE OPERATION MAY FAIL FOR ANOTHER

That is the whole point of the project: not to be told which laws hold, but to find out by testing.

Testing them

OPERATION	COMMUTATIVE?	ASSOCIATIVE?	TEST USED
addition	yes	yes	$3 + 5 = 5 + 3$
multiplication	yes	yes	$3 \cdot 5 = 5 \cdot 3$
subtraction	no	no	$3 - 5 \neq 5 - 3$
division	no	no	$6 \div 2 \neq 2 \div 6$

ONE EXAMPLE PROVES NOTHING; ONE COUNTER-EXAMPLE PROVES EVERYTHING

$2 + 2 = 2 \cdot 2$ does not make addition and multiplication the same. But $3 - 5 \neq 5 - 3$ settles for ever that subtraction is not commutative.

Counter-examples

CLAIM	COUNTER - EXAMPLE	VERDICT
subtraction is commutative	$3 - 5 = -2, 5 - 3 = 2$	false
division is associative	$(12 \div 6) \div 2 = 1, 12 \div (6 \div 2) = 4$	false
subtraction is associative	$(9 - 4) - 2 = 3, 9 - (4 - 2) = 7$	false
multiplication distributes over addition	no counter-example exists	true
division distributes from the right	$(6 + 4) \div 2 = 5 = 3 + 2$	true
division distributes from the left	$12 \div (6 + 2) = 1.5, \text{ but } 2 + 6 = 8$	false

A COUNTER-EXAMPLE MUST BE A SINGLE, COMPLETE CASE

“Sometimes it does not work” is not a counter-example. Two numbers, both calculations and the two different answers — that is what settles a claim.

Presenting the findings

PART OF THE REPORT	WHAT IT CONTAINS
the question	which laws hold for which operations
the method	the numbers you tested and why you chose them
the table	four operations against three laws, with ✓ or ✗
the evidence	one counter-example for every ✗
the conclusion	two sentences on what the pattern is

THE PATTERN WORTH NOTICING

Addition and multiplication obey the laws; subtraction and division do not. The two that fail are exactly the two that undo the other two — and that is not a coincidence.

02 Worked examples

EXAMPLE 1 Is subtraction commutative? Justify your answer.

- 1 Test 3 and 5.
- 2 $3 - 5 = -2$
- 3 $5 - 3 = 2$
- 4 The two differ, so no — and one counter-example is enough.

ANSWER

No

EXAMPLE 2 Is division associative?

① Test 12, 6, 2.

② $(12 \div 6) \div 2 = 1$

③ $12 \div (6 \div 2) = 4$

④ Different, so no.

ANSWER

No

EXAMPLE 3 Show that $a(b + c) = ab + ac$ with $a = 4$, $b = 5$, $c = 3$.

① Left: $4(5 + 3) = 4 \cdot 8 = 32$.

② Right: $4 \cdot 5 + 4 \cdot 3 = 20 + 12 = 32$.

③ They agree — as the distributive law says they must.
One case, not a proof.

ANSWER

Both give 32

03 Interactive model

Ask each group to try to break one law and to bring the counter-example to the board; the groups that fail have found the true laws, which is the more interesting result.

Which laws hold?

Addition is commutative:

Subtraction is commutative:

Multiplication is commutative:

Division is commutative:

Addition is associative:

Division is associative:

true

false

$a(b + c) = ab + ac$:

true

false

One example is enough to prove a law:

true

false

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Commutative	O'rin almashtirish xossasi	Переместительный
Associative	Guruhlash xossasi	Сочетательный
Distributive	Taqsimot xossasi	Распределительный
Law	Qonun	Закон
Counter-example	Qarshi misol	Контрпример
To test	Sinab ko'rmoq	Проверить
Conclusion	Xulosa	Вывод
Project	Loyiha	Проект

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Commutative means:

the order does not matter

the grouping does not matter

brackets expand

nothing changes

Associative means:

the order does not matter

the grouping does not matter

signs cancel

the answer is zero

Subtraction is:

commutative

not commutative

associative

distributive

A claim is disproved by:

one example

one counter-example

many examples

a table

The two operations obeying all the laws are:

addition and subtraction

addition and multiplication

multiplication and division

subtraction and division

$(6 + 4) \div 2$ equals:

5

8

3

1.5

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is addition commutative?

ANSWER Yes

2. Is subtraction commutative?

ANSWER No

3. Is multiplication commutative?

ANSWER Yes

4. Is division commutative?

ANSWER No

5. Is addition associative?

ANSWER Yes

6. Is division associative?

ANSWER No

7. State the distributive law

ANSWER $a(b + c) = ab + ac$

Medium · the standard question

7 PROBLEMS

1. A counter-example to “subtraction is commutative”

ANSWER $3 - 5 \neq 5 - 3$

2. A counter-example to “division is associative”

ANSWER $(12 \div 6) \div 2 \neq 12 \div (6 \div 2)$

3. A counter-example to “subtraction is associative”

ANSWER $(9 - 4) - 2 \neq 9 - (4 - 2)$

4. Check $a(b + c) = ab + ac$ at 4, 5, 3

ANSWER Both 32

5. $(6 + 4) \div 2$ and $6 \div 2 + 4 \div 2$

ANSWER Both 5

6. $12 \div (6 + 2)$ and $12 \div 6 + 12 \div 2$

ANSWER 1.5 and 8 — different

7. How many counter-examples disprove a claim?

ANSWER One

Hard · stretch and reason

7 PROBLEMS

1. Does division distribute over addition from the right?

ANSWER Yes — $(a + b) \div c = a \div c + b \div c$

2. From the left?

ANSWER No — $c \div (a + b) \neq c \div a + c \div b$

3. Is $2 + 2 = 2 \cdot 2$ evidence that the operations agree?

ANSWER No — one case proves nothing

4. Find another pair with $a + b = ab$

ANSWER 0 and 0

5. Why do subtraction and division fail the laws?

ANSWER They are the inverses of the operations that obey them

6. Is $a - (b - c) = (a - b) + c$ true?

ANSWER Yes — check with numbers

7. Which of the three laws holds for all four operations?

ANSWER None of them — subtraction and division fail every one

07 Homework

Homework — the project

One page: the table, one counter-example for every cross, and two sentences of conclusion.

1. Draw the table of four operations against the commutative and associative laws.
2. Fill it in with ✓ and ✗, testing each cell with numbers of your own.
3. Write one counter-example beside every ✗.
4. Test the distributive law with three numbers of your choice.

5. Write two sentences saying which operations obey the laws, and what the two exceptions have in common.

Revision — fractions and their division

The four operations on fractions gathered before the control work.

LESSON 34–35

2 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 7 CONSOLIDATION

LESSON CLOCK

20'

20'

25'

12'

3

The four operations side by side	20 min
Mixed numbers	20 min
Word problems	25 min
The errors that remain	12 min
Homework	3 min

LEARNING OBJECTIVES

- Choose the right method for each of the four operations.
- Work fluently with mixed numbers.
- Solve a word problem in three lines.
- Identify and repair the errors that remain.

TEXTBOOK

National	pp. 94–98
Cambridge	Stage 7, pp. 66–80
Workbook	Revision 7

01

Explanation

The four operations side by side

OPERATION	WHAT IT NEEDS	EXAMPLE
addition	a common denominator	$\frac{2}{3} + \frac{1}{4} = \frac{11}{12}$
subtraction	a common denominator	$\frac{5}{6} - \frac{1}{3} = \frac{1}{2}$
multiplication	nothing — multiply across	$\frac{2}{3} \cdot \frac{3}{5} = \frac{2}{5}$
division	the reciprocal of the divisor	$\frac{2}{3} \div \frac{4}{5} = \frac{5}{6}$

ONLY + AND − NEED A COMMON DENOMINATOR

Finding one before multiplying or dividing is not wrong, only wasted work — and it is the single commonest habit that slows this topic down.

Mixed numbers

OPERATION	BEST METHOD	EXAMPLE
adding	wholes and fractions separately, or convert	$1\frac{1}{2} + 2\frac{1}{3} = 3\frac{5}{6}$
subtracting	convert if borrowing is needed	$3\frac{1}{4} - 1\frac{1}{2} = 1\frac{3}{4}$
multiplying	always convert first	$1\frac{1}{2} \cdot 2 = 3$
dividing	always convert first	$2\frac{1}{4} \div 1\frac{1}{2} = 1\frac{1}{2}$

NEVER MULTIPLY THE WHOLE PARTS AND THE FRACTIONS SEPARATELY

$1\frac{1}{2} \cdot 2\frac{1}{2}$ is not $2\frac{1}{4}$. Convert to $\frac{3}{2} \cdot \frac{5}{2}$, which is $\frac{15}{4} = 3\frac{3}{4}$.

Word problems

PROBLEM	OPERATION	ANSWER
$\frac{3}{4}$ of 24 pupils are present	multiply	18
9 m of rope in $\frac{3}{4}$ m pieces	divide	12
$\frac{2}{3}$ of a tank is 20 l	divide	30 l
$\frac{1}{2}$ l shared between 4	divide	$\frac{1}{8}$ l
$\frac{1}{4}$ added to $\frac{1}{3}$ of a day	add	$\frac{7}{12}$

THREE LINES EVERY TIME

Name the operation, do the arithmetic, answer in words with units. The middle line is the easiest of the three and carries the fewest marks.

The errors that remain

ERROR	LOOKS LIKE	CORRECT
adding denominators	$\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$	$\frac{5}{6}$
common denominator before ×	extra work	multiply across
turning over the wrong fraction	$\frac{2}{3} \div \frac{4}{5} = \frac{3}{2} \cdot \frac{4}{5}$	$\frac{2}{3} \cdot \frac{5}{4}$
mixed number multiplied in parts	$1\frac{1}{2} \cdot 2\frac{1}{2} = 2\frac{1}{4}$	$3\frac{3}{4}$
answer not simplified	$\frac{12}{18}$	$\frac{2}{3}$

02 Worked examples

EXAMPLE 1 Compute $1\frac{1}{2} \cdot 2\frac{1}{2}$.

1

Convert both: $\frac{3}{2}$ and $\frac{5}{2}$.

2

$\frac{3}{2} \cdot \frac{5}{2} = \frac{15}{4}$

3

$= 3\frac{3}{4}$

ANSWER

$3\frac{3}{4}$

EXAMPLE 2 Compute $3\frac{1}{4} - 1\frac{1}{2}$.

① Convert: $\frac{13}{4} - \frac{3}{2}$.

② Common denominator 4: $\frac{13}{4} - \frac{6}{4}$.

③ $= \frac{7}{4} = 1\frac{3}{4}$

ANSWER

$$1\frac{3}{4}$$

EXAMPLE 3 $\frac{2}{3}$ of a tank is 20 litres. Find its capacity.

① A part is given, the whole is wanted: divide.

② $20 \div \frac{2}{3} = 20 \cdot \frac{3}{2}$

③ $= 30$ litres.

ANSWER

30 litres

03 Interactive model

Put one addition, one multiplication and one division on the board unlabelled and ask which needs a common denominator; the answer separates fluency from habit.

Which method does this one need?

$\frac{2}{3} + \frac{1}{4}$ needs:

a common denominator

the reciprocal

nothing

converting

$\frac{2}{3} \cdot \frac{3}{5}$ needs:

a common denominator

the reciprocal

multiplying across

converting

$\frac{2}{3} \div \frac{4}{5}$ needs:

a common denominator

the reciprocal of the divisor

nothing

squaring

$\frac{1}{2} + \frac{1}{3}$ equals:

$$\frac{2}{5}$$

$$\frac{5}{6}$$

$$\frac{1}{6}$$

$$\frac{2}{6}$$

$1\frac{1}{2} \cdot 2\frac{1}{2}$ equals:

$$2\frac{1}{4}$$

$$3\frac{3}{4}$$

$$3$$

$$2\frac{1}{2}$$

$3\frac{1}{4} - 1\frac{1}{2}$ equals:

$$1\frac{3}{4}$$

$$2\frac{1}{4}$$

$$1\frac{1}{4}$$

$$2\frac{3}{4}$$

2 3 of a tank is 20 l; the tank holds:

$$13\frac{1}{3} \text{ l}$$

30 l

40 l

60 l

12 18 as an answer is:

finished

not simplified

wrong

improper

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Revision	Takrorlash	Повторение
Common denominator	Umumiy maxraj	Общий знаменатель
Reciprocal	Teskari son	Обратное число
Mixed number	Aralash son	Смешанное число
To cancel	Qisqartirish	Сокращать
Lowest terms	Qisqarmas ko'rinish	Несократимый вид
Word problem	Matnli masala	Текстовая задача
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Which operations need a common denominator?

all four

+ and –

× and ÷

none

To divide by a fraction:

find a common denominator

multiply by the reciprocal

multiply across

subtract

Before multiplying mixed numbers you should:

multiply the wholes

convert to improper fractions

add them

round them

$\frac{1}{2} + \frac{1}{3}$ equals:

$$\frac{2}{5}$$

$$\frac{5}{6}$$

$$\frac{1}{5}$$

$$\frac{2}{6}$$

9 m of rope in $\frac{3}{4}$ m pieces gives:

6.75

12

9

3

An answer of $\frac{12}{18}$ should be written:

$$\frac{12}{18}$$

$$\frac{2}{3}$$

$$\frac{6}{9}$$

0.67

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up**7 PROBLEMS**

1. $\frac{2}{3} + \frac{1}{4}$

ANSWER $\frac{11}{12}$

2. $\frac{5}{6} - \frac{1}{3}$

ANSWER $\frac{1}{2}$

3. $\frac{2}{3} \cdot \frac{3}{5}$

ANSWER $\frac{2}{5}$

4. $\frac{2}{3} \div \frac{4}{5}$

ANSWER $\frac{5}{6}$

5. $\frac{1}{2} + \frac{1}{3}$

ANSWER $\frac{5}{6}$

6. $\frac{3}{4}$ of 24

ANSWER 18

7. 9 m in $\frac{3}{4}$ m pieces

ANSWER 12

Medium · the standard question

7 PROBLEMS

1. $1\frac{1}{2} + 2\frac{1}{3}$

ANSWER $3\frac{5}{6}$

2. $3\frac{1}{4} - 1\frac{1}{2}$

ANSWER $1\frac{3}{4}$

3. $1\frac{1}{2} \cdot 2\frac{1}{2}$

ANSWER $3\frac{3}{4}$

4. $2\frac{1}{4} \div 1\frac{1}{2}$

ANSWER $1\frac{1}{2}$

5. $\frac{2}{3}$ of a tank is 20 l: the tank

ANSWER 30 l

6. $\frac{1}{2}$ l shared between 4

ANSWER $\frac{1}{8}$ l

7. $\frac{1}{4} + \frac{1}{3}$

ANSWER $\frac{7}{12}$

Hard · stretch and reason

7 PROBLEMS

1. $2\frac{1}{3} \cdot 1\frac{2}{7}$

ANSWER 3

2. $4\frac{1}{2} \div \frac{3}{4}$

ANSWER 6

3. $\frac{3}{4} + \frac{2}{3} - \frac{1}{2}$

ANSWER $\frac{11}{12}$

4. $\frac{5}{8} \div \frac{5}{6} \cdot \frac{2}{3}$

ANSWER $\frac{1}{2}$

5. A recipe needs $\frac{3}{4}$ kg for 6 people: for 8

ANSWER 1 kg

6. Which is larger, $\frac{2}{3} \div \frac{3}{4}$ or $\frac{2}{3} \cdot \frac{3}{4}$?

ANSWER The division

7. Why is a common denominator not needed for $\times$?

ANSWER The parts are recut by the multiplication itself

07 Homework

Homework — 5 tasks

Name the operation before starting, and simplify every answer.

1. Compute $\frac{3}{5} + \frac{1}{4}$ and $\frac{7}{8} - \frac{1}{2}$.

2. Compute $\frac{4}{9} \cdot \frac{3}{8}$ and $\frac{4}{9} \div \frac{8}{3}$.
3. Compute $2\frac{1}{2} \cdot 1\frac{1}{5}$.
4. Compute $3\frac{3}{4} \div 1\frac{1}{4}$.
5. $\frac{5}{8}$ of a barrel is 25 litres. Find its capacity.

Control work 2 — division of fractions, and work on the mistakes

The four operations on fractions in one paper, then the repair.

LESSON 36–37

2 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 2

S7 7 REVIEW

LESSON CLOCK

3

35'

12'

25'

5'

Instructions	3 min
The paper	35 min
Answers	12 min
Diagnosis and rewrite	25 min
The map	5 min

LEARNING OBJECTIVES

- Work accurately with all four operations on fractions.
- Handle mixed numbers correctly.
- Solve a word problem, naming the operation.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 54–98
Cambridge	Stage 7, pp. 66–80
Workbook	Control paper 2

01

Explanation

The paper — 25 marks, 35 minutes

Q	TASK	MARKS	FROM
1	$\frac{3}{4} + \frac{2}{5}$ and $\frac{7}{8} - \frac{1}{4}$	4	L21
2	$\frac{4}{9} \cdot \frac{3}{8}$	3	L21
3	$\frac{5}{6} \div 5$ and $8 \div \frac{2}{3}$	4	L22–26
4	$\frac{2}{3} \div \frac{8}{9}$	4	L27–29
5	$2\frac{1}{4} \div 1\frac{1}{2}$	4	L27–29
6	$\frac{3}{5}$ of a tank is 24 litres: find its capacity	3	L30–32
7	How many $\frac{3}{4}$ m planks come from 9 m?	3	L30–32

THE ANSWERS

$1\frac{3}{20}$ and $\frac{5}{8}$; $\frac{1}{6}$; $\frac{1}{6}$ and 12; $\frac{3}{4}$; $1\frac{1}{2}$; 40 litres; 12 planks.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
denominators added	$\frac{3}{4} + \frac{2}{5} = \frac{5}{9}$	use the LCD 20
cancelled after multiplying	$\frac{12}{72}$	cancel first: $\frac{1}{6}$
numerator multiplied when dividing by a whole number	$\frac{25}{6}$	$\frac{1}{6}$
wrong fraction inverted	$\frac{3}{2} \cdot \frac{8}{9}$	$\frac{2}{3} \cdot \frac{9}{8}$
mixed number not converted	$2\frac{1}{4} \div 1\frac{1}{2} = 2\frac{1}{6}$	$\frac{9}{4} \div \frac{3}{2}$
multiplied instead of divided	$24 \cdot \frac{3}{5}$	$24 \div \frac{3}{5}$
answer not simplified	$\frac{9}{12}$	$\frac{3}{4}$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

CASE	METHOD	ANSWER COMPARED WITH THE FIRST NUMBER
fraction $\div$ whole number	multiply the denominator	smaller
whole number $\div$ fraction	multiply by the reciprocal	larger
fraction $\div$ fraction under 1	multiply by the reciprocal	larger
fraction $\div$ fraction over 1	multiply by the reciprocal	smaller
mixed numbers	convert first, then as above	depends on the divisor

LOOKING FORWARD

Next come three short Cambridge lessons — common multiples, divisibility and roots — and then ratio, where dividing fractions returns as “how many times bigger is this than that”.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $\frac{5}{6} \div 5$ and $8 \div \frac{2}{3}$.

$$\textcircled{1} \quad \frac{5}{6} \cdot \frac{1}{5}$$

Cancel the 5s.

$$\textcircled{2} \quad = \frac{1}{6}$$

Smaller ✓

$$\textcircled{3} \quad 8 \cdot \frac{3}{2}$$

Turn over the divisor.

$$\textcircled{4} \quad = 12$$

Larger ✓

ANSWER

$\frac{1}{6}$ and 12

EXAMPLE 2 Model answer, Q5: $2\frac{1}{4} \div 1\frac{1}{2}$.

① Convert: $\frac{9}{4} \div \frac{3}{2}$.

Never divide mixed numbers directly.

② $\frac{9}{4} \cdot \frac{2}{3}$

③ Cancel and multiply.

④ $= \frac{3}{2} = 1\frac{1}{2}$

ANSWER

$$1\frac{1}{2}$$

EXAMPLE 3 Model answer, Q6: $\frac{3}{5}$ of a tank is 24 litres.

① A part is given and the whole is wanted: divide.

② $24 \div \frac{3}{5} = 24 \cdot \frac{5}{3}$

③ $= 40$ litres.

Larger than 24 ✓

ANSWER

40 litres

03 Interactive model

Return Q6 and Q7 first and ask which operation each needed; almost every lost mark there is a choice of operation, not arithmetic.

The chapter in eight questions

3 4 + 2 5 equals:

$$\frac{5}{9}$$

$$1\frac{3}{20}$$

$$\frac{6}{20}$$

$$1\frac{1}{20}$$

4 9 · 3 8 equals:

$$\frac{1}{6}$$

$$\frac{12}{72} \text{ unsimplified}$$

$$\frac{7}{17}$$

$$\frac{32}{27}$$

5 6 ÷ 5 equals:

$$\frac{1}{6}$$

$$\frac{25}{6}$$

$$\frac{5}{30} \text{ unsimplified}$$

$$\frac{6}{5}$$

8 ÷ 2 3 equals:

$$\frac{16}{3}$$

$$12$$

$$\frac{4}{3}$$

$$5\frac{1}{3}$$

2 3 ÷ 8 9 equals:

$$\frac{16}{27}$$

$$\frac{3}{4}$$

$$\frac{4}{3}$$

$$\frac{27}{16}$$

$2\frac{1}{4} \div 1\frac{1}{2}$ equals:

$$1\frac{1}{2}$$

$$2\frac{1}{6}$$

$$3\frac{3}{8}$$

$$\frac{2}{3}$$

$\frac{3}{5}$ of a tank is 24 l; the tank holds:

$$14.4 \text{ l}$$

$$40 \text{ l}$$

$$30 \text{ l}$$

$$72 \text{ l}$$

9 m in $\frac{3}{4}$ m planks gives:

$$6.75$$

$$12$$

$$9$$

$$3$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Division of fractions	Kasrlarni bo'lish	Деление дробей
Reciprocal	Teskari son	Обратное число
Mixed number	Aralash son	Смешанное число
Common denominator	Umumiy maxraj	Общий знаменатель
Lowest terms	Qisqarmas ko'rinish	Несократимый вид
Word problem	Matnli masala	Текстовая задача
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 needs:

Q2 is quickest if you:

Q3 first part gives an answer:

Q3 second part gives an answer:

larger

smaller

equal

negative

Q5 must begin with:

turning over

converting the mixed numbers

a common denominator

cancelling

Q6 needs:

multiplication

division

addition

subtraction

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{3}{4} + \frac{2}{5}$

ANSWER $1\frac{3}{20}$

2. $\frac{7}{8} - \frac{1}{4}$

ANSWER $\frac{5}{8}$

3. $\frac{4}{9} \cdot \frac{3}{8}$

ANSWER $\frac{1}{6}$

4. $\frac{5}{6} \div 5$

ANSWER $\frac{1}{6}$

5. $8 \div \frac{2}{3}$

ANSWER 12

6. $\frac{2}{3} \div \frac{8}{9}$

ANSWER $\frac{3}{4}$

7. $2\frac{1}{4} \div 1\frac{1}{2}$

ANSWER $1\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. $\frac{3}{5}$ of a tank is 24 l

ANSWER 40 l

2. 9 m in $\frac{3}{4}$ m planks

ANSWER 12

3. $1\frac{1}{2} + 2\frac{3}{4}$

ANSWER $4\frac{1}{4}$

4. $5 \div \frac{5}{8}$

ANSWER 8

5. $\frac{7}{10} \div \frac{7}{5}$

ANSWER $\frac{1}{2}$

6. $\frac{3}{8} \cdot 4$

ANSWER $1\frac{1}{2}$

7. $\frac{9}{10} \div 3$

ANSWER $\frac{3}{10}$

Hard · stretch and reason

7 PROBLEMS

1. $3\frac{1}{3} \div 2\frac{2}{9}$

ANSWER $1\frac{1}{2}$

2. $\frac{5}{6} + \frac{3}{4} \div \frac{3}{2}$

ANSWER $1\frac{1}{3}$

3. A tank $\frac{7}{8}$ full holds 35 l: its capacity

ANSWER 40 l

4. How many $\frac{2}{5}$ kg bags from 6 kg?

ANSWER 15

5. $1 \div \frac{3}{7}$

ANSWER $2\frac{1}{3}$

6. After using $\frac{3}{8}$, 30 l is left: the original

ANSWER 48 l

7. Why is the answer to Q3 second part larger than 8?

ANSWER The divisor is smaller than one

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.

2. Compute $\frac{5}{6} + \frac{3}{8}$ and $\frac{5}{6} - \frac{1}{3}$.
3. Compute $\frac{3}{4} \div 6$ and $10 \div \frac{5}{7}$.
4. Compute $3\frac{1}{2} \div 1\frac{3}{4}$.
5. $\frac{4}{9}$ of a road is 12 km. Find the whole road.

Lowest common multiples and highest common factors

A Cambridge insert: prime factors, and the two numbers they build.

LESSON 38

1 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 1.3–1.4

LESSON CLOCK

10'

12'

12'

6'

Prime factors	10 min
The HCF	12 min
The LCM	12 min
Where each is used	6 min

LEARNING OBJECTIVES

- Write a number as a product of prime factors.
- Find the HCF and the LCM of two numbers.
- Use the HCF to simplify a fraction and the LCM to add fractions.
- Recognise where each is needed in a word problem.

TEXTBOOK

National	pp. 99–102
Cambridge	Stage 7, pp. 18–24
Workbook	Exercise 1.3

01

Explanation

Prime factors

Every whole number above 1 is a product of primes, in exactly one way.

NUMBER	PRIME FACTORS	IN INDEX FORM
12	$2 \cdot 2 \cdot 3$	$2^2 \cdot 3$
18	$2 \cdot 3 \cdot 3$	$2 \cdot 3^2$
24	$2 \cdot 2 \cdot 2 \cdot 3$	$2^3 \cdot 3$
60	$2 \cdot 2 \cdot 3 \cdot 5$	$2^2 \cdot 3 \cdot 5$

DIVIDE BY THE SMALLEST PRIME THAT FITS, REPEATEDLY

$60 \div 2 = 30$, $30 \div 2 = 15$, $15 \div 3 = 5$, $5 \div 5 = 1$. The divisors used, in order, are the prime factorisation.

The HCF

The **highest common factor** is the product of the primes both numbers share, each to the lower of its two powers.

NUMBERS	FACTORISED	SHARED	HCF
12, 18	$2^2 \cdot 3, 2 \cdot 3^2$	$2 \cdot 3$	6
24, 60	$2^3 \cdot 3, 2^2 \cdot 3 \cdot 5$	$2^2 \cdot 3$	12
8, 15	$2^3, 3 \cdot 5$	nothing	1

AN HCF OF 1 MEANS THE NUMBERS ARE COPRIME

8 and 15 share no prime factor, so the fraction $\frac{8}{15}$ is already in its lowest terms.

The LCM

The **lowest common multiple** takes every prime that appears in either number, each to the higher of its powers.

NUMBERS	FACTORISED	TAKEN	LCM
12, 18	$2^2 \cdot 3, 2 \cdot 3^2$	$2^2 \cdot 3^2$	36
24, 60	$2^3 \cdot 3, 2^2 \cdot 3 \cdot 5$	$2^3 \cdot 3 \cdot 5$	120
8, 15	$2^3, 3 \cdot 5$	$2^3 \cdot 3 \cdot 5$	120

HCF × LCM = the product of the two numbers

A CHECK THAT COSTS NOTHING

$6 \cdot 36 = 216 = 12 \cdot 18 \checkmark$. If the HCF and LCM you found do not multiply to the product of the two numbers, one of them is wrong.

Where each is used

PROBLEM	WHICH	WHY
simplifying $\frac{12}{18}$	HCF	divide both by 6
adding $\frac{1}{12} + \frac{1}{18}$	LCM	the denominator 36
the largest identical parcels from 24 pens and 60 pencils	HCF	12
two buses leaving every 12 and 18 minutes together again	LCM	36

“CUTTING UP” NEEDS THE HCF; “MEETING AGAIN” NEEDS THE LCM

The HCF is never bigger than either number; the LCM is never smaller. Comparing your answer with the two numbers tells you at once whether you found the right one.

02

Worked examples

EXAMPLE 1 Find the HCF and LCM of 12 and 18.

1

$12 = 2^2 \cdot 3$ and $18 = 2 \cdot 3^2$.

2

HCF: shared primes at the lower power — $2 \cdot 3 = 6$.

3

LCM: every prime at the higher power — $2^2 \cdot 3^2 = 36$.

4

Check: $6 \cdot 36 = 216 = 12 \cdot 18$ ✓

ANSWER

HCF 6, LCM 36

EXAMPLE 2 24 pens and 60 pencils are made into identical parcels with nothing left over. What is the largest number of parcels?

- 1 “Largest identical parcels” means the HCF.
- 2 $24 = 2^3 \cdot 3$, $60 = 2^2 \cdot 3 \cdot 5$.
- 3 $\text{HCF} = 2^2 \cdot 3 = 12$.
- 4 Twelve parcels, each with 2 pens and 5 pencils.

ANSWER

12 parcels

EXAMPLE 3 Two buses leave at 8:00, one every 12 minutes and one every 18. When do they next leave together?

- 1 “Together again” means the LCM.
- 2 LCM of 12 and 18 is 36.
- 3 36 minutes later, at 8:36.

ANSWER

8:36

03 Interactive model

Ask for the HCF and LCM of 12 and 18 by listing factors and multiples first; the prime-factor method then arrives as a shortcut for something already understood.

Building the LCM from prime factors

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
12	$2^2 \cdot 3$	3
18	$2 \cdot 3^2$	2

LOWEST COMMON DENOMINATOR

$$2^2 \cdot 3^2 = 36$$

HCF = $2 \cdot 3 = 6$, and $6 \cdot 36 = 12 \cdot 18$ ✓

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Factor	Bo'luvchi	Делитель
Multiple	Karrali son	Кратное
Prime number	Tub son	Простое число
Prime factorisation	Tub ko'paytuvchilarga ajratish	Разложение на простые множители
Highest common factor	EKUB	НОД
Lowest common multiple	EKUK	НОК
Common	Umumiy	Общий
Product	Ko'paytma	Произведение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

12 in prime factors is:

$2 \cdot 6$

$2^2 \cdot 3$

$3 \cdot 4$

$2 \cdot 3^2$

The HCF of 12 and 18 is:

2

3

6

36

The LCM of 12 and 18 is:

6

36

54

216

The HCF of 8 and 15 is:

1

2

3

120

HCF $\times$ LCM equals:

the sum

the product of the numbers

the difference

nothing special

Two buses meeting again needs the:

HCF

LCM

product

difference

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 12 in prime factors

ANSWER $2^2 \cdot 3$

2. 18 in prime factors

ANSWER $2 \cdot 3^2$

3. 24 in prime factors

ANSWER $2^3 \cdot 3$

4. HCF of 12 and 18

ANSWER 6

5. LCM of 12 and 18

ANSWER 36

6. HCF of 8 and 15

ANSWER 1

7. LCM of 8 and 15

ANSWER 120

Medium · the standard question

7 PROBLEMS

1. 60 in prime factors

ANSWER $2^2 \cdot 3 \cdot 5$

2. HCF of 24 and 60

ANSWER 12

3. LCM of 24 and 60

ANSWER 120

4. Simplify $\frac{12}{18}$ using the HCF

ANSWER $\frac{2}{3}$

5. Add $\frac{1}{12} + \frac{1}{18}$ using the LCM

ANSWER $\frac{5}{36}$

6. 24 pens and 60 pencils in identical parcels

ANSWER 12 parcels

7. Buses every 12 and 18 minutes from 8:00

ANSWER 8:36

Hard · stretch and reason

7 PROBLEMS

1. HCF and LCM of 36 and 48

ANSWER 12 and 144

2. Check that $\text{HCF} \times \text{LCM} = 36 \cdot 48$

ANSWER $12 \cdot 144 = 1728$ ✓

3. HCF of 105 and 154

ANSWER 7

4. LCM of 4, 6 and 10

ANSWER 60

5. The largest square tile that tiles a 60 by 84 cm floor exactly

ANSWER 12 cm

6. Two lights flash every 15 and 25 seconds: together again after

ANSWER 75 s

7. If the HCF of two numbers is one of them, what does that mean?

ANSWER The smaller divides the larger

07 Homework

Homework — 5 tasks

Check every pair with $\text{HCF} \times \text{LCM} = \text{the product of the numbers}$.

- Write 45 and 75 as products of prime factors.
- Find their HCF and LCM, and check with the product rule.
- Simplify $\frac{45}{75}$ using the HCF.
- Find the LCM of 6, 8 and 12.

5. Two runners lap a track in 40 and 60 seconds. After how long are they together at the start again?

Tests for divisibility

A Cambridge insert: deciding what divides a number without doing the division.

LESSON 39

1 × 40 MIN

MATEMATIKA 6, QO'SHIMCHA MAVZU

S7 1.5

LESSON CLOCK

12'

10'

12'

6'

The tests	12 min
Using them together	10 min
Factorising quickly	12 min
Why the test for 3 works	6 min

LEARNING OBJECTIVES

- State and apply the tests for 2, 3, 4, 5, 6, 8, 9 and 10.
- Combine tests to check divisibility by a composite number.
- Use the tests to factorise quickly.
- Explain why the test for 3 works.

TEXTBOOK

National	pp. 103–105
Cambridge	Stage 7, pp. 24–27
Workbook	Exercise 1.5

01

Explanation

The tests

DIVISOR	TEST	EXAMPLE
2	last digit even	374
3	digit sum divisible by 3	471 → 12
4	last two digits divisible by 4	1316 → 16
5	last digit 0 or 5	385
6	divisible by 2 and by 3	474
8	last three digits divisible by 8	3128 → 128
9	digit sum divisible by 9	576 → 18
10	last digit 0	4370

THE DIGIT-SUM TESTS CAN BE REPEATED

For 99 999 the digit sum is 45, whose digit sum is 9 — divisible by 9, so the original is too.

Using them together

For a composite divisor, test its coprime factors separately.

DIVISOR	TEST	WARNING
6	2 and 3	—
12	3 and 4	not 2 and 6
15	3 and 5	—
18	2 and 9	not 3 and 6

THE TWO FACTORS MUST SHARE NOTHING

$12 = 2 \cdot 6$, but passing the 2 and 6 tests is not enough: 18 passes both and is not a multiple of 12. Use 3 and 4, which are coprime.

Factorising quickly

NUMBER	TESTS IT PASSES	FACTORISATION
180	2, 3, 4, 5, 9, 10	$2^2 \cdot 3^2 \cdot 5$
252	2, 3, 4, 9	$2^2 \cdot 3^2 \cdot 7$
385	5	$5 \cdot 7 \cdot 11$
1001	none of the small ones	$7 \cdot 11 \cdot 13$

THE TESTS SAVE THE TRIAL DIVISIONS

Knowing at a glance that 180 is divisible by 4 and by 9 gives $180 = 36 \cdot 5$ immediately, without any long division at all.

Why the test for 3 works

Every power of ten is one more than a multiple of 3: $10 = 9 + 1$, $100 = 99 + 1$, $1000 = 999 + 1$.

STEP	FOR 471
split by place value	$4 \cdot 100 + 7 \cdot 10 + 1$
rewrite the powers	$4(99 + 1) + 7(9 + 1) + 1$
separate	$(4 \cdot 99 + 7 \cdot 9) + (4 + 7 + 1)$
the first bracket	always a multiple of 3
so the test	depends only on the digit sum

THE SAME ARGUMENT WORKS FOR 9

99 and 999 are multiples of 9 as well, which is why the digit-sum test serves both numbers — and why no such simple test exists for 7.

02

Worked examples

EXAMPLE 1 Is 471 divisible by 3? By 9?

1

Digit sum: $4 + 7 + 1 = 12$.

2

12 is divisible by 3 — so 471 is.

3

12 is not divisible by 9 — so 471 is not.

ANSWER

By 3 yes, by 9 no

EXAMPLE 2 Is 1316 divisible by 4?

① Look at the last two digits: 16.

② $16 \div 4 = 4$.

③ So yes.

ANSWER

Yes

EXAMPLE 3 Factorise 180 using the tests.

① Last two digits 80 — divisible by 4.

$$180 = 4 \cdot 45$$

② Digit sum 9 — divisible by 9.

$$45 = 9 \cdot 5$$

③ $180 = 4 \cdot 9 \cdot 5$

④ $= 2^2 \cdot 3^2 \cdot 5$

ANSWER

$$2^2 \cdot 3^2 \cdot 5$$

03 Interactive model

Call out five-digit numbers and have the class shout which of 2, 3, 5 and 9 divide them; the tests become automatic in a single lesson.

Which numbers divide it?

Is 374 divisible by 2?

Is 471 divisible by 3?

Is 471 divisible by 9?

Is 1316 divisible by 4?

Is 385 divisible by 5?

Is 474 divisible by 6?

yes

no

Is 576 divisible by 9?

yes

no

To test for 12 use:

2 and 6

3 and 4

12 itself only

6 and 6

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Divisible	Bo'linadi	Делится
Test for divisibility	Bo'linish alomati	Признак делимости
Digit sum	Raqamlar yig'indisi	Сумма цифр
Last digit	Oxirgi raqam	Последняя цифра
Even number	Juft son	Чётное число
Odd number	Toq son	Нечётное число
Composite number	Murakkab son	Составное число
Remainder	Qoldiq	Остаток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A number is divisible by 3 when:

it is even

its digit sum is

it ends in 3

it is odd

By 4 when:

it is even

its last two digits are

its digit sum is

it ends in 4

By 6 when it is divisible by:

2 and 3

3 only

2 only

6 alone

By 9 when:

it ends in 9

its digit sum is

it is odd

it is even

Testing for 12 by 2 and 6 is:

correct

not enough

too strong

impossible

Which has no simple digit test?

3

7

9

5

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is 374 divisible by 2?

ANSWER Yes

2. Is 471 divisible by 3?

ANSWER Yes

3. Is 471 divisible by 9?

ANSWER No

4. Is 1316 divisible by 4?

ANSWER Yes

5. Is 385 divisible by 5?

ANSWER Yes

6. Is 4370 divisible by 10?

ANSWER Yes

7. Is 576 divisible by 9?

ANSWER Yes

Medium · the standard question

7 PROBLEMS

1. Is 474 divisible by 6?

ANSWER Yes

2. Is 3128 divisible by 8?

ANSWER Yes

3. Which of 2, 3, 5, 9 divide 180?

ANSWER All four

4. Which divide 385?

ANSWER 5 only

5. Factorise 180

ANSWER $2^2 \cdot 3^2 \cdot 5$

6. Factorise 252

ANSWER $2^2 \cdot 3^2 \cdot 7$

7. Test for 15: use

ANSWER 3 and 5

Hard · stretch and reason

7 PROBLEMS

1. Is 99 999 divisible by 9?

ANSWER Yes — digit sum 45

2. Why is testing 12 with 2 and 6 wrong?

ANSWER 18 passes both but is not a multiple of 12

3. Factorise 1001

ANSWER $7 \cdot 11 \cdot 13$

4. Explain the test for 3

ANSWER Every power of ten is one more than a multiple of 3

5. The smallest digit d making $47d$ divisible by 9

ANSWER $d = 7$

6. Which digits d make $12d$ divisible by 4?

ANSWER 0, 4, 8

7. Is $2 \cdot 3 \cdot 5 \cdot 7 + 1$ divisible by any of 2, 3, 5, 7?

ANSWER No — each leaves remainder 1

07 Homework

Homework — 5 tasks

Write which test you used beside every answer.

1. Test 2 358 for divisibility by 2, 3, 4, 6 and 9.
2. Test 7 425 for divisibility by 3, 5, 9 and 15.
3. Factorise 360 using the tests.
4. Find the digit d making $61d$ divisible by 3, listing every possibility.
5. Explain in two sentences why the digit-sum test works for 9.

Square roots and cube roots

A Cambridge insert: undoing a square and a cube, and knowing the small ones by heart.

LESSON 40

1 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 1.6

LESSON CLOCK

10'

12'

12'

6'

Squares and cubes	10 min
The roots	12 min
Estimating	12 min
In area and volume	6 min

LEARNING OBJECTIVES

- Find squares and cubes of small whole numbers.
- Find square roots and cube roots exactly where they are whole.
- Estimate a square root between two whole numbers.
- Use roots in area and volume problems.

TEXTBOOK

National	pp. 106–108
Cambridge	Stage 7, pp. 27–31
Workbook	Exercise 1.6

01

Explanation

Squares and cubes

<i>N</i>	1	2	3	4	5	6	7	8	9	10
<i>n</i> ²	1	4	9	16	25	36	49	64	81	100
<i>n</i> ³	1	8	27	64	125	216	343	512	729	1000

LEARN THESE TWO ROWS

They are needed for the rest of school mathematics — in area, in volume, in Pythagoras and in every quadratic equation from Grade 8 onwards. Ten minutes now saves hours later.

The roots

$\sqrt{25} = 5$ because $5^2 = 25$; $\sqrt[3]{27} = 3$ because $3^3 = 27$

ROOT	VALUE	CHECK
$\sqrt{49}$	7	$7^2 = 49$
$\sqrt{81}$	9	$9^2 = 81$
$\sqrt{100}$	10	$10^2 = 100$
$\sqrt[3]{8}$	2	$2^3 = 8$
$\sqrt[3]{125}$	5	$5^3 = 125$
$\sqrt[3]{1000}$	10	$10^3 = 1000$

A ROOT UNDOES A POWER; IT DOES NOT HALVE IT

$\sqrt{36}$ is 6, not 18. The question a root asks is “what number, multiplied by itself, gives this?” — never “what is half of it?”

Estimating

Most numbers are not perfect squares. Trap the root between the two nearest ones.

ROOT	BETWEEN	NEARER TO
$\sqrt{50}$	7 and 8	7
$\sqrt{30}$	5 and 6	5
$\sqrt{90}$	9 and 10	9
$\sqrt[3]{30}$	3 and 4	3

$\sqrt{50}$ IS A LITTLE OVER 7

Because 49 is just below 50 and 64 is far above it. Comparing the gaps tells you which whole number the root is nearer, without any calculation.

In area and volume

PROBLEM	WORKING	ANSWER
a square of area 64 cm^2 : its side	$\sqrt{64}$	8 cm
a square of area 121 cm^2 : its perimeter	$4 \cdot \sqrt{121}$	44 cm
a cube of volume 125 cm^3 : its edge	$\sqrt[3]{125}$	5 cm
a cube of edge 4 cm : its volume	4^3	64 cm^3
a cube of volume 64 cm^3 : its surface	$6 \cdot 4^2$	96 cm^2

SQUARES BELONG TO AREA, CUBES TO VOLUME

Two directions give cm^2 and a square; three give cm^3 and a cube. The units say which root is needed before any number is written.

02 Worked examples

EXAMPLE 1 Find $\sqrt{81}$ and $\sqrt[3]{64}$.

① $9^2 = 81$, so $\sqrt{81} = 9$.

② $4^3 = 64$, so $\sqrt[3]{64} = 4$.

Note that $\sqrt{64} = 8$ — different roots, different answers.

ANSWER

9 and 4

EXAMPLE 2 Between which two whole numbers does $\sqrt{50}$ lie?

- ① $7^2 = 49$ and $8^2 = 64$.
- ② $49 < 50 < 64$
- ③ So $7 < \sqrt{50} < 8$, and much nearer 7.

ANSWER

Between 7 and 8

EXAMPLE 3 A cube has volume 125 cm^3 . Find its edge and its surface area.

- ① $\sqrt[3]{125} = 5$, so the edge is 5 cm .
- ② Each face is $5^2 = 25 \text{ cm}^2$.
- ③ Six faces: 150 cm^2 .

ANSWER 5 cm and 150 cm^2 **03 Interactive model**

Build a cube from 27 unit blocks and ask for its edge; the cube root is picked up in the hand rather than from a table.

A number, its square and its square root

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value³ **64**

a¹ **64**

Raising the answer to the power 3 gives back a¹ — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Square	Kvadrat	Квадрат
Cube	Kub	Куб
Square root	Kvadrat ildiz	Квадратный корень
Cube root	Kub ildiz	Кубический корень
Perfect square	To'la kvadrat	Полный квадрат
Perfect cube	To'la kub	Полный куб
To estimate	Baholash	Оценить
Inverse operation	Teskari amal	Обратное действие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** $\sqrt{49}$ equals: $\sqrt{100}$ equals: $\sqrt[3]{27}$ equals:

$\sqrt[3]{64}$ equals:

4

8

16

32

$\sqrt{50}$ lies between:

6 and 7

7 and 8

8 and 9

24 and 25

A square of area 64 cm^2 has side:

8 cm

16 cm

32 cm

4 cm

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 6^2

ANSWER 36

2. 9^2

ANSWER 81

3. 4^3

ANSWER 64

4. 5^3

ANSWER 125

5. $\sqrt{49}$

ANSWER 7

6. $\sqrt{100}$

ANSWER 10

7. $\sqrt[3]{27}$

ANSWER 3

Medium · the standard question

7 PROBLEMS

1. $\sqrt[3]{64}$

ANSWER 4

2. $\sqrt{64}$

ANSWER 8

3. $\sqrt[3]{1000}$

ANSWER 10

4. $\sqrt{50}$ lies between

ANSWER 7 and 8

5. $\sqrt[3]{30}$ lies between

ANSWER 3 and 4

6. A square of area 121 cm^2 : its perimeter

ANSWER 44 cm

7. A cube of volume 125 cm^3 : its edge

ANSWER 5 cm

Hard · stretch and reason

7 PROBLEMS

1. A cube of volume 64 cm^3 : its surface area

ANSWER 96 cm^2

2. A cube of volume 125 cm^3 : its surface area

ANSWER 150 cm^2

3. $\sqrt{144} + \sqrt[3]{8}$

ANSWER 14

4. $\sqrt{(36 + 64)}$

ANSWER 10

5. Is $\sqrt{36} + \sqrt{64}$ the same as $\sqrt{(36 + 64)}$?

ANSWER No — 14 against 10

6. A square of area 90 cm^2 : its side, to the nearest cm

ANSWER 9 cm (a little over 9.4)

7. Which is larger, $\sqrt{81}$ or $\sqrt[3]{512}$?

ANSWER $\sqrt{81} = 9$ beats $\sqrt[3]{512} = 8$

07 Homework

Homework — 5 tasks

Learn the squares to 12 and the cubes to 10 by heart.

- Write the squares of 1 to 12 .
- Write the cubes of 1 to 10 .
- Find $\sqrt{169}$, $\sqrt{225}$ and $\sqrt[3]{216}$.
- Between which whole numbers does $\sqrt{200}$ lie?
- A cube has volume 343 cm^3 . Find its edge and its surface area.

Recall — ratio

What a ratio is, how it differs from a fraction, and how it simplifies.

LESSON 41

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 12 ENTRY

LESSON CLOCK

10'

12'

12'

6'

What a ratio says	10 min
Simplifying	12 min
Ratio against fraction	12 min
Reading a ratio back	6 min

LEARNING OBJECTIVES

- Write a ratio from a description and simplify it.
- Convert units before comparing.
- Say what each number of a ratio counts.
- Distinguish a ratio from a fraction of the whole.

TEXTBOOK

National	pp. 109–112
Cambridge	Stage 7, pp. 118–122
Workbook	Exercise 12.1

01

Explanation

What a ratio says

A **ratio** compares two quantities of the same kind. In a class of 12 boys and 18 girls the ratio of boys to girls is 12 : 18, which simplifies to 2 : 3.

SITUATION	RATIO	SIMPLIFIED
12 boys, 18 girls	12 : 18	2 : 3
20 cm to 50 cm	20 : 50	2 : 5
4 red, 6 blue, 10 green	4 : 6 : 10	2 : 3 : 5

THE ORDER MATTERS

2 : 3 is boys to girls; 3 : 2 is girls to boys. Writing the ratio the wrong way round reverses every answer that follows.

Simplifying

Divide every part by their highest common factor — exactly as with a fraction. And convert to the same units first.

RATIO	FIRST STEP	SIMPLIFIED
12 : 18	divide by 6	2 : 3
50 cm : 2 m	50 : 200	1 : 4
30 min : 2 h	30 : 120	1 : 4
0.5 : 2	5 : 20	1 : 4
$\frac{1}{2} : \frac{1}{3}$	multiply by 6	3 : 2

DIFFERENT UNITS MAKE A RATIO MEANINGLESS

50 cm : 2 m is not 25 : 1 — it is 1 : 4. Converting first is not optional; it changes the answer completely.

Ratio against fraction

A ratio compares two parts with each other; a fraction compares one part with the whole.

STATEMENT	MEANING
boys to girls = 2 : 3	for every 2 boys there are 3 girls
boys are $\frac{2}{5}$ of the class	2 of every 5 pupils
girls are $\frac{3}{5}$ of the class	the parts add to 5
boys are $\frac{2}{3}$ as many as the girls	the ratio again, as a fraction

ADD THE PARTS TO GET THE WHOLE

A ratio of 2 : 3 has 5 parts altogether. Every fraction-of-the-whole question starts by adding the parts.

Reading a ratio back

RATIO	TOTAL PARTS	IF THE TOTAL IS 30
2 : 3	5	12 and 18
1 : 4	5	6 and 24
1 : 2 : 3	6	5, 10 and 15
3 : 7	10	9 and 21

ONE PART FIRST, THEN MULTIPLY OUT

Total ÷ number of parts gives one part; each share is that times its own number. Sharing in a ratio, four lessons from now, is exactly this table read forwards.

02 Worked examples

EXAMPLE 1 Simplify the ratio 50 cm : 2 m.

- 1 Convert to the same unit: 50 cm : 200 cm.
- 2 Divide both by 50.
- 3 = 1 : 4

ANSWER

1 : 4

EXAMPLE 2 A class has 12 boys and 18 girls. Write the ratio of boys to girls and the fraction of the class that is boys.

- ① Ratio: $12 : 18 = 2 : 3$.
- ② Total parts: $2 + 3 = 5$.
- ③ Boys are $\frac{2}{5}$ of the class.

ANSWER

$2 : 3$ and $\frac{2}{5}$

EXAMPLE 3 Simplify $\frac{1}{2} : \frac{1}{3}$.

- ① Multiply both parts by 6.
A ratio is unchanged by that.
- ② $3 : 2$

ANSWER

$3 : 2$

03 Interactive model

Count the boys and girls in the room and write the ratio on the board; the class simplifies its own numbers, which is more convincing than any textbook example.

Ratio, or fraction of the whole?

12 : 18 simplifies to:

2 : 3

3 : 2

6 : 9

1 : 2

50 cm : 2 m simplifies to:

25 : 1

1 : 4

50 : 2

1 : 40

With *12* boys and *18* girls, boys are what fraction of the class?

$\frac{2}{3}$

$\frac{2}{5}$

$\frac{3}{5}$

$\frac{1}{2}$

A ratio of 2 : 3 has how many parts?

2

3

5

6

2 : 3 and 3 : 2 are:

the same

different

both wrong

equal to $\frac{2}{3}$

$\frac{1}{2} : \frac{1}{3}$ simplifies to:

1 : 1

3 : 2

2 : 3

1 : 6

30 shared in $1 : 4$ gives:

6 and 24

10 and 20

5 and 25

15 and 15

30 min : 2 h simplifies to:

15 : 1

1 : 4

30 : 2

1 : 2

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Ratio	Nisbat	Отношение
Part	Ulush	Часть
To simplify a ratio	Nisbatni qisqartirish	Сократить отношение
Equivalent ratios	Teng nisbatlar	Равные отношения
Order of the terms	Hadlar tartibi	Порядок членов
Same units	Bir xil birlik	Одинаковые единицы
Total parts	Jami ulushlar	Всего частей
Comparison	Taqqoslash	Сравнение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A ratio compares:

a part with the whole

two quantities of the same kind

two units

nothing

Before simplifying, the units must be:

different

the same

metric

ignored

$20 : 50$ simplifies to:

$2 : 5$

$5 : 2$

$4 : 10$

$1 : 3$

In $2 : 3$ the first quantity is:

A ratio of $1 : 2 : 3$ has:

Boys to girls $2 : 3$ means girls are:

\frac{2}{5}
"/>

\frac{3}{5}
"/>

\frac{2}{3}
"/>

\frac{1}{2}
"/>

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify $12 : 18$

ANSWER $2 : 3$

2. Simplify $20 : 50$

ANSWER $2 : 5$

3. Simplify $4 : 6 : 10$

ANSWER $2 : 3 : 5$

4. Simplify $50 \text{ cm} : 2 \text{ m}$

ANSWER $1 : 4$

5. Simplify $30 \text{ min} : 2 \text{ h}$

ANSWER $1 : 4$

6. The parts of $2 : 3$

ANSWER 5

7. The parts of $1 : 2 : 3$

ANSWER 6

Medium · the standard question

7 PROBLEMS

1. 12 boys and 18 girls: the ratio

ANSWER 2 : 3

2. And the fraction that is boys

ANSWER $\frac{2}{5}$

3. Simplify 0.5 : 2

ANSWER 1 : 4

4. Simplify $\frac{1}{2} : \frac{1}{3}$

ANSWER 3 : 2

5. 30 shared in 1 : 4

ANSWER 6 and 24

6. 30 shared in 2 : 3

ANSWER 12 and 18

7. 30 shared in 1 : 2 : 3

ANSWER 5, 10, 15

Hard · stretch and reason

7 PROBLEMS

1. Simplify $750\text{ g} : 1.5\text{ kg}$

ANSWER $1 : 2$

2. Simplify $45\text{ s} : 3\text{ min}$

ANSWER $1 : 4$

3. If boys to girls is $3 : 7$, what fraction are girls?

ANSWER $\frac{7}{10}$

4. A ratio $2 : 3$ with the first quantity 14: the second

ANSWER 21

5. Simplify $\frac{2}{3} : \frac{4}{9}$

ANSWER $3 : 2$

6. Why must units match before simplifying?

ANSWER Otherwise the two numbers count different things

7. A ratio of $1 : 1$ means

ANSWER the two quantities are equal

07 Homework

Homework — 5 tasks

Convert to the same unit before you simplify anything.

1. Simplify $16 : 24$, $35 : 49$ and $18 : 27 : 45$.
2. Simplify $80\text{ cm} : 2\text{ m}$ and $20\text{ min} : 1\text{ h}$.
3. A class has 14 boys and 21 girls. Write the ratio and the fraction that is girls.
4. Share 40 in the ratio $3 : 5$.

5. Explain the difference between “boys to girls is 2 : 3” and “boys are $\frac{2}{3}$ of the class”.

Ratio and fraction

Moving between “three parts to five” and “three eighths of the whole”.

LESSON 42–43

2 × 40 MIN

MATEMATIKA 6, §9

S7 12

LESSON CLOCK

20'

25'

25'

8'

2

From ratio to fraction	20 min
From fraction to ratio	25 min
One quantity from the other	25 min
Mixed problems	8 min
Homework	2 min

LEARNING OBJECTIVES

- Write each part of a ratio as a fraction of the whole.
- Write a ratio from two fractions of a whole.
- Find one quantity from the other using the ratio.
- Solve problems mixing ratios and fractions.

TEXTBOOK

National	pp. 113–120
Cambridge	Stage 7, pp. 118–125
Workbook	Exercise 12.2

01

Explanation

From ratio to fraction

Add the parts of the ratio; each part is that many of the total.

RATIO	TOTAL PARTS	FIRST AS A FRACTION	SECOND AS A FRACTION
2 : 3	5	$\frac{2}{5}$	$\frac{3}{5}$
3 : 5	8	$\frac{3}{8}$	$\frac{5}{8}$
1 : 4	5	$\frac{1}{5}$	$\frac{4}{5}$
2 : 3 : 5	10	$\frac{2}{10} = \frac{1}{5}$	$\frac{3}{10}$ and $\frac{5}{10}$

THE DENOMINATOR IS THE TOTAL, NOT THE OTHER PART

In 2 : 3 the first quantity is $\frac{2}{5}$ of the whole — but $\frac{2}{3}$ of the second quantity. Both statements are true and they mean different things.

From fraction to ratio

If one part is $\frac{3}{8}$ of the whole, the rest is $\frac{5}{8}$, so the ratio is 3 : 5.

GIVEN	THE REST	RATIO
$\frac{3}{8}$ are boys	$\frac{5}{8}$	3 : 5
$\frac{2}{7}$ are red	$\frac{5}{7}$	2 : 5
$\frac{1}{3}$ is water	$\frac{2}{3}$	1 : 2
$\frac{3}{4}$ is full	$\frac{1}{4}$	3 : 1

A COMMON DENOMINATOR TURNS FRACTIONS INTO A RATIO

If $\frac{1}{2}$ is juice and $\frac{1}{3}$ is water, write both over 6: $\frac{3}{6}$ and $\frac{2}{6}$, so the ratio is 3 : 2 — with a sixth left over for something else.

One quantity from the other

PROBLEM	WORKING	ANSWER
boys to girls 2 : 3, 14 boys	one part = 7	21
ratio 3 : 5, the smaller is 12	one part = 4	20
ratio 4 : 7, the larger is 35	one part = 5	20
ratio 2 : 3, 14 boys: the class	$5 \cdot 7$	35

FIND ONE PART FIRST, EVERY TIME

Divide the known quantity by its own number in the ratio. That single number then answers every other question in the problem.

Mixed problems

PROBLEM	ANSWER
a drink is 1 part syrup to 4 water; syrup as a fraction	$\frac{1}{5}$
600 ml of that drink: the syrup in it	120 ml
$\frac{3}{8}$ of a class of 32 are boys: the ratio of boys to girls	3 : 5
and the number of girls	20

READ WHICH QUANTITY THE GIVEN NUMBER BELONGS TO

“Fourteen boys” is one part of the ratio; “a class of thirty-five” is the total. The two lead to different first steps, and misreading which is given accounts for most lost marks here.

02

Worked examples

EXAMPLE 1 Boys to girls is 2 : 3 and there are 14 boys. How many girls, and how many pupils?

1

14 boys is 2 parts, so one part is 7.

2

Girls: $3 \cdot 7 = 21$.

3

Class: $5 \cdot 7 = 35$.
Check: $14 + 21 = 35$ ✓

ANSWER

21 girls, 35 pupils

EXAMPLE 2 In a class $\frac{3}{8}$ are boys. Write the ratio of boys to girls.

① Girls are $1 - \frac{3}{8} = \frac{5}{8}$.

② Both are eighths, so compare the numerators.

③ $3 : 5$

ANSWER

$3 : 5$

EXAMPLE 3 A drink is 1 part syrup to 4 parts water. How much syrup is in 600 ml?

① Total parts: 5.

② Syrup is $\frac{1}{5}$ of the drink.

③ $600 \div 5 = 120$ ml.

ANSWER

120 ml

03 Interactive model

Mix a jug of squash in front of the class at 1 : 4 and ask for the fraction that is syrup; the fifth is visible in the jug.

Ratio to fraction and back

In 2 : 3 the first part is what fraction of the whole?

$$\frac{2}{3}$$

$$\frac{2}{5}$$

$$\frac{3}{5}$$

$$\frac{1}{2}$$

In 3 : 5 the second part is:

$$\frac{5}{8}$$

$$\frac{5}{8}$$

$$\frac{3}{8}$$

$$\frac{5}{3}$$

If 3 8 are boys, the ratio of boys to girls is:

3 : 8

3 : 5

5 : 3

8 : 3

Boys to girls 2 : 3 with 14 boys gives girls:

7

21

28

35

And a class of:

21

28

35

42

A drink 1 : 4 syrup to water is what fraction syrup?

$$\frac{1}{4}$$

$$\frac{1}{5}$$

$$\frac{4}{5}$$

$$\frac{1}{3}$$

600 ml of it contains syrup:

100 ml

120 ml

150 ml

480 ml

In 2 : 3, the first is what fraction of the second?

$$\frac{2}{5}$$

$$\frac{2}{3}$$

$$\frac{3}{2}$$

$$\frac{3}{5}$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Ratio	Nisbat	Отношение
Fraction of the whole	Butunning qismi	Часть от целого
Total parts	Jami ulushlar	Всего частей
One part	Bir ulush	Одна часть
To convert	O'girmoq	Преобразовать
Share	Ulush	Доля
Proportion	Proporsiya	Пропорция
Equivalent	Teng kuchli	Равносильный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To turn a ratio into fractions you:

multiply the parts

add the parts

subtract them

divide them

In $1 : 4$ the first part is:

$$\frac{1}{4}$$

$$\frac{1}{5}$$

$$\frac{4}{5}$$

$$\frac{1}{3}$$

If 2 7 are red, the ratio red to the rest is:

$2 : 7$

$2 : 5$

$5 : 2$

$7 : 2$

Ratio 3 : 5 with the smaller 12 gives the larger:

15

20

24

36

Ratio 4 : 7 with the larger 35 gives the smaller:

15

20

25

28

The first step in every such problem is:

add the parts

find one part

divide the total

multiply

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. In $2 : 3$ the first as a fraction

ANSWER $\frac{2}{5}$

2. In $3 : 5$ the second as a fraction

ANSWER $\frac{5}{8}$

3. In $1 : 4$ the first as a fraction

ANSWER $\frac{1}{5}$

4. If $\frac{3}{8}$ are boys, the ratio boys : girls

ANSWER $3 : 5$

5. If $\frac{2}{7}$ are red, the ratio red : rest

ANSWER $2 : 5$

6. The parts of $2 : 3 : 5$

ANSWER 10

7. In that ratio the last as a fraction

ANSWER $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. Boys to girls 2 : 3, 14 boys: girls

ANSWER 21

2. And the class

ANSWER 35

3. Ratio 3 : 5, smaller 12: larger

ANSWER 20

4. Ratio 4 : 7, larger 35: smaller

ANSWER 20

5. A drink 1 : 4: syrup in 600 ml

ANSWER 120 ml

6. $\frac{3}{8}$ of 32 are boys: the girls

ANSWER 20

7. And the ratio boys : girls

ANSWER 3 : 5

Hard · stretch and reason

7 PROBLEMS

1. If $\frac{1}{2}$ is juice and $\frac{1}{3}$ water, the ratio juice : water

ANSWER 3 : 2

2. Ratio 5 : 3, the difference is 16: the total

ANSWER 64

3. Ratio 2 : 5, the total is 63: the parts

ANSWER 18 and 45

4. In 2 : 3, the first is what fraction of the second?

ANSWER $\frac{2}{3}$

5. A mixture 2 : 3 : 5 of 400 g: the three parts

ANSWER 80, 120, 200 g

6. If boys are $\frac{4}{9}$ of a class, the ratio boys : girls

ANSWER 4 : 5

7. Why is the denominator the total and not the other part?

ANSWER A fraction of the whole counts every pupil

07 Homework

Homework — 5 tasks

Find one part first; every other number in the problem follows from it.

- In 3 : 7 write each part as a fraction of the whole.
- If $\frac{2}{5}$ of a class are boys, write the ratio of boys to girls.
- Boys to girls is 3 : 4 and there are 18 boys. Find the girls and the class size.

4. A drink is 2 parts syrup to 7 water. How much syrup is in 900 ml?
5. A ratio is 5 : 8 and the larger quantity is 40. Find the smaller and the total.

Word problems on ratio

Sharing an amount, scaling a recipe, and finding a quantity from a difference.

LESSON 44–46

3 × 40 MIN

MATEMATIKA 6, §10

S7 12

LESSON CLOCK

25'

30'

30'

25'

10

Sharing an amount	25 min
From one share to the total	30 min
From a difference	30 min
Recipes and mixtures	25 min
Homework	10 min

LEARNING OBJECTIVES

- Share a quantity in a given ratio.
- Find the total from one share or from the difference of two shares.
- Scale a recipe or a mixture up and down.
- Check that the shares rebuild the total.

TEXTBOOK

National	pp. 121–130
Cambridge	Stage 7, pp. 118–125
Workbook	Exercise 12.3

01

Explanation

Sharing an amount

STEP	SHARING 120 IN 2 : 3	
add the parts	$2 + 3 = 5$	
one part	$120 \div 5 = 24$	
the shares	48 and 72	
check	$48 + 72 = 120 \checkmark$	

AMOUNT	RATIO	SHARES
120	2 : 3	48, 72
200	3 : 7	60, 140
450	4 : 5	200, 250
96	1 : 3	24, 72

THE CHECK TAKES TWO SECONDS

Adding the shares must give the original amount. Nearly every arithmetic slip in this topic shows up in that one line.

From one share to the total

PROBLEM	ONE PART	TOTAL
ratio 2 : 3, the first share is 18	9	45
ratio 3 : 4, the second share is 28	7	49
ratio 5 : 2, the second share is 16	8	56
ratio 1 : 6, the first share is 9	9	63

DIVIDE THE SHARE BY ITS OWN NUMBER, NOT BY THE TOTAL PARTS

With ratio 2 : 3 and a first share of 18, one part is $18 \div 2 = 9$ — not $18 \div 5$. The share belongs to the 2, so the 2 is the divisor.

From a difference

The difference between two shares is the difference of their parts, times one part.

PROBLEM	PARTS OF DIFFERENCE	ONE PART	TOTAL
ratio 5 : 3, difference 16	2	8	64
ratio 7 : 4, difference 12	3	4	44
ratio 3 : 1, difference 30	2	15	60

ONE PART ANSWERS EVERYTHING

Whether the problem gives a total, one share or a difference, the first move is the same: turn what you are given into the value of one part.

Recipes and mixtures

PROBLEM	WORKING	ANSWER
concrete 1 : 2 : 4 cement, sand, gravel in 700 kg	one part = 100	100, 200, 400 kg
a recipe for 4 uses 600 g flour: for 6	150 · 6	900 g
paint 3 : 2 blue to white, 12 l of blue	one part = 4	8 l white
a drink 1 : 5 in 1.8 l	one part = 0.3	0.3 l syrup

KEEP THE UNITS WITH EVERY NUMBER

A recipe scaled from kilograms to grams by mistake is out by a factor of a thousand. Writing the unit beside each share is the cheapest protection there is.

02

Worked examples

EXAMPLE 1 Share 450 in the ratio 4 : 5.

1

4 + 5 = 9 parts.

2

450 ÷ 9 = 50 for one part.

3

4 · 50 = 200 and 5 · 50 = 250.
Check: 200 + 250 = 450 ✓

ANSWER

200 and 250

EXAMPLE 2 Two shares are in the ratio $5 : 3$ and differ by 16. Find both.

① The difference is $5 - 3 = 2$ parts.

② $16 \div 2 = 8$ for one part.

③ $5 \cdot 8 = 40$ and $3 \cdot 8 = 24$.

Check: $40 - 24 = 16$ ✓

ANSWER

40 and 24

EXAMPLE 3 Concrete is mixed $1 : 2 : 4$ by mass. How much of each is in 700 kg?

① $1 + 2 + 4 = 7$ parts.

② $700 \div 7 = 100$ kg for one part.

③ 100, 200 and 400 kg.

Check: they total 700 ✓

ANSWER

100, 200, 400 kg

03 Interactive model

Use a real recipe card and scale it for the whole class; the ratio work is identical to the textbook and the purpose is never in doubt.

Total, share or difference?

120 in 2 : 3 gives:

40, 80

48, 72

50, 70

60, 60

200 in 3 : 7 gives:

60, 140

70, 130

80, 120

100, 100

Ratio 2 : 3 with first share 18: one part is:

3.6

6

9

18

So the total is:

27

45

30

90

Ratio 5 : 3 with difference 16: one part is:

2

8

16

4

So the shares are:

40 and 24

30 and 14

50 and 34

20 and 4

Concrete 1 : 2 : 4 in 700 kg has cement:

50 kg

100 kg

175 kg

200 kg

The check on any sharing problem is:

the ratio

the shares add to the total

the difference

the units

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To share in a ratio	Nisbatda taqsimlash	Разделить в отношении
Share	Ulush	Доля
Difference	Ayirma	Разность
Recipe	Retsept	Рецепт
To scale	Miqyoslash	Масштабировать
Mixture	Aralashma	Смесь
Total	Jami	Всего
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first step in sharing is:

divide by the first part

add the parts

multiply

subtract

96 in $1 : 3$ gives:

24 and 72

32 and 64

48 and 48

16 and 80

Given one share, divide it by:

the total parts

its own number in the ratio

the other number

the total

Ratio 7 : 4 with difference 12: the total is:

A recipe for 4 scaled to 6 multiplies by:

Paint 3 : 2 with 12 l of blue needs white:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Share 120 in 2 : 3

ANSWER 48 and 72

2. Share 200 in 3 : 7

ANSWER 60 and 140

3. Share 450 in 4 : 5

ANSWER 200 and 250

4. Share 96 in 1 : 3

ANSWER 24 and 72

5. Ratio 2 : 3, first share 18: one part

ANSWER 9

6. And the total

ANSWER 45

7. Ratio 3 : 4, second share 28: the total

ANSWER 49

Medium · the standard question

7 PROBLEMS

1. Ratio 5 : 3, difference 16

ANSWER 40 and 24

2. Ratio 7 : 4, difference 12

ANSWER 28 and 16

3. Concrete 1 : 2 : 4 in 700 kg

ANSWER 100, 200, 400 kg

4. A recipe for 4 with 600 g flour: for 6

ANSWER 900 g

5. Paint 3 : 2, 12 l blue: the white

ANSWER 8 l

6. A drink 1 : 5 in 1.8 l: the syrup

ANSWER 0.3 l

7. Ratio 5 : 2, second share 16: the total

ANSWER 56

Hard · stretch and reason

7 PROBLEMS

1. Share 540 in 2 : 3 : 4

ANSWER 120, 180, 240

2. Ratio 4 : 9, difference 35: the total

ANSWER 91

3. A recipe for 6 uses 750 g: for 10

ANSWER 1250 g

4. Money shared 3 : 5 so that the smaller share is 45 000

ANSWER Total 120 000

5. Concrete 1 : 2 : 4 using 150 kg of cement: the total

ANSWER 1050 kg

6. Two numbers in ratio 4 : 7 add to 132

ANSWER 48 and 84

7. Why divide a known share by its own number?

ANSWER That share is exactly that many parts

07 Homework

Homework — 5 tasks

Add your shares back to the total before writing the answer down.

- Share 360 in the ratio 5 : 4.
- Share 840 in the ratio 1 : 2 : 4.
- Two shares are in the ratio 7 : 3 and differ by 24. Find both.
- A recipe for 5 people uses 400 g of rice. How much for 8?
- Paint is mixed 4 : 3 blue to white. With 20 litres of blue, how much white is needed?

The ratio of three quantities

Chaining two ratios into one, and sharing among three.

LESSON 47–49

3 × 40 MIN

MATEMATIKA 6, §11

S7 12

LESSON CLOCK

25'

30'

30'

25'

10

Sharing among three

25 min

Chaining two ratios

30 min

From one quantity to all three

30 min

Mixtures

25 min

Homework

10 min

LEARNING OBJECTIVES

• Share a quantity among three in a given ratio.

• Combine $a : b$ and $b : c$ into $a : b : c$.

• Find all three quantities from one of them.

• Solve mixture problems with three ingredients.

TEXTBOOK

National

pp. 131–140

Cambridge

Stage 7, pp. 118–125

Workbook

Exercise 12.4

01

Explanation

Sharing among three

Nothing new: add all three parts, find one part, multiply out.

AMOUNT	RATIO	PARTS	SHARES
300	1 : 2 : 3	6	50, 100, 150
700	1 : 2 : 4	7	100, 200, 400
540	2 : 3 : 4	9	120, 180, 240
480	3 : 5 : 8	16	90, 150, 240

THE CHECK IS THE SAME

All three shares must add to the original amount. With three numbers to add there is more to go wrong, so the check matters more, not less.

Chaining two ratios

Given $A : B = 2 : 3$ and $B : C = 4 : 5$, the two B s must be made equal before the ratios can be joined.

STEP	WORKING
the two ratios	$A : B = 2 : 3, B : C = 4 : 5$
make the B s match — LCM of 3 and 4 is 12	$A : B = 8 : 12, B : C = 12 : 15$
join them	$A : B : C = 8 : 12 : 15$

$A : B$	$B : C$	$A : B : C$
$2 : 3$	$3 : 4$	$2 : 3 : 4$
$2 : 3$	$4 : 5$	$8 : 12 : 15$
$1 : 2$	$3 : 7$	$3 : 6 : 14$

ONLY THE SHARED QUANTITY IS MATCHED

Scaling $A : B$ by 4 and $B : C$ by 3 changes the numbers but not the ratios — that is exactly why it is allowed. Scaling only one of the two would break them.

From one quantity to all three

RATIO	GIVEN	ONE PART	ALL THREE
$2 : 3 : 5$	the first is 12	6	12, 18, 30
$1 : 4 : 6$	the second is 20	5	5, 20, 30
$3 : 4 : 5$	the third is 35	7	21, 28, 35
$2 : 3 : 4$	the total is 180	20	40, 60, 80

ALWAYS ONE PART FIRST

Whatever is given — a share, the total, or a difference — the first line of working is the value of one part. Every three-quantity problem then finishes in one more line.

Mixtures

PROBLEM	WORKING	ANSWER
concrete $1 : 2 : 4$ using 150 kg of cement	one part = 150	1050 kg in all
a salad $3 : 2 : 1$ of 600 g	one part = 100	$300, 200, 100$ g
brass $7 : 3$ copper to zinc in 500 g	one part = 50	350 g copper
a mix $2 : 3 : 5$ with 60 g of the largest part	one part = 12	120 g in all

READ WHICH PART THE GIVEN AMOUNT IS

“Using 150 kg of cement” names the 1 ; “with 60 g of the largest part” names the 5 . The same number leads to very different totals depending on which part it is.

02

Worked examples

EXAMPLE 1 Share 540 in the ratio $2 : 3 : 4$.

1

$2 + 3 + 4 = 9$ parts.

2

$540 \div 9 = 60$ for one part.

3

$120, 180$ and 240 .
Check: they total 540 ✓

ANSWER

$120, 180, 240$

EXAMPLE 2 If $A : B = 2 : 3$ and $B : C = 4 : 5$, find $A : B : C$.

① The B s are 3 and 4; their LCM is 12.

② Scale the first by 4: $8 : 12$.

③ Scale the second by 3: $12 : 15$.

④ $A : B : C = 8 : 12 : 15$

ANSWER

$8 : 12 : 15$

EXAMPLE 3 Concrete is $1 : 2 : 4$ cement, sand and gravel. A builder uses 150 kg of cement. What is the total mass?

① Cement is 1 part, so one part is 150 kg.

② Total parts: 7.

③ $7 \cdot 150 = 1050$ kg.

ANSWER

1050 kg

03 Interactive model

Give the class two ratios sharing a middle quantity and ask them to join them without being shown how; most reach the LCM idea themselves.

Matching the shared quantity

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
3	<i>B in the first ratio</i>	$\times 4$
4	<i>B in the second</i>	$\times 3$

LOWEST COMMON DENOMINATOR

$$A : B : C = 8 : 12 : 15$$

The LCM of 3 and 4 is 12.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Three-part ratio	Uch hadli nisbat	Отношение трёх величин
To combine ratios	Nisbatlarni birlashtirish	Объединить отношения
Common term	Umumiy had	Общий член
Equivalent ratio	Teng nisbat	Равное отношение
Share	Ulush	Доля
Mixture	Aralashma	Смесь
Total parts	Jami ulushlar	Всего частей
Scaling	Miqyoslash	Масштабирование

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

300 in $1 : 2 : 3$ gives:

50, 100, 150

60, 100, 140

100, 100, 100

30, 60, 210

To join $A : B$ and $B : C$ you make equal:

the As

the Bs

the Cs

the totals

$2 : 3$ and $4 : 5$ join to:

$2 : 3 : 5$

$8 : 12 : 15$

$2 : 12 : 5$

$6 : 12 : 20$

Ratio $2 : 3 : 5$ with the first 12 gives the third:

18

24

30

36

Concrete $1 : 2 : 4$ with 150 kg cement totals:

450 kg

750 kg

1050 kg

1200 kg

The first line of working is always:

the total

one part

the ratio

the check

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Share 300 in 1 : 2 : 3

ANSWER 50, 100, 150

2. Share 700 in 1 : 2 : 4

ANSWER 100, 200, 400

3. Share 540 in 2 : 3 : 4

ANSWER 120, 180, 240

4. Share 480 in 3 : 5 : 8

ANSWER 90, 150, 240

5. The parts of 2 : 3 : 5

ANSWER 10

6. Ratio 2 : 3 : 5, first 12: one part

ANSWER 6

7. And all three

ANSWER 12, 18, 30

Medium · the standard question

7 PROBLEMS

1. $A : B = 2 : 3, B : C = 3 : 4$

ANSWER $2 : 3 : 4$

2. $A : B = 2 : 3, B : C = 4 : 5$

ANSWER $8 : 12 : 15$

3. $A : B = 1 : 2, B : C = 3 : 7$

ANSWER $3 : 6 : 14$

4. Ratio $1 : 4 : 6$, second 20

ANSWER $5, 20, 30$

5. Ratio $3 : 4 : 5$, third 35

ANSWER $21, 28, 35$

6. Concrete $1 : 2 : 4$ with 150 kg cement

ANSWER 1050 kg

7. A salad $3 : 2 : 1$ of 600 g

ANSWER $300, 200, 100$ g

Hard · stretch and reason

7 PROBLEMS

1. Brass 7 : 3 in 500 g: the copper

ANSWER 350 g

2. A mix 2 : 3 : 5 with 60 g of the largest part: the total

ANSWER 120 g

3. $A : B = 3 : 4$, $B : C = 6 : 7$: find $A : B : C$

ANSWER 9 : 12 : 14

4. Share 2 400 in 1 : 3 : 4

ANSWER 300, 900, 1200

5. Ratio 2 : 3 : 4, the difference between largest and smallest is 30

ANSWER 30, 45, 60

6. If $A : B = 2 : 5$ and $B : C = 10 : 3$, find $A : C$

ANSWER 4 : 3

7. Why must the shared quantity be matched before joining?

ANSWER Otherwise the two ratios measure B in different units

07 Homework

Homework — 5 tasks

One part first; then every share is one multiplication away.

- Share 720 in the ratio 2 : 3 : 7.
- If $A : B = 3 : 5$ and $B : C = 2 : 3$, find $A : B : C$.
- A mixture is 1 : 3 : 6 and uses 40 g of the first ingredient. Find the total mass.
- Three shares are in the ratio 4 : 5 : 6 and the largest is 42. Find the other two.
- Concrete 1 : 2 : 4 is needed to a total of 1 400 kg. Find each ingredient.

Variable ratios — direct and inverse proportion

When one quantity grows with another, and when it shrinks instead.

LESSON 50–52

3 × 40 MIN

MATEMATIKA 6, §12

S7 12

LESSON CLOCK

25'

30'

30'

25'

10

Direct proportion	25 min
The unitary method	30 min
Inverse proportion	30 min
Which is which?	25 min
Homework	10 min

LEARNING OBJECTIVES

- Recognise direct proportion and use the unitary method.
- Recognise inverse proportion and use the constant product.
- Decide which kind a situation is before calculating.
- Read a proportional relationship from a table.

TEXTBOOK

National	pp. 141–150
Cambridge	Stage 7, pp. 118–125
Workbook	Exercise 12.5

01

Explanation

Direct proportion

Two quantities are in **direct proportion** when doubling one doubles the other. Their quotient stays the same.

$$\frac{y}{x} = k, \text{ a constant}$$

PENS	1	2	5	8
cost, sum	3000	6000	15 000	24 000
cost ÷ pens	3000	3000	3000	3000

THE CONSTANT IS THE PRICE OF ONE

In every direct proportion the constant has a meaning: the cost of one pen, the distance in one hour, the mass of one metre. Finding it is the whole method.

The unitary method

PROBLEM	ONE UNIT	ANSWER
5 pens cost 15 000: 8 pens	3000	24 000
4 m of cloth costs 60 000: 7 m	15 000	105 000
a car does 180 km in 3 h: in 5 h	60 km	300 km
6 identical books weigh 2.4 kg: 10 books	0.4 kg	4 kg

DIVIDE, THEN MULTIPLY

Down to one, then up to what is asked. Two operations, in that order, solve every direct-proportion problem in the course.

Inverse proportion

Two quantities are in **inverse proportion** when doubling one halves the other. Their product stays the same.

$xy = k, \text{ a constant}$

WORKERS	2	3	4	6
days	12	8	6	4
product	24	24	24	24

THE CONSTANT IS THE SIZE OF THE JOB

24 worker-days is the work itself, whoever does it. Once that number is found, any number of workers can be answered by dividing.

Which is which?

SITUATION	KIND	CONSTANT
cost of n identical pens	direct	the price of one
distance in t hours at a fixed speed	direct	the speed
workers and days for one job	inverse	worker-days
speed and time for a fixed distance	inverse	the distance
taps and time to fill a tank	inverse	tap-minutes
a person's age and height	neither	—

NOT EVERYTHING THAT CHANGES TOGETHER IS PROPORTIONAL

A child twice as old is not twice as tall, and eight workers may get in each other's way. Proportion is a model, and the question is always whether it fits.

02

Worked examples

EXAMPLE 1 5 pens cost 15 000 sum. What do 8 cost?

1

More pens, more money — direct.

2

One pen: $15\,000 \div 5 = 3000$.

3

Eight pens: $8 \cdot 3000 = 24\,000$.

ANSWER

24 000 sum

EXAMPLE 2 4 workers finish a job in 6 days. How long would 8 take?

① More workers, fewer days — inverse.

② $4 \cdot 6 = 24$ worker-days.

③ $24 \div 8 = 3$ days.

ANSWER

3 days

EXAMPLE 3 A car covers 180 km in 3 hours at a steady speed. How far in 5 hours?

① Direct: more time, more distance.

② One hour: 60 km.

③ Five hours: 300 km.

ANSWER

300 km

03 Interactive model

Ask how long six taps take if three take twenty minutes; the wrong answer of forty is worth hearing aloud before the constant product is introduced.

Direct, inverse, or neither?

The cost of n identical pens:

Workers and days for one job:

5 pens cost 15 000; 8 cost:

4 workers take 6 days; 8 take:

12

3

2

4

Speed and time for a fixed distance:

direct

inverse

neither

constant

In direct proportion the constant is:

xy

$\frac{y}{x}$

$x + y$

$x - y$

In inverse proportion the constant is:

$$xy$$

$$\frac{y}{x}$$

$$x + y$$

$$y - x$$

A person's age and height are:

directly proportional

inversely proportional

not proportional

constant

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Direct proportion	To'g'ri proporsionallik	Прямая пропорциональность
Inverse proportion	Teskari proporsionallik	Обратная пропорциональность
Unitary method	Birlik usuli	Метод приведения к единице
Constant	O'zgarmas	Постоянная
Product	Ko'paytma	Произведение
Rate	Tezlik, narx	Норма
To increase	Ortmoq	Увеличиваться
To decrease	Kamaymoq	Уменьшаться

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Direct proportion keeps constant:

the product

the quotient

the sum

the difference

Inverse proportion keeps constant:

the product

the quotient

the sum

the difference

4 m of cloth costs 60 000; 7 m costs:

90 000

105 000

120 000

84 000

6 taps fill a tank in 4 h; 8 taps take:

3 h

5 h

2 h

6 h

The unitary method means:

multiply then divide

divide to one, then multiply

add

take the product

Doubling one quantity halves the other: this is:

direct

inverse

neither

impossible

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 5 pens cost 15 000: one pen

ANSWER 3000

2. So 8 pens cost

ANSWER 24 000

3. 4 m of cloth costs 60 000: 7 m

ANSWER 105 000

4. 180 km in 3 h: in 5 h

ANSWER 300 km

5. 6 books weigh 2.4 kg: 10 books

ANSWER 4 kg

6. Cost against number of pens is

ANSWER direct

7. Workers against days is

ANSWER inverse

Medium · the standard question

7 PROBLEMS

1. 4 workers take 6 days: 8 workers

ANSWER 3 days

2. 6 taps fill a tank in 4 h: 8 taps

ANSWER 3 h

3. 3 taps take 20 min: 6 taps

ANSWER 10 min

4. Speed and time for a fixed distance

ANSWER inverse

5. Distance and time at a fixed speed

ANSWER direct

6. Age and height

ANSWER neither

7. 12 workers take 5 days: 4 workers

ANSWER 15 days

Hard · stretch and reason

7 PROBLEMS

1. If y is inversely proportional to x and $y = 12$ at $x = 2$: y at $x = 8$

ANSWER 3

2. If y is directly proportional to x and $y = 12$ at $x = 2$: y at $x = 8$

ANSWER 48

3. 9 workers take 8 days; how many for 6 days?

ANSWER 12

4. Food lasts 30 people 12 days: for 40 people

ANSWER 9 days

5. 15 m of wire weighs 600 g: the mass of 4 m

ANSWER 160 g

6. Why is “more workers, more days” never the model?

ANSWER The job is fixed, so the product must be

7. A tap fills a tank in 12 min: two such taps together

ANSWER 6 min

07 Homework

Homework — 5 tasks

Name the kind of proportion before doing any arithmetic.

- 7 notebooks cost 42 000 sum. Find the cost of 11.
- A car travels 240 km in 4 hours. How far in 7 hours?
- 5 workers finish a job in 12 days. How long would 10 take?
- 4 taps fill a tank in 15 minutes. How long would 6 take?

5. Say which of these is direct, which inverse and which neither: cost and quantity; speed and time; a pupil's age and marks.

Control work 3 — ratio and proportion

Simplifying, sharing, chaining and proportion in one short paper.

LESSON 53

1 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 3

S7 12 REVIEW

LESSON CLOCK

2'

28'

8'

2'

Instructions

2 min

The paper

28 min

Answers and diagnosis

8 min

What comes next

2 min

LEARNING OBJECTIVES

- Simplify ratios, converting units first.
- Share a quantity in a two- and three-part ratio.
- Use direct and inverse proportion correctly.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National

pp. 109–150

Cambridge

Stage 7, pp. 118–125

Workbook

Control paper 3

01

Explanation

The paper — 20 marks, 28 minutes

Q	TASK	MARKS	FROM
1	Simplify $24 : 36$ and $75\text{ cm} : 3\text{ m}$	3	L41
2	In $3 : 5$ write each part as a fraction of the whole	3	L42–43
3	Share 480 in the ratio $5 : 7$	3	L44–46
4	Two shares are in the ratio $7 : 3$ and differ by 24: find both	3	L44–46
5	Share 540 in the ratio $2 : 3 : 4$	3	L47–49
6	7 books cost 49 000: find the cost of 11	3	L50–52
7	5 workers take 12 days: how long for 10?	2	L50–52

THE ANSWERS

$2 : 3$ and $1 : 4$; $\frac{3}{8}$ and $\frac{5}{8}$; 200 and 280; 42 and 18; 120, 180, 240; 77 000 sum; 6 days.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
units not converted	$75 : 3$	$75 : 300 = 1 : 4$
denominator taken as the other part	$\frac{3}{5}$ of the whole	$\frac{3}{8}$
total divided by the wrong number	$480 \div 5$	$480 \div 12$
difference treated as the total	$24 \div 10$	$24 \div 4$
three-part total miscounted	$2 + 3 + 4 = 8$	9
inverse solved as direct	24 days	6 days
no check	shares that do not add to the total	add them back

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	IT RETURNS IN
Q1–Q2	percentages, lesson 55 — a percentage is a ratio out of 100
Q3–Q5	the pie chart, lesson 160, where 360° is shared in a ratio
Q6	speed, lesson 117 — distance is directly proportional to time
Q7	every “more workers, fewer days” problem in the year

QUARTER I ENDS HERE

Directed numbers, algebra, the division of fractions and ratio are done. Quarter II opens with percentages, which uses all four.

02 Worked examples**EXAMPLE 1** Model answer, Q4: two shares in ratio 7 : 3 differ by 24.

① The difference is $7 - 3 = 4$ parts.
Not 10.

② $24 \div 4 = 6$ for one part.

③ $7 \cdot 6 = 42$ and $3 \cdot 6 = 18$.
Check: $42 - 18 = 24$ ✓

ANSWER

42 and 18

EXAMPLE 2 Model answer, Q6: 7 books cost 49 000; find 11.

① Direct proportion.

② One book: $49\,000 \div 7 = 7000$.

③ Eleven: $11 \cdot 7000 = 77\,000$.

ANSWER

77 000 sum

EXAMPLE 3 Model answer, Q7: 5 workers take 12 days; how long for 10?

① Inverse proportion — more workers, fewer days.

② $5 \cdot 12 = 60$ worker-days.

③ $60 \div 10 = 6$ days.

ANSWER

6 days

03 Interactive model

Return Q7 first and ask who doubled the days; naming the mistake as “solved an inverse problem as a direct one” makes it memorable.

The block in eight questions

24 : 36 simplifies to:

2 : 3

3 : 2

4 : 6

1 : 2

75 cm : 3 m simplifies to:

25 : 1

1 : 4

75 : 3

1 : 40

In 3 : 5 the first part is:

$\frac{3}{5}$

$\frac{3}{8}$

$\frac{5}{8}$

$\frac{5}{3}$

480 in 5 : 7 gives:

200, 280

240, 240

180, 300

220, 260

Ratio 7 : 3 with difference 24 gives one part:

2.4

4

6

8

540 in 2 : 3 : 4 gives:

120, 180, 240

100, 200, 240

135, 180, 225

90, 180, 270

7 books cost 49 000; 11 cost:

66 000

77 000

70 000

84 000

5 workers take 12 days; 10 take:

24

6

10

2.4

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Ratio	Nisbat	Отношение
To share	Taqsimlash	Разделить
One part	Bir ulush	Одна часть
Direct proportion	To'g'ri proporsionallik	Прямая пропорциональность
Inverse proportion	Teskari proporsionallik	Обратная пропорциональность
Units	O'lchov birligi	Единицы
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 requires first:

dividing

converting units

adding

multiplying

Q2 denominator is:

the other part

the total of the parts

the first part

10

Q3 divides 480 by:

5

7

12

35

Q4 divides 24 by:

10

7

3

4

Q6 is:

direct

inverse

neither

a ratio

Q7 is:

direct

inverse

neither

a ratio

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify $24 : 36$

ANSWER $2 : 3$

2. Simplify $75\text{ cm} : 3\text{ m}$

ANSWER $1 : 4$

3. In $3 : 5$, the first as a fraction

ANSWER $\frac{3}{8}$

4. Share 480 in $5 : 7$

ANSWER 200 and 280

5. Ratio $7 : 3$, difference 24

ANSWER 42 and 18

6. Share 540 in $2 : 3 : 4$

ANSWER 120, 180, 240

7. 7 books cost 49 000: 11 books

ANSWER 77 000

Medium · the standard question

7 PROBLEMS

1. 5 workers take 12 days: 10 workers

ANSWER 6 days

2. Simplify 40 min : 2 h

ANSWER 1 : 3

3. Share 320 in 3 : 5

ANSWER 120 and 200

4. Ratio 5 : 2, difference 18

ANSWER 30 and 12

5. 9 m of cloth costs 72 000: 5 m

ANSWER 40 000

6. 6 taps fill a tank in 10 min: 4 taps

ANSWER 15 min

7. In 2 : 3 : 5, the last as a fraction

ANSWER $\frac{1}{2}$

Hard · stretch and reason

7 PROBLEMS

1. $A : B = 2 : 3$, $B : C = 4 : 5$: find $A : B : C$

ANSWER $8 : 12 : 15$

2. Share 1 200 in $1 : 2 : 3 : 6$

ANSWER 100, 200, 300, 600

3. Food for 20 people lasts 9 days: for 15 people

ANSWER 12 days

4. A ratio $4 : 9$ with the total 91

ANSWER 28 and 63

5. Why is a ratio of $75 : 3$ wrong for $75 \text{ cm} : 3 \text{ m}$?

ANSWER The two numbers count different units

6. If 12 workers take 10 days, how many take 8?

ANSWER 15

7. Which of Q6 and Q7 keeps a constant product?

ANSWER Q7, the inverse one

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Simplify $45 : 60$ and $90 \text{ cm} : 2.7 \text{ m}$.
3. Share 660 in the ratio $4 : 7$.
4. 8 pencils cost 24 000 sum. Find the cost of 14.
5. 6 workers take 15 days. How long would 9 take?

Think — ratio in context

Scales on maps and plans, best value in a shop, and ratios that describe the world.

LESSON 54

1 × 40 MIN

MATEMATIKA 6, O'YLAB KO'R

S7 12 APPLICATION

LESSON CLOCK

12'

12'

10'

6'

Scales	12 min
Best value	12 min
Ratios in the world	10 min
Is the model reasonable?	6 min

LEARNING OBJECTIVES

- Read and use the scale of a map or plan.
- Compare prices by unit cost to find the better value.
- Interpret a ratio given in a real context.
- Judge whether a proportional model is reasonable.

TEXTBOOK

National	pp. 151–153
Cambridge	Stage 7, pp. 118–125
Workbook	Exercise 12.6

01

Explanation

Scales

A scale is a ratio: $1 : 100\,000$ means one centimetre on the map is $100\,000$ centimetres — one kilometre — on the ground.

SCALE	1 CM REPRESENTS	5 CM REPRESENTS
$1 : 100\,000$	1 km	5 km
$1 : 25\,000$	250 m	1.25 km
$1 : 50$	50 cm	2.5 m
$1 : 200$	2 m	10 m

CONVERT THE UNITS AT THE END, NOT IN THE MIDDLE

A scale of $1 : 25\,000$ turns 3 cm into $75\,000\text{ cm}$; only then divide by $100\,000$ to get 0.75 km . Converting halfway through is where the factors of ten go missing.

Best value

Compare the cost of one unit — or the amount for one sum — and the better buy is obvious.

OFFER	COST	PER UNIT	VERDICT
3 pens	12 000	4000	—
5 pens	18 000	3600	better
500 g of rice	9 000	18 a gram	—
2 kg of rice	32 000	16 a gram	better

THE BIGGER PACK USUALLY WINS – BUT NOT ALWAYS

Working out the unit cost takes ten seconds and settles it. Shops rely on customers assuming rather than dividing.

Ratios in the world

CONTEXT	RATIO	WHAT IT TELLS YOU
a map	1 : 50 000	how much the ground is shrunk
concrete	1 : 2 : 4	the recipe for a strong mix
a school	1 : 20 teachers to pupils	the size of a class
a photograph enlarged	1 : 3	every length trebled
gears on a bicycle	2 : 5	turns of the wheel per pedal turn

A RATIO COMPRESSES A WHOLE SENTENCE

“For every teacher there are twenty pupils” is one line of English; 1 : 20 is four characters, and it can be calculated with.

Is the model reasonable?

CLAIM	PROPORTIONAL?	COMMENT
3 pens cost three times one pen	yes	if the price is fixed
100 pens cost a hundred times one	usually not	bulk discounts
2 painters take half the time	roughly	up to a point
50 painters take a fiftieth of the time	no	they cannot all reach the wall
a child twice as old is twice as tall	no	growth is not proportional

EVERY MODEL HAS A RANGE WHERE IT WORKS

Proportion is an excellent description of prices and speeds, and a poor one of growth and crowded workplaces. Saying where a model stops working is part of using it well.

02

Worked examples

EXAMPLE 1 A map has scale 1 : 25 000. Two towns are 8 cm apart on it. How far apart are they really?

1

$8 \cdot 25\,000 = 200\,000$ cm.
Multiply by the scale.

2

$200\,000 \div 100\,000 = 2$.
Convert at the end.

3

2 km.

ANSWER

2 km

EXAMPLE 2 Which is better value: 3 pens for 12 000 or 5 for 18 000?

① $12\,000 \div 3 = 4000$ each.

② $18\,000 \div 5 = 3600$ each.

③ The pack of five is better value.

ANSWER

5 for 18 000

EXAMPLE 3 A plan of a room has scale 1 : 50. The room is 4 m long. How long is it on the plan?

① $4\text{ m} = 400\text{ cm}$.

② $400 \div 50 = 8$.

Divide, because the plan is smaller.

③ 8 cm.

ANSWER

8 cm

03 Interactive model

Bring a real map and a supermarket receipt; both are ratio problems, and neither looks like a textbook exercise.

Scales and best value

At $1 : 100\,000$, 1 cm is:

100 m

1 km

10 km

100 km

At $1 : 25\,000$, 8 cm is:

2 km

20 km

200 m

25 km

At $1 : 50$, a 4 m room is drawn:

8 cm

80 cm

2 cm

20 cm

3 pens for 12 000 cost each:

3000

4000

6000

36 000

5 for 18 000 cost each:

3600

4000

3000

9000

So the better value is:

3 for 12 000

5 for 18 000

the same

cannot say

50 painters taking a fiftieth of the time is:

reasonable

unreasonable

certain

proportional

A scale of $1 : 200$ means 1 cm is:

200 cm

2 m

both of these

200 m

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Scale	Masshtab	Масштаб
Map	Xarita	Карта
Plan	Reja	План
Unit cost	Birlik narxi	Цена за единицу
Best value	Eng foydali	Наиболее выгодно
Real distance	Haqiqiy masofa	Реальное расстояние
Model	Model	Модель
Reasonable	Maqbul	Разумный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A map scale is a:

fraction

ratio

percentage

proportion of area

To go from map to ground you:

divide by the scale

multiply by the scale

add it

square it

To go from ground to map you:

divide by the scale

multiply by the scale

subtract it

do nothing

Best value is found by comparing:

total prices

unit costs

pack sizes

shops

At 1 : 25 000, 4 cm is:

1 km

10 km

100 m

25 km

Proportion describes growth in height:

well

badly

exactly

always

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. At $1 : 100\,000$, 1 cm represents

ANSWER 1 km

2. At $1 : 25\,000$, 1 cm represents

ANSWER 250 m

3. At $1 : 50$, 1 cm represents

ANSWER 50 cm

4. At $1 : 100\,000$, 5 cm represents

ANSWER 5 km

5. 3 pens for $12\,000$: one pen

ANSWER 4000

6. 5 pens for $18\,000$: one pen

ANSWER 3600

7. Which is better value?

ANSWER The pack of five

Medium · the standard question

7 PROBLEMS

1. At $1 : 25\,000$, 8 cm represents

ANSWER 2 km

2. At $1 : 50$, a 4 m room is drawn

ANSWER 8 cm

3. 500 g for $9\,000$ or 2 kg for $32\,000$

ANSWER The 2 kg pack

4. A photo enlarged $1 : 3$: a 5 cm side becomes

ANSWER 15 cm

5. At $1 : 200$, 7 cm represents

ANSWER 14 m

6. A 12 m wall at $1 : 200$

ANSWER 6 cm

7. Teachers to pupils $1 : 20$ with 300 pupils

ANSWER 15 teachers

Hard · stretch and reason

7 PROBLEMS

1. At $1 : 50\,000$, a real distance of 3.5 km is drawn

ANSWER 7 cm

2. A map at $1 : 25\,000$ and one at $1 : 100\,000$: which shows more detail?

ANSWER The $1 : 25\,000$

3. 750 g for 15 000 or 1.2 kg for 22 800

ANSWER The 1.2 kg pack, at 19 a gram

4. A model car at $1 : 43$ is 10 cm long: the real car

ANSWER 4.3 m

5. Why is “100 pens cost 100 times one” often false?

ANSWER Bulk prices are usually discounted

6. Gears $2 : 5$: wheel turns for 30 pedal turns

ANSWER 75

7. At what scale is 1 cm equal to 1 m?

ANSWER $1 : 100$

07 Homework

Homework — 5 tasks

Convert units only at the end of the calculation.

- At a scale of $1 : 25\,000$, find the real distance for 6 cm.
- At a scale of $1 : 50$, find the plan length of a 7 m wall.
- Which is better value: 4 notebooks for 20 000 or 7 for 33 600?
- A map shows two towns 4.5 cm apart at $1 : 200\,000$. Find the real distance.
- Give one situation where proportion describes the world well and one where it does not.